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Measuring and Closing the Gap Between Declared and Real Urgency in a Supply Chain Digital Twin

*Cognitive adequacy scoring, forecast-assisted declaration,
and an opening toward satellite-observed capacity*

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Confidentiality and Use of Software Tools

Confidentiality. This thesis contains commercially sensitive information belonging to the ALTEN Group, including internal scoring parameters and the architecture of an unreleased software product. It is submitted for examination on a confidential basis and is marked *Company Confidential* throughout.

Use of software tools. Numerical results were produced with the Python codebase described in Section 3.3, using NumPy, SciPy, Matplotlib, Shapely, PyProj and Ultralytics. Large language models were used in two distinct and clearly separated capacities, both declared in full. First, as an *object of study*: the eight role-playing agents of the HÉLIOS campaign are language models, and this is an experimental design choice reported and discussed as such in Sections 3.4.3 and 4.7. Second, as a *writing aid* for language correction and $\text{\LaTeX}$ formatting. All scientific claims, all numerical results and all interpretations are my own, and every figure in this thesis was produced either by the project’s own code or by hand.

Abstract

In mutualised logistics networks the urgency an operator declares often diverges from the urgency computable from the operational state. Over-declaration wastes shared capacity; under-declaration hides rupture risk. No existing digital twin measures that divergence or propagates it across a supplier network.

This thesis builds and tests one. Real urgency is aggregated at each node from six KPI blocks through a weighted probabilistic OR and propagated upstream across a directed acyclic supply graph; declared urgency is elicited through a consistency-gated pairwise questionnaire and propagated downstream; their mismatch is scored on $[0, 100]$ with an asymmetric Prospect-Theory penalty. The system was evaluated through a pre-registered campaign replaying the 2020 to 2022 semiconductor shortage over nineteen turns and eight nodes, with a blind control arm and a forecast-assisted treatment arm on a frozen scenario.

The results are negative and diagnostically precise. The forecast layer shows no discrimination above chance and is severely miscalibrated, at a Brier skill between -3.4 and -6.9 against a base-rate forecaster. Forecasts below 0.1 are well calibrated, while twenty-one announcing a mean probability of 0.948 were followed by the predicted outcome exactly zero times. One block explains it: the time block reads a lead-time distribution that no shock ever reaches. In the treatment arm, on the turn the forecast first became visible, all eight declarants revised their urgency downward or held it and none revised upward, two turns before the scenario's critical event. Both arms carry identical forecasts, so the effect belongs to the display. Substituting a plausible but naive outcome definition improves the apparent skill fivefold while leaving discrimination unchanged.

Keywords: supply chain digital twin, urgency adequacy, cognitive bias, multi-criteria decision analysis, forecast calibration, automation bias, serious game.

Résumé

Dans les réseaux logistiques mutualisés, l'urgence qu'un opérateur déclare diverge souvent de celle que l'on peut calculer à partir de l'état opérationnel. La sur-déclaration gaspille de la capacité partagée ; la sous-déclaration dissimule un risque de rupture. Aucun jumeau numérique existant ne mesure cette divergence ni ne la propage à l'échelle d'un réseau de fournisseurs.

Ce mémoire en construit un et le met à l'épreuve. L'urgence réelle est agrégée à chaque nœud à partir de six blocs de KPI par un OU probabiliste pondéré, puis propagée vers l'aval sur un graphe orienté acyclique ; l'urgence déclarée est élicitée par un questionnaire de comparaisons par paires soumis à un contrôle de cohérence, puis propagée vers l'amont ; leur écart est noté sur $[0, 100]$ par une pénalité asymétrique issue de la théorie des perspectives. Le système a été évalué par une campagne pré-enregistrée jouant la pénurie de semi-conducteurs de 2020 à 2022 sur dix-neuf tours et huit nœuds, avec un bras témoin en aveugle et un bras de traitement assisté par prévision, sur un scénario gelé.

Les résultats sont négatifs et précis sur le plan diagnostique. La couche de prévision ne discrimine pas mieux que le hasard et se révèle fortement mal calibrée, avec un *skill* de Brier compris entre $-3,4$ et $-6,9$ face à un prévisionniste annonçant le taux de base. Les prévisions inférieures à $0,1$ sont bien calibrées, tandis que vingt-et-une annonçant une probabilité moyenne de $0,948$ n'ont été suivies de l'issue prédite exactement zéro fois. Un seul bloc l'explique : le bloc temporel lit une distribution de délai qu'aucun choc n'atteint jamais. Dans le bras de traitement, au tour où la prévision devient visible pour la première fois, les huit déclarants ont tous revu leur urgence à la baisse ou l'ont maintenue, aucun ne l'a revue à la hausse, deux tours avant l'événement critique du scénario. Les deux bras portent des prévisions identiques : l'effet tient donc à l'affichage. Substituer une définition d'issue plausible mais naïve multiplie par cinq le *skill* apparent sans modifier la discrimination.

Mots-clés : jumeau numérique de chaîne d'approvisionnement, adéquation d'urgence, biais cognitif, aide multicritère à la décision, calibration de prévisions, biais d'automatisation, jeu sérieux.

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1 Introduction

1.1 Industrial and Technical Context

Supply chain planning has become a multi-dimensional assessment problem. An operator deciding which of two late shipments to expedite is simultaneously weighing cost, deadline pressure, supplier reliability and, increasingly, carbon footprint, while most monitoring and planning tools still process one criterion at a time (Pinedo, 2012; Chakraborty et al., 2020). That mismatch between the dimensionality of the decision and the dimensionality of the tooling is precisely why practitioners fall back on judgment-based prioritisation, and why the quality of that judgment has become an engineering concern rather than a purely human one.

This thesis was produced during an R&D placement in the Direction Innovation Numérique (DIN) of the ALTEN Group, which conducts applied research where digital twin technology, logistics optimisation and sustainable transport meet. Its Smart Green Supply Chain programme (Ben Zakour, 2024) pursues three aims at once. The lean aim attacks waste, meaning cost, delay and over-stock, through pull-flow logic and real-time scheduling. The green aim attacks the carbon footprint of each transport leg, working to the Global Logistics Emissions Council (GLEC) Framework v3.2 (Smart Freight Centre, 2023) and ISO 14083:2023 (International Organization for Standardization, 2023). The predictive aim is the one this thesis sits inside: anticipating disruption through probabilistic modelling, in the manner of a digital supply chain twin (Ivanov; Dolgui, 2020).

Behind all three is Industry 4.0 (Kagermann et al., 2013), and its promise that cyber-physical systems, the Internet of Things (IoT) and digital twins can turn a reactive supply chain into a self-organising one. The Physical Internet (Montreuil, 2011) supplies the setting for the simulation environment used here, since it imagines logistics networks that route shipments through shared infrastructure the way a network routes packets.

1.2 The DISCO Digital Twin and Prior Research at ALTEN-DIN

DISCO, the Supply Chain Illustrator, is a serious game and digital twin developed by the DIN to simulate a complete supply chain for a fictitious satellite product. It involves 22 component suppliers across structural assemblies, optical systems and embedded electronics (Baxas, 2025). Players take the role of supply chain buyers and decide on supplier selection, transport routes and factory allocation. Figure 1 shows the supplier network they work with and the three dimensions their decisions are scored on, and Table 1 summarises the technology stack underneath it.

Its scoring engine applies Min-Max normalisation over the values of the current game session, illustrated here on the carbon dimension:

$$\text{ScoreCO}_2 = 1 + 9 \times \frac{\text{realCO}_2 - \text{co}_{2,\min}}{\text{co}_{2,\max} - \text{co}_{2,\min}}, \quad (1)$$

with analogous formulae for cost and risk, on a scale from 1 for the best achievable performance to 10 for the worst.

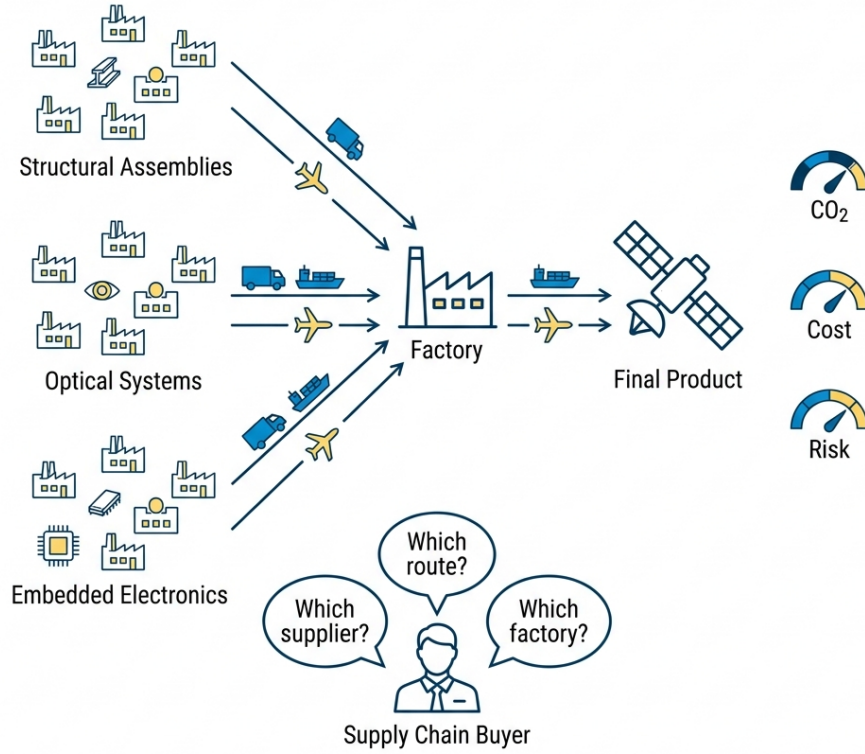


Figure 1: The DISCO serious game network

Table 1: DISCO technology stack

Component	Technology / Value
Backend API	Spring Boot + Java 21
Database	PostgreSQL (Azure)
Frontend	Angular
Authentication	Player key / Azure AD SSO OAuth2
Scores computed	CO ₂ , Cost, Risk per supplier and itinerary, on $[1, 10]$

This work sits within a body of prior DIN research. Baxas (2025) defines the DISCO software architecture including the Time To Service (TTS) and Time To Recovery (TTR) formalisation. Baulain et al. (2024) developed the Physical Internet routing module, providing transit times through the HERE Maps API with Vincenty distances and mandatory Heavy Goods Vehicle (HGV) pause compliance, together with the Itinerary Performance Index (Itinerary Performance Index (IPG)) normalised on $[0, 1]$. Kabouche (2025) developed the supplier risk scorer combining delay history, geographic vulnerability and financial fragility. Cordova (2024) developed the per-route CO₂ quantification module using the GLEC and VECTO methodology. Mondelice (2025) developed the Transport Management System interface providing real-time transit-time extraction.

These works form the organisational and thematic context of the present research. The system delivered here, named SupplyScore and described in Section 3.3, is a standalone Python application that runs alongside these modules rather than inside them. Their coupling is a stated perspective of the programme, not a delivered capability, and this thesis makes no

claim to the contrary.

1.3 Problem Statement

This thesis addresses the layer that comes *before* any planning or scheduling decision, and which the classical literature has not fully investigated: the urgency signal itself is systematically biased at the source, and must be measured and corrected before it can safely inform any downstream decision (M. Zhu et al., 2018; Rusou et al., 2020).

The consequences are worse where capacity is shared. In an interconnected network, a biased urgency signal does not stay local: it propagates, and misallocation grows as it travels. Over-triage, declaring more urgency than the situation warrants, takes capacity from whoever needed it more. Under-triage does the opposite and is more dangerous, because a risk nobody has declared is a risk nobody is watching (Li et al., 2020; Ivanov; Dolgui, 2020).

Digital twin architectures and real-time decision support systems have demonstrated strong capabilities in operational monitoring, disruption detection, resource reallocation and order synchronisation (Ding et al., 2023; Leung et al., 2022; Hrušovský et al., 2021; Karam et al., 2022; X. Zhu et al., 2025). They do so primarily by computing the physical operational state of the system. To the best of the review presented in Section 2, existing systems do not explicitly model the gap between a declared urgency and a computed operational urgency.

Figure 2 shows what that gap looks like over the life of a task. Two curves are plotted against time. The step line is declared urgency, which moves only when the operator submits a review, and the smooth curve is real urgency, which moves as the operational state does. Reading left to right, the figure passes through four phases: the two signals agree, then a first adequacy defect opens, then a second defect of the opposite sign, then agreement returns. Shading tells the two defects apart. Where the step line sits above the curve the operator is claiming more urgency than the state justifies, and the enclosed area is false urgency. Where the curve sits above the step line the state is worse than anyone has declared, and the enclosed area is hidden risk. Leader lines pointing at a curve label that signal; leader lines pointing into a shaded area label the gap between them.

Formally, let $U_d \in [0, 1]$ denote the declared urgency and $U_r(t) \in [0, 1]$ the real urgency computed from operational data in the digital twin. The mismatch between the two signals is

$$\Delta U(t) = U_d - U_r(t) \in [-1, 1]. \quad (2)$$

A tolerance band $\varepsilon \in [0, 1]$ is introduced so that small deviations are not treated as operationally meaningful. Over-declaration then corresponds to $\Delta U(t) > \varepsilon$, under-declaration to $\Delta U(t) < -\varepsilon$, and approximate agreement to $|\Delta U(t)| \leq \varepsilon$. Table 2 gives the operational reading of each condition.

The two conditions need not cost the same. Weighting them asymmetrically, with $\lambda_{\text{under}} \geq \lambda_{\text{over}}$, prices under-prioritisation above over-prioritisation. Two bodies of work support doing so: operations research showing that decision errors carry unequal downstream consequences (Kim et al., 2020; DiwasSingh et al., 2020), and scheduling work that models differentiated tardiness penalties instead of a uniform delay cost (Baker et al., 1982; Chauhan et al., 2022). How much asymmetry is right is a calibration question, answerable only in a specific setting.

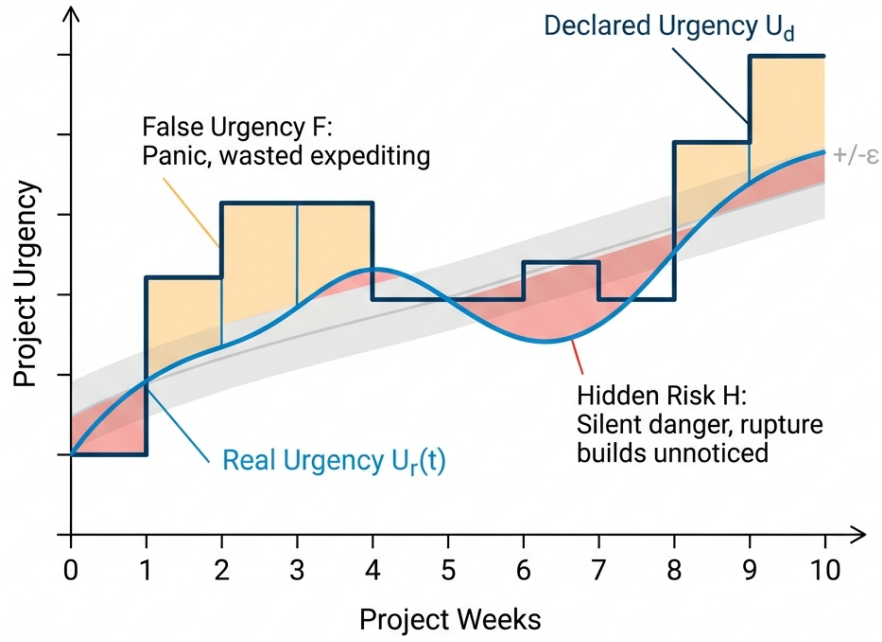


Figure 2: Declared and real urgency, and the two mismatch pathologies

Table 2: Urgency pathologies and their consequences

Condition	Interpretation	Potential consequence
$\Delta U(t) > \varepsilon$	Over-declaration of urgency, or false urgency	Excess expediting, avoidable cost, possible delay for other flows
$\Delta U(t) < -\varepsilon$	Under-declaration of urgency, or hidden risk	Hidden lateness or stockout risk, delayed intervention, downstream disruption
$ \Delta U(t) \leq \varepsilon$	Acceptable alignment	No material urgency mismatch

1.4 Aim, Objectives and Research Questions

The aim of this research is to establish whether the divergence between declared and computed urgency is measurable to a standard that would justify acting on it.

That wording is deliberate on two counts. It commits to a falsifiable target rather than to building a system: a score that cannot beat a trivial baseline against recorded outcomes has not met the standard, however well engineered the system producing it. And it makes measurability the question rather than assuming it, because the two signals being compared are not independent by construction, and an architecture that lets the same person supply both sides is measuring its own input.

Three objectives follow from that aim:

- O1.** Define a multi-block, DAG-propagated real urgency $U_r(t)$ and a cognitive adequacy score $A(t) \in [0, 100]$ measuring the gap between the planner's declared urgency U_d and the digital twin's operational state.
- O2.** Develop and implement a network propagation and criticality-driven prioritisation layer,

combining PROMETHEE II outranking with systematic shock simulation, to identify structurally critical nodes.

- O3.** Validate the resulting signal empirically, against recorded outcomes, and measure what happens to human declaration behaviour when the system's own forecast is placed in front of the declarant.

Four research questions structure the scientific approach:

- RQ1.** *Multi-block real urgency.* How can a physically and operationally grounded real urgency $U_r(t)$ be computed dynamically in a multi-tenant digital twin from heterogeneous KPI blocks, covering time, capacity, performance, risk, cost and environmental dimensions, and propagated coherently across a multi-tier supply chain graph?
- RQ2.** *Cognitive adequacy formulation.* How should an adequacy score $A(t)$ be formulated to measure the alignment between declared urgency U_d and computed real urgency $U_r(t)$ while asymmetrically penalising under-triage against over-triage, and does that score demonstrate predictive validity against recorded operational outcomes?
- RQ3.** *Network prioritisation and criticality.* How can pairwise elicitation methods, namely AHP and the Fuzzy Best-Worst Method, be combined with outranking aggregation through PROMETHEE II and systematic shock simulation to identify which nodes of a supply network are structurally critical?
- RQ4.** *Operational validation and the effect of showing the forecast.* How can this framework be operationalised and empirically validated within a persistent digital twin exercised as a weekly-cadence serious game, and what happens to declared urgency once the declarant is shown the system's own prediction?

The second half of RQ4 is not an afterthought. The campaign produced a result about the declarant rather than only about the model, and reporting the model's scores without it would be misleading (Section 4.7).

1.5 Research Hypotheses

These six hypotheses were pre-registered. They were written, with their numeric thresholds, and frozen before the campaign of Section 3.4 produced any data, which is why they appear here rather than next to the results they decide. They are reproduced unchanged. Where the campaign later showed a threshold to be badly chosen, or a hypothesis to be untestable on the scenario as scripted, that is reported as a finding in Section 5.2 and not repaired by rewriting the hypothesis.

Each states an observable outcome with a threshold the delivered system either meets or does not, and each serves one of the research questions above.

H1 (Cognitive latency): Declared urgency follows computed real urgency with a delay of at least one weekly cycle on the majority of nodes. Test: the lagged correlation between U_d and U_r peaks at a lag of one week or more for at least five of the eight campaign nodes.

H2 (Early detection of hidden risk): The hidden-risk signal $H = \max(U_r - U_d, 0)$ exceeds 0.3 at the crisis-origin node several weeks before the rupture becomes publicly visible.

H3 (Structural criticality): The systematic shock simulation ranks the true crisis origin first, by impact on the final customer, on at least 80% of the informative, non-saturated

weekly rounds.

H4 (Predictive validity): At threshold $H \geq 0.5$ with a four-week horizon, the hidden-risk signal predicts recorded negative outcomes, meaning a missed milestone, a critical event or a node abandonment, with precision and recall both above 0.5.

H5 (Urgency contagion): On weeks when crisis news becomes public, operators of nodes that are not physically affected still raise their declared urgency, reproducing the mere-urgency effect (M. Zhu et al., 2018) at network scale.

H6 (Perception hysteresis): Once the physical situation improves, declared urgency decays at less than half the rate of real urgency.

Table 3 maps each hypothesis to the question it serves and states what its failure would mean, so that a negative result carries a consequence rather than just a verdict.

Table 3: Each hypothesis, its research question, and the cost of failure

Hypothesis	Serves	If it fails
H1 cognitive latency	RQ2	Declared urgency tracks the physical signal without delay, so there is no lag to correct and the adequacy layer has nothing to add.
H2 early detection	RQ2	The hidden-risk signal arrives no earlier than the rupture it is meant to anticipate, which makes it a description rather than a warning.
H3 structural criticality	RQ3	The network ranking cannot be trusted to identify the bottleneck, so criticality-driven prioritisation is unsafe to act on.
H4 predictive validity	RQ2	The signal does not predict recorded outcomes, and the score is a diagnostic curiosity rather than a decision aid. This is the hypothesis the aim of Section 1.4 stands or falls on.
H5 urgency contagion	RQ1, RQ4	Operators do not import urgency from context, so declaration bias is individual rather than systemic and does not propagate.
H6 perception hysteresis	RQ2	Over-declaration corrects itself once the situation eases, so the asymmetric penalty is mis-specified and λ_{under} needs recalibrating.

One omission from this register is worth flagging in advance, because Section 4.6 measures its cost. The thresholds were pre-registered but the definition of an adverse outcome was not, and that definition turns out to move the measured result by more than the thresholds do.

Distinct from these testable hypotheses, the framework rests on stated modelling assumptions, each verifiable in the delivered code: the network is a directed acyclic graph, so there are no feedback loops and a linear topological sort is valid; reviews follow a strict ISO weekly cycle; declared urgency is damped as it descends toward suppliers while real risk is amplified as it ascends toward the customer, each scaled by a per-arc coefficient; the six KPI blocks are treated as independent so that a probabilistic aggregation is admissible; task statuses drive explicit override rules; and the cognitive loss is deliberately asymmetric, with under-declaration priced at roughly twice the cost of panic. These are design commitments, not

empirical claims, and the limitations they induce are catalogued in Section 5.3.

1.6 Scope and Delimitations

Four boundaries are set explicitly, because each of them shapes what the results in Section 4 can and cannot support.

The declarants are language-model agents, not human consultants. The campaign reported here was run with eight isolated language-model agents, one per role, each confined to its own role card and turn briefings. This is an experimental design choice with real consequences for external validity, discussed in Sections 3.4.3 and 4.8. The human campaign for which the protocol was originally frozen has not been run.

The validation is a single scenario. All results come from one pre-registered scenario, a semiconductor shortage propagating through an eight-node satellite supply chain over nineteen weekly turns. Statements about how the framework behaves are statements about its behaviour on that scenario.

SupplyScore is standalone. It does not integrate with DISCO, with an ERP, or with the routing and carbon modules of Section 1.2. Every KPI reaching the model arrives either through a manual weekly review or through a scripted campaign event.

The satellite observation work is an opening, not a second contribution. Section 4.9 reports a separate system that estimates warehouse capacity from public satellite imagery. It is included because it addresses the freshness limitation that the rest of this thesis documents about itself, and it is reported at the depth that claim requires, without any assertion that the two systems have been connected.

1.7 Thesis Structure

Four sections follow. Section 2 reviews the streams that bear on the problem and states the gap each one leaves. Section 3 keeps three things apart that are easily confused: the mathematical model, the choices made in building it into software, and the experiment designed to test it. It closes with the method's own limits, its reproducibility provisions and its ethics. Section 4 reports what was measured, then diagnoses it. Section 5 states what the work contributes, what each hypothesis did, where the work stops, and what it opens.

2 Literature Review

2.1 Introduction and Key Definitions

This review answers a single question: which existing methods can be assembled to measure, correct and propagate a biased urgency signal, and which part of that assembly is missing from the literature? The relevant work is broad but fragmented across operations research, behavioural science and digital twin engineering. Read individually, each stream solves one piece of the problem. Read together, they leave a specific slot empty. No retrieved work combines, within a single supply chain decision architecture, the five elements this thesis requires: a multi-block operational urgency function, a structured client-side urgency input, an adequacy layer comparing stated and operational urgency, network propagation of that urgency across a multi-tier DAG, and empirical validation of the adequacy signal against recorded outcomes (Sawik, 2013; Sawik, 2014; Sawik, 2016; Cohen et al., 2019; Hosseini; Ivanov, 2020).

The review is organised by the role each stream plays in that argument rather than by chronology or method family. Figure 3 maps the streams onto that organisation and shows where the empty slot falls. Each section closes with a statement of the gap it leaves, and Section 2.14 collects those gaps.

Six terms are used throughout with the following meanings.

Urgency is how far a task needs prompt action to avoid a material loss of operational performance. Two neighbouring words mean something else. *Importance* is about strategic value; *priority* is about current execution order. A task can be important and not urgent, and it can be first in the queue for reasons that have nothing to do with either (J. Chen et al., 2025; M. Zhu et al., 2018).

Real-time scheduling refers to the class of algorithms that assign execution order to tasks given temporal constraints, resource constraints and optimality criteria (Pinedo, 2012). This literature is drawn on only as historical grounding for deadline-sensitive value functions; no scheduling algorithm is delivered here.

Digital twin is a synchronised digital representation of a physical system supporting monitoring, simulation and decision-making from real-time and historical data (Ivanov; Dolgui, 2020).

Multi-criteria decision-making (Multi-Criteria Decision-Making (MCDM)) covers methods that combine multiple, potentially conflicting performance dimensions into a decision (Saaty, 1980; Brans; Vincke, et al., 1986).

Physical Internet (Physical Internet (PI)) is the model of open, mutualised logistics networks routing shipments through shared infrastructure using standardised modular containers (Montreuil, 2011).

Serious game is a simulation environment designed to train decision-makers through realistic scenarios with immediate feedback and scoring (Sterman, 1989; Prensky, 2001).

2.2 Risk Propagation and Deadline-Sensitive Value

Within real-time systems, deadline-sensitive scheduling theory models task value as a function of completion or response time, an approach known as time-value or utility-accrual

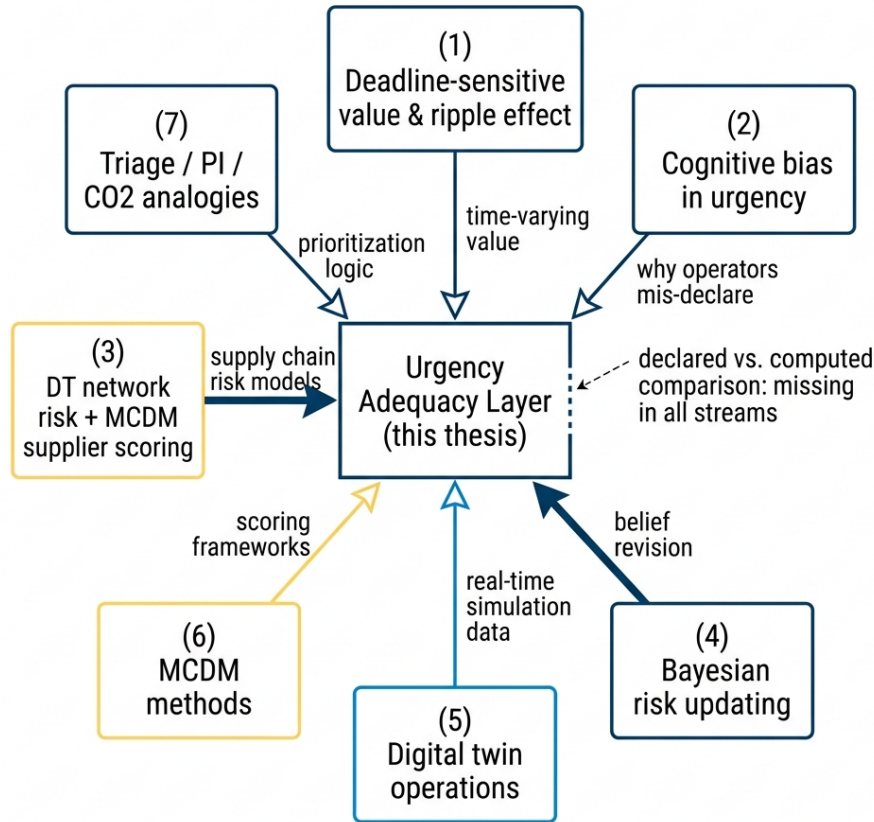


Figure 3: Map of the literature streams

scheduling (K. Chen et al., 1996; J. Chen et al., 2025). Chen and Mühlethaler (K. Chen et al., 1996) show that task utility is typically constant up to a deadline and then non-increasing with delay, and a broader survey confirms that deadline-sensitive literature models task value as a function of response time with bounded, structured degradation after critical thresholds (J. Chen et al., 2025). This establishes the precedent for treating urgency as time-varying, but it stops at a single task in isolation.

A complementary line of evidence comes from network risk propagation. Ivanov and colleagues formalise the ripple effect, whereby a local disruption at one supplier cascades through downstream tiers and amplifies as it travels (Ivanov; Dolgui, 2020; Ivanov; Dolgui, 2019; Li et al., 2020). Probabilistic risk inference for supply chain systems (Zheng et al., 2020a) further shows that aggregating heterogeneous risk indicators into a single node-level score is itself an open modelling question, distinct from how that score should then be propagated.

Table 4 compares four candidate aggregation and growth models found across these two lines.

The two block-level families differ in what they hide. A weighted average is fully compensatory, so a catastrophic reading on one indicator can be offset by acceptable readings elsewhere. A noisy-OR is not, because any factor approaching its maximum drives the aggregate toward its own maximum regardless of the others. Which behaviour is desirable depends on whether

Table 4: Aggregation and growth models found in the literature

Model	Formula	Key property	Where it appears
Linear in time	$U_0 + \lambda t$	Constant pressure	Deadline-sensitive scheduling (K. Chen et al., 1996)
Hyperbolic in time	$K/(t_c - t)$, bounded	Finite-time singularity, discounting	Temporal discounting (Steel et al., 2006); crash precursors (Johansen et al., 2001)
Weighted average of blocks	$\sum_m \omega_m u_m$	Fully compensatory: one weak block is hidden by strong ones	Composite indicators (Greco et al., 2018)
Noisy-OR over blocks	$1 - \prod_m (1 - u_m)^{\omega_m}$	Non-compensatory: one extreme block dominates	Combining independent causal evidence (Pearl, 1988)

the aggregate is meant to describe average condition or exposure to the worst active cause, and the composite-indicator literature treats that as a modelling decision rather than a solved question (Greco et al., 2018; Becker et al., 2017).

The hyperbolic family carries a further caveat. A structurally similar acceleration appears in Log-Periodic Power Law (Log-Periodic Power Law (LPPL)) models of financial crash precursors (Johansen et al., 2001; Johansen et al., 2000), but empirical analyses find LPPL-family models highly parameter-sensitive and unsuited to use as production risk models (Feigenbaum, 2001; Brée et al., 2010).

Gaps in risk propagation and deadline-sensitive value. Neither stream specifies how to aggregate heterogeneous, block-structured operational risk indicators at a single node before propagating them. Neither addresses declared urgency, or the mismatch between what is stated and what is computable. And the aggregation literature stops at the observation that the choice matters, without establishing whether equal values on structurally different indicators carry equal risk meaning, which is the question any multi-block aggregate has to answer before it can be interpreted.

2.3 Cognitive Bias in Urgency Declaration

Behavioural operations research suggests that human decision-makers systematically misrepresent urgency, a finding with direct implications for any system relying on operator-declared urgency as an input.

Two convergent lines show this from different angles. M. Zhu et al. (2018) report the Mere-Urgency Effect in five controlled experiments: participants consistently over-prioritised imminent low-value tasks over distant high-value tasks, and the bias persisted even when participants were informed of its existence. Steel et al. (2006) provides a complementary formal model through Temporal Motivational Theory (Temporal Motivational Theory (TMT)):

$$V(t) = \frac{U \times E}{1 + \Gamma \times D(t)^k}, \quad (3)$$

where U is objective utility, E is the expectancy of success, $D(t)$ is the delay to deadline, and Γ and k are individual discounting parameters. As the delay to deadline shrinks toward zero, the motivational value $V(t)$ converges to the full product of utility and expectancy, so immediacy becomes more salient as the deadline approaches. Near-term tasks can therefore attract disproportionate attention even when their broader objective value is lower (Steel et al., 2006). Table 5 gives the operational role of each parameter.

Table 5: Temporal Motivational Theory parameters

Symbol	Concept	Operational role
$V(t)$	Temporal motivational value	Dynamic task salience
U	Objective utility	Task strategic importance
E	Expectancy of success	Probability of task completion
$D(t)$	Delay to deadline	Remaining safe execution window
Γ	Impulsiveness parameter	Operator sensitivity to delay
k	Sensitivity scaling parameter	Non-linear scaling of delay discounting

These laboratory findings are supported by evidence from operational settings. Rusou et al. (2020) show that operators under workload preferentially complete small, easy tasks, the operational analogue of under-triage for large-scope supply chain activities. DiwasSingh et al. (2020) confirm in a large field study that completing easy tasks first significantly hurts overall performance. Hospital admission control under occupancy uncertainty (Kim et al., 2020) produces over- and under-triage patterns analogous to those seen in logistics operators facing uncertain lead times. Human-centred decision support systems such as that of Rosenthal et al. (2020) use visual feedback to show the mismatch between human plans and system temporal constraints, helping users calibrate their schedules.

Gaps in cognitive bias research. Behavioural research diagnoses the problem and quantifies it in controlled settings, but the magnitude of the bias in industrial supply chain contexts remains a matter of calibration, and this literature does not prescribe a real-time operational correction.

2.4 Indicators of Delivery Rupture Risk

If urgency is to be aggregated from several indicators, the literature has to supply the indicators. This section reviews the body of work that identifies which observable quantities precede a delivery rupture, meaning a failure to deliver at the agreed time, place, quantity and quality. It is the source from which Section 3.2.2 later selects.

Heckmann, Comes and Nickel establish rupture as the terminal event a supply risk measure should anticipate, and criticise the field for measuring exposure without connecting it to that event (Heckmann et al., 2015). Indicator-based supplier monitoring proceeds by identifying observable precursors rather than modelling the event directly (Mokhtar et al., 2019).

Five families of precursor recur across that work. Slack, meaning the margin between a realistic lead time and the committed deadline, is the one indicator whose exhaustion is irreversible, since a task with no remaining slack cannot be recovered (Cohen et al., 2019; Bojko et al., 2021; Sun et al., 2020; Sawik, 2013). Capacity saturation adds waiting time that a nominal lead time does not show, so a loaded system carries a rupture risk its announced schedule conceals (Tomlin, 2006; Hosseini; Tajik, et al., 2019; Taghavi et al., 2024). Resource performance drift is monitored through composite proxies, most often Overall Equipment Effectiveness, because direct performance data is rarely available on the supplier side (Mokhtar et al., 2019; Hlioui et al., 2017). Adverse-event exposure is treated probabilistically, as the product of a failure probability, a recovery duration and an impact severity (Hosseini; Ivanov; Dolgui, 2020; Liu et al., 2021; Hosseini; Ivanov, 2020). Cost signals such as expediting premiums are read as symptoms of strain rather than causes of rupture: paying for air freight to protect a deadline is evidence the system is already under stress, and it also bounds which corrective actions stay affordable (Shen et al., 2019; Tomlin, 2006). Emissions constraints enter the same way, as a restriction on the admissible set of responses rather than as a rupture mechanism, and become decisive mainly in the trade-off between slow sea freight and fast air freight (Hoen et al., 2014; Smart Freight Centre, 2023; International Organization for Standardization, 2023).

Gaps in rupture-risk indicators. This literature names precursors and monitors them individually. It does not establish a selection criterion for deciding which precursors belong in a single composite, nor does it address what happens when several are combined: whether the indicators are independent enough to be multiplied, and whether a given numerical level on one carries the same risk meaning as the same level on another. Those two questions are left open, and any multi-block urgency model inherits them.

2.5 Closest Antecedents: Network Risk Twins and MCDM Supplier Scoring

The works most structurally similar to this thesis combine multi-criteria supplier scoring with disruption-aware network monitoring. Sawik (2013) develops mixed-integer programming models for resilient supplier order assignment integrating cost, service level and disruption risk, extended in later work to joint and integrated formulations (Sawik, 2014; Sawik, 2016). Cohen et al. (2019) and Afrasiabi et al. (2022) extend this to MCDM supplier prioritisation under disruption, combining criteria weighting with allocation optimisation. Chakraborty et al. (2020) confirm that rule-based heuristics remain competitive against exact methods in dynamic industrial environments. These works are cited for the closeness of their subject matter, not because this thesis reuses their optimisation machinery, which addresses allocation and sequencing rather than urgency measurement.

Gaps in the closest antecedents. Two gaps distinguish this stream from the present work. First, none introduces a network-propagated urgency signal computed independently at every node and combined across tiers; risk is typically evaluated once, at the point of decision. Second, none compares a client-stated urgency against a model-derived urgency: where

the declared signal exists at all, it is used directly, without validation against live operational proxies.

2.6 Enabling Method: Bayesian and Probabilistic Risk Updating

Bayesian methods are a well-supported approach for revising risk estimates from new evidence in supply chain research (Pearl, 1988; Koller et al., 2009). Rather than a full Bayesian network with an explicit conditional-probability-table structure, this study draws on the narrower and more tractable case of Beta-Bernoulli conjugate updating: a failure probability p is treated as a Beta-distributed belief with an equivalent prior sample size n_0 , revised from a small number of newly observed equivalent trials (k, n) each time a calibrated operational event is declared.

Three convergent findings support this choice. Hosseini; Ivanov (2020) survey Bayesian methods applied to supply chain risk and resilience, finding that probabilistic belief-revision methods are often preferred to deterministic scoring when sparse new evidence must update an existing estimate. Huang et al. (2020) and Deng et al. (2020) develop Bayesian models for disruption and delay probability estimation. Lockamy et al. (2011) and Qazi et al. (2017) support Bayesian revision as a practical tool under sparse, event-driven evidence, consistent with the small-sample setting of a weekly-cadence digital twin. Plain frequency counts, by contrast, are unstable when only a handful of events are declared per node per project (Zheng et al., 2020b).

Gaps in Bayesian risk updating. Bayesian belief revision quantifies disruption exposure but does not, by itself, compare it against a client-declared urgency, and it says nothing about whether the resulting probabilities are well calibrated once they reach a human reader.

2.7 Enabling Method: Digital Twins and Real-Time Decision Support

The digital twin concept was formalised by Grieves (2014) as a virtualised representation of a physical process. Tao et al. (2018) extended it to manufacturing, and Fuller et al. (2020) surveyed the enabling technologies. In supply chain management, Ivanov; Dolgui, et al. (2021) define the Supply Chain (SC) Digital Twin (DT) as integrating simulation, optimisation and machine learning. Ivanov; Dolgui (2020) address digital supply chain twins for disruption risk and resilience management. Wang et al. (2020) demonstrate twin-driven planning in a manufacturing context analogous to DISCO's satellite component network. In the Industry 5.0 setting, these systems have evolved toward human-in-the-loop Cognitive Digital Twins, where semantic models and reasoning methods capture human feedback and synchronise mental models with simulated physical operations (Nillofar et al., 2023).

Real-time freight planning work shows that operational urgency proxies, meaning slack, lateness risk and shortage exposure, are already present within these systems (Hrušovský et al., 2021; Perez et al., 2022). Model-predictive control for vehicle routing (Karam et al., 2022) and twin-based disruption response (Ding et al., 2023; Leung et al., 2022; X. Zhu et al., 2025) confirm the state of the art in live operational replanning.

Gaps in digital twin decision support. In the retrieved literature, no system combines a declared urgency input, a computed operational urgency, and their propagated mismatch

within one architecture. A second gap matters for this thesis specifically: these systems are evaluated on the accuracy of their physical state estimation, and their inputs are assumed to arrive. Where an input depends on the same operator whose judgment the system exists to support, that circularity is not usually treated as a measurement problem.

2.8 Enabling Methods: Multi-Criteria Decision Methods

This research builds on two distinct domains within multi-criteria decision-making: pairwise comparison methods for eliciting weights, and outranking aggregation for turning those weights into a defensible ranking of nodes.

Pairwise elicitation. To construct a structured representation of declared urgency U_d , criteria weights must be systematically elicited. The Analytic Hierarchy Process (Analytic Hierarchy Process (AHP)) (Saaty, 1980) derives weights as the principal eigenvector of a pairwise comparison matrix, validating logical consistency through the Consistency Ratio ($ConsistencyRatio(CR) \leq 0.10$). While widely applied in logistics, AHP suffers from rank reversal, where introducing a new alternative alters the relative ranking of existing ones, and is sensitive to cognitive noise during elicitation (Dyer, 1990; Triantaphyllou; Yanase, 2024). To reduce operator load and improve consistency, the Best-Worst Method (Best-Worst Method (BWM)) (Rezaei, 2015) reduces the number of comparisons from $n(n-1)/2$ to $2(n-1)$ by anchoring around chosen best and worst criteria. The Fuzzy Best-Worst Method (Fuzzy Best-Worst Method (F-BWM)) (S. Guo et al., 2017; Yucesan et al., 2019; Amiri et al., 2020) extends this with Triangular Fuzzy Numbers (Triangular Fuzzy Numbers (TFNs)) to absorb judgment uncertainty, and consistency conditions for the fuzzy variant have been established (Liang et al., 2020).

Outranking aggregation and the composite indicator critique. For prioritising which nodes need attention, direct weighted averages risk fully compensatory aggregation, where an excellent score on one criterion completely masks a critical failure on another. Outranking methods such as PROMETHEE II (Brans; Vincke, et al., 1986; Brans; Mareschal, 2005) establish pairwise preference flows ϕ^+ and ϕ^- , allowing a non-compensatory ordering. A cross-industry survey (Behzadian et al., 2010) confirms its suitability for supplier and node evaluation. Composite-index literature nonetheless warns that nominal weights often diverge from effective importance because of correlation, normalisation and aggregation dynamics (Greco et al., 2018; Becker et al., 2017; Borgonovo et al., 2021), and the choice of normalisation is itself non-neutral, since it can produce rank reversal (Triantaphyllou, 2000).

Gaps in multi-criteria decision methods. These methods weight criteria and rank alternatives, but they are static elicitation and aggregation tools. None is routinely compared against a computed operational counterpart. And the composite-indicator critique is generally raised as a caveat at the end of a paper rather than turned into something a running system can check on its own data.

2.9 Forecast-Assisted Decision and Automation Bias

The streams above concern how to compute a signal. This one concerns what happens when the signal is shown to the person whose judgment it was built to support, and it is included

because the campaign reported in Section 4.7 produced a result in exactly that register.

Parasuraman et al. (1997) set out the canonical taxonomy of misuse, disuse and abuse of automation, establishing that the failure modes of automated aids are as much about human reliance as about machine accuracy. Skitka et al. (1999) demonstrate automation bias directly: participants supported by an automated aid committed both omission errors, failing to react to a problem the aid did not flag, and commission errors, following the aid against contradicting evidence. Cummings (2004) extends the finding to time-critical decision support, the family to which a weekly supply chain review belongs. Goddard et al. (2012) provide a systematic review in clinical decision support, and report that automation bias is frequent, that it is mediated by the confidence with which advice is presented, and that it is only partially mitigated by training.

The direction of the effect is not uniform. Dietvorst et al. (2015) document algorithm aversion, where people abandon an algorithm after seeing it err even when it outperforms them, while Logg et al. (2019) document algorithm appreciation, where people weight algorithmic advice more heavily than human advice. Bansal et al. (2021) show that adding explanations to a model's output can increase acceptance of its recommendations whether or not those recommendations are correct, so that explanation improves persuasion more reliably than it improves joint accuracy.

Two consequences follow for the present work. First, an advisory signal changes the behaviour it measures, which means a system that both elicits declared urgency and displays a forecast to the declarant is not observing an independent signal. Second, a confidently presented but poorly calibrated probability is a specific hazard rather than a cosmetic defect, since the confidence with which advice is presented is itself a mediator of reliance (Goddard et al., 2012).

Gaps in forecast-assisted decision research. This literature is dominated by single-decision laboratory tasks and clinical decision support. It does not address a repeated, multi-turn setting in which the same operator is shown a forecast about their own node, week after week, while their declaration feeds back into the model's state. Nor does it address the case where the displayed forecast is systematically miscalibrated in a known direction, which is the situation measured in Section 4.5.

2.10 Validation Methodology: Serious Games

The Beer Game (Sterman, 1989) is the foundational supply chain serious game. Sterman shows that players systematically misperceive the feedback in their own supply line, underestimating orders already placed but not yet received, and over-order with a lag once a shortage appears, amplifying the resulting oscillation as it travels upstream, the mechanism known as the bullwhip effect (Lee et al., 1997). This decision bias is analogous to the false-urgency pathology, where an operator over-reacts to a signal after a delay rather than adjusting smoothly. Prensky (2001) establishes the theoretical basis for digital game-based learning, and Deterding et al. (2011) formalise gamification design principles.

Mohaddesi et al. (2020) introduce the Gamette methodology, a bite-sized approach to capturing decision-making patterns for behavioural modelling, directly applicable to a weekly-review protocol in which operators repeatedly submit urgency assessments. Bellotti et al. (2013)

provide an evaluation framework for serious game learning outcomes. Lafkihi et al. (2025) use a freight transportation game to study carrier behaviour on online spot-market platforms, a setting connected to the Physical Internet’s vision of mutualised capacity allocation.

Gaps in serious game validation. Serious games are established as instruments for observing decision bias, but they are usually scored on learning outcomes or on aggregate system performance. They are rarely used as a controlled comparison in which the same frozen scenario is replayed with and without a treatment applied to the declarant, which is the design adopted in Section 3.4.3.

2.11 Probabilistic Forecast Verification

Because this thesis scores a probabilistic forecast against recorded outcomes, it inherits the vocabulary of forecast verification. Brier (1950) introduced the mean squared error of probabilistic forecasts, and Murphy (1973) decomposed it into reliability, resolution and uncertainty, establishing the distinction this thesis relies on: a forecast can order outcomes correctly while announcing the wrong probabilities, and the two failures require different remedies. Gneiting et al. (2007) formalise strictly proper scoring rules, under which a forecaster minimises expected loss only by reporting their true belief.

The distinction has been rediscovered in machine learning. Niculescu-Mizil et al. (2005) show that several strong classifiers produce distorted probabilities that require post-hoc correction, and C. Guo et al. (2017) show that modern neural networks are systematically over-confident, with accuracy improving while calibration degrades. Both report that the correction is a separate, cheap, post-hoc layer rather than a change to the model.

Gaps in forecast verification practice. The verification literature assumes the outcome being predicted is unambiguously defined. In an operational supply chain setting it is not: whether a node-week counts as an adverse outcome depends on choices about which events qualify and over what horizon. Section 4.6 shows that this choice, and nothing else, moves the measured skill of the same forecasts by a factor of five.

2.12 Analogical and Contextual Literature

Three streams provide vocabulary and context rather than foundations, and are flagged as non-foundational wherever they are used.

Medical triage as a conceptual analogy. Medical triage provides an analogy for prioritising under severe capacity constraints and cognitive pressure. Protocols such as the Manchester Triage System (Manchester Triage System (MTS)) (Mackway-Jones et al., 2013) categorise patients into five urgency levels using flowcharts, whereas the Emergency Severity Index integrates predicted resource consumption. Comparative clinical studies show substantial classification discrepancies between the two systems, at a weighted $\kappa = 0.51$, so even trained professionals disagree on urgency assessment (Kuligowski et al., 2025). The clinical literature also establishes an asymmetry that logistics work rarely states explicitly: under-triage, meaning a patient classified as less urgent than they are, is more dangerous than over-triage, because the first costs a life and the second costs a queue (Kim et al., 2020; Mackway-Jones et al., 2013).

Physical Internet and synchromodality. The Physical Internet is a model of interconnected logistics networks in which routing, storage and transport assets are mutualised across actors (Montreuil, 2011). In such networks urgency stops being a local concern and becomes a network-level quality of service variable, because the capacity one actor claims is capacity another loses.

A related but separate concept is synchromodality. Where the Physical Internet describes who shares the infrastructure, synchromodality describes when the mode is chosen: a shipment can be moved between road, rail and waterway in real time as conditions change, instead of being committed to one mode when it is booked. The two can operate together and neither presupposes the other. Dynamic routing under synchromodality needs priority signals that stay valid across rolling planning windows (W. Guo et al., 2020), and disruptions travel quickly through interconnected networks, producing cascading ripple effects (Li et al., 2020).

Regulatory carbon accounting. Corporate sustainability reporting requirements have made transport emissions a reported quantity rather than an optional one (Cordova, 2024). The GLEC Framework v3.2 (Smart Freight Centre, 2023) and Vehicle Energy Consumption calculation TOol (VECTO) (European Commission, 2017) define emissions baselines for heavy goods vehicles, and emissions factors relate non-linearly to vehicle speed and cargo load, so a carbon figure cannot be recovered from cost or distance alone (ADEME, 2014).

Earth observation of built and logistics assets. The opening reported in Section 4.9 rests on a separate literature. Overhead object detection has a standard benchmark in DOTA (Xia et al., 2018), whose oriented bounding boxes and vehicle classes make it the reference for overhead vehicle detection, while building footprint extraction is benchmarked by Inria (Maggiori et al., 2017) and SpaceNet (Van Etten et al., 2018). Promptable segmentation foundation models (Kirillov et al., 2023) have made it possible to sample candidate sites from imagery alone rather than from a cadastral database. Separately, a body of economics work establishes that satellite observation can serve as an independent proxy for activity that official statistics record slowly or not at all, from night-time lights as a growth proxy (Henderson et al., 2012) to daytime imagery as a poverty predictor (Jean et al., 2016).

Gap in Earth observation for supply chain state. That economics literature measures aggregate activity at regional scale. It does not produce the per-facility quantities a supply chain twin needs, meaning storable volume and transfer capacity at a named site, and the detection literature stops at footprints and vehicles without converting them into an operational capacity figure.

2.13 Source Review Grid

Following the structured review method required by the programme's writing guidelines, each source analysed above has been synthesised in an evaluation grid recording its main argument, methodological approach, conclusions, thematic relevance to this thesis, and a critical appraisal of its limitations. The grid is given in Appendix B. It provides a comparative frame for assessing the contribution of each source and makes explicit which claims in this thesis rest on foundational work and which rest on analogy.

2.14 Identified Gaps

Each stream reviewed above closes on the limit of what it settles. Collected, those limits describe one absence rather than several.

Operational monitoring evaluates one criterion at a time and does not capture the multi-block character of supply chain state. The rupture-risk literature names precursors but supplies no criterion for choosing which belong in one composite, and leaves open whether equal values on structurally different indicators mean equal risk. Pairwise comparison methods elicit weights reliably but are static, and are not compared against a computed counterpart. Out-ranking prevents compensatory aggregation but is not combined with a network-propagated signal. Digital twins and Bayesian revision monitor disruption well, and have no mechanism for comparing an operator's input against the physical state they maintain. The automation bias literature shows that displaying a forecast changes the judgment it supports, without examining a repeated loop in which that judgment feeds back into the model. Forecast verification separates ordering from calibration, and assumes the outcome is unambiguously defined.

The reviewed work therefore covers multi-block risk aggregation, network propagation, twin operations, multi-criteria elicitation, and the human response to advisory signals. No retrieved work combines them with a structured declared-urgency input and an empirically tested adequacy layer. That missing combination is the gap, and closing it is the contribution this thesis claims: a measurement of the divergence between declared and computed urgency, propagated across a supplier network, and scored against recorded outcomes rather than asserted.

Three of the gaps above become testable questions rather than design choices, and they are the ones this thesis carries into Section 3: whether the divergence can be measured at all once both signals come from independent sources, whether the resulting signal predicts anything, and what happens to the declared side when the computed side is shown to the person declaring.

3 Methods

3.1 Methodological Framework

This research is structured as an agile test-and-learn programme rather than a linear sequence, operating through repeated iterations of literature analysis, mathematical specification, software implementation and scenario-based simulation (Figure 5). That structure is mapped onto the IMRaD form of this thesis as follows: the specification and implementation iterations are reported in this section as the method, and the simulation and campaign iterations are reported in Section 4 as the results.

The programme is organised into four operational areas:

- **Literature review and gap analysis**, establishing the psychological and mathematical foundations of multi-block urgency aggregation and cognitive bias (Section 2).
- **Mathematical framework specification**, deriving the multi-block real urgency function, its DAG propagation and the asymmetric adequacy score from first principles, so that the result is dimensionally and probabilistically consistent (Section 3.2).
- **Pairwise comparison elicitation**, formulating crisp AHP and Fuzzy Best-Worst Method weight vectors to build a structured representation of declared urgency (Section 3.3.2).
- **Implementation, simulation and validation**, building a modular, tested Python library exercised through a multi-page Dash web application, running controlled shock and scenario simulations on a persistent digital twin, and confronting the resulting signal with recorded outcomes through a dedicated calibration service and a serious game campaign (Sections 3.3 and 3.4).

The remainder of this section keeps three things apart, because mixing them is what makes a framework of this kind hard to read. Section 3.2 states the mathematical model on its own terms, saying nothing about tooling. Section 3.3 gives the application choices, meaning how each object in that model is elicited, computed and exposed. Section 3.4 gives the use case built to test it. The method's own limitations, its reproducibility provisions and its ethical considerations close the section.

3.1.1 The Proposed Structure

Section 2.14 identified one absence rather than several. The response designed here is a two-layer loop, and stating it before the mathematics makes the equations that follow easier to place.

The first layer is a cognitive adequacy check. It captures the operator's declared urgency U_d through consistency-gated pairwise comparisons, computes a real urgency $U_r(t)$ from live operational state, and scores their mismatch through an asymmetric penalty. The second layer propagates and prioritises. Local urgency travels across the supply network, declared urgency descending toward suppliers and real risk ascending toward the final customer, and the resulting network state feeds an outranking and a shock-simulation criticality index that together indicate which nodes warrant attention. Figure 4 shows how the two layers sit relative to each other and where the declared and computed signals meet.

Two choices inside that design are worth naming immediately, since Section 2.2 left both open. The aggregation over blocks is a noisy-OR rather than a weighted average, because a

complete capacity failure should not be offset by acceptable cost and carbon readings, and non-compensatory behaviour is the property that guarantees it. The penalty over the mismatch is asymmetric, pricing under-declaration above over-declaration, on the clinical analogy of Section 2.12, where under-triage costs more than a queue. Both are design commitments open to challenge, and Section 5.3 records what each one costs.

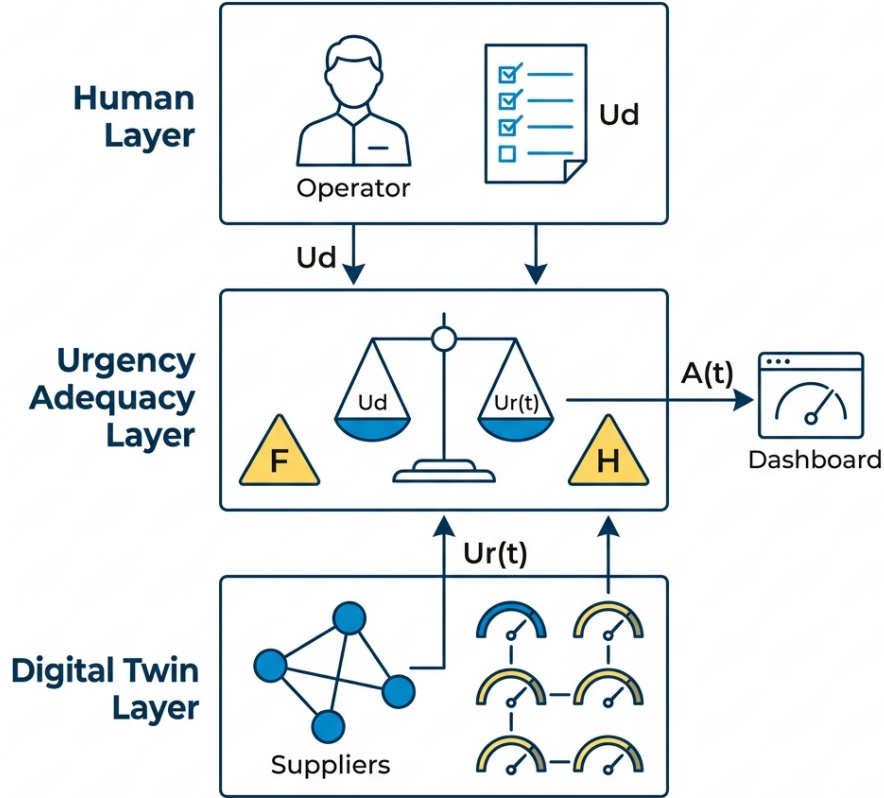


Figure 4: The layered adequacy architecture

3.2 Mathematical Model

This section states what real urgency and declared urgency are, how each propagates across the network, and how their mismatch is scored. It deliberately leaves out tooling, elicitation mechanics and operational cadence, which are application decisions addressed in Section 3.3.

3.2.1 Local Real Urgency

Real urgency is not read from a single deadline-proximity signal. It is aggregated at each node from six independent KPI blocks, namely time, capacity, performance, risk, cost and carbon, each producing a partial urgency $u_m \in [0, 1]$, or no value at all when the underlying KPI data is unavailable. The aggregation is a weighted probabilistic OR, so that a single saturated block is sufficient to drive local urgency toward its maximum regardless of how well the others score:

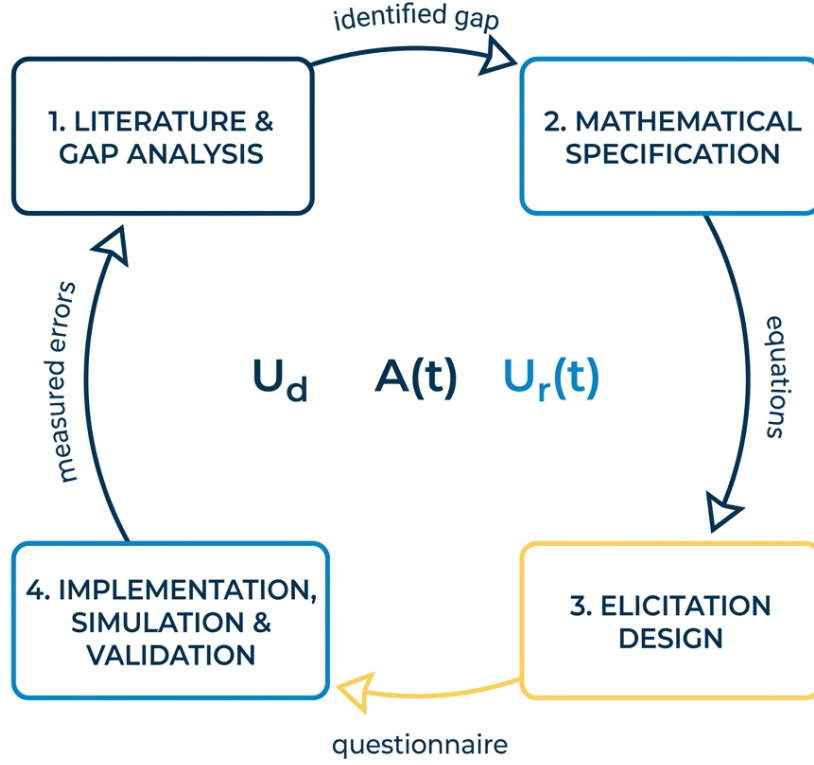


Figure 5: The test-and-learn iteration loop

$$U_{r,\text{local}} = 1 - \prod_m (1 - u_m)^{\omega_m}, \quad (4)$$

where $\omega_m \geq 0$ is the aggregation weight of block m and the product runs over blocks with available data. Table 6 gives each block's schematic formula and what it measures. The notation $[x]_+ = \max(x, 0)$ denotes the positive part, and the per-term weights α_i and β_i are renormalised over the terms whose data is actually available.

A bare formula does not say what its bounds mean, so three properties of Equation 4 are stated explicitly. They are also where the model is most open to challenge.

What the endpoints mean. Each raw KPI carries physical units, whether days, euros, kilograms or percent, and passes through a transfer function onto the dimensionless interval $[0, 1]$. At $u_m = 0$ the block is nominal and contributes nothing: its factor $(1 - 0)^{\omega_m}$ equals 1 and leaves the product unchanged. At $u_m = 1$ the factor is 0, so the product collapses and $U_{r,\text{local}} = 1$ whatever the other blocks read. The upper endpoint therefore has a precise operational meaning: the block has reached the state past which it cannot make the node any more urgent, whether that state is a deadline already passed, a fully depleted buffer or a near-certain failure. The word used for it in this thesis, saturation, is a label adopted for this

Table 6: The six real urgency blocks

Block	Formula	Explanation
u_{time}	$P(L > d^* - t) + \kappa_{\text{retard}}[r]_+ - \kappa_{\text{avance}}[-r]_+$	Probability that the lead time L , modelled as Normal(μ , σ), exceeds the time left before the next active deadline d^* ; forced to 1 once the deadline has passed. The planning term r compares theoretical with reported progress.
u_{cap}	$\alpha_1 \left(1 - \frac{V_{\text{rem}}}{V_{\text{max}}}\right) + \alpha_2 \left(1 - \frac{W_{\text{rem}}}{W_{\text{max}}}\right) + \alpha_3 \left[\frac{D - \Phi}{D}\right]_+$	Saturation of the node's capacity: how little storage volume and weight remain, plus the fraction of demand D not covered by the current flow rate Φ .
u_{perf}	$1 - \text{OEE}$	Complement of Overall Equipment Effectiveness, itself availability $\times$ performance $\times$ quality.
u_{risk}	$1 - \exp(-p_{\text{fail}} \times t_{\text{recov}} \times \text{sev}/T_{\text{ref}})$	Expected unavailability: failure probability times recovery time times impact severity, normalised by a reference horizon, then combined by probabilistic OR with environmental and political exposure.
u_{cost}	$\beta_1 \frac{c_{\text{op}} - c_{\text{nom}}}{c_{\text{nom}}} + \beta_2 [\text{tariff} - 1]_+ + \beta_3 \frac{c_{\text{stock}}}{c_{\text{ref}}}$	Financial stress: operating cost drift above nominal, tariff multiplier above neutral, and storage cost relative to a reference.
u_{co2}	$\left[\frac{e - e_{\text{target}}}{e_{\text{max}} - e_{\text{target}}}\right]_+$	Normalised overshoot of measured emissions above target, saturating at the maximum threshold.

model and not a concept imported from the reviewed literature, which is why the endpoint is defined here by its effect on the product rather than by citation.

How a missing block is handled. When a KPI is unavailable the block is given weight $\omega_m = 0$, which sets its factor to $(1 - u_m)^0 = 1$ and leaves the aggregate untouched. Absence is therefore not represented by $u_m = 0$, which would be a positive claim that the block is healthy, and the two cases are distinguished in the data model as well as in the mathematics. The remaining weights are renormalised so they still sum to their intended total, which keeps the aggregate comparable between a node with six live blocks and a node with three.

Whether the blocks are exchangeable. The product in Equation 4 is symmetric in its factors, so it treats the blocks as interchangeable up to their weights. That is the strongest assumption in the model and it does not hold cleanly. A value of $u_{\text{time}} = 0.3$ means a 30% probability of missing the next deadline, a quantity with a frequentist reading. A value of $u_{\text{cost}} = 0.3$ means operating cost has drifted three tenths of the way along a normalising scale toward a reference, which is an ordinal statement dressed in a cardinal number. The two are not equally informative about rupture, and no transfer function makes them so.

Three consequences follow, and they are carried forward rather than resolved. The weights ω_m are the only mechanism the model has for expressing that some blocks are more diagnostic than others, so they absorb a distinction that arguably belongs in the transfer functions instead. Any interpretation of the aggregate as a probability is therefore approximate, and the aggregate is used in this thesis for ranking and for comparison against U_d , never as a calibrated probability in its own right. And the assumption is testable: if two blocks carry the

same information, the correlation between them is observable, which is exactly the check reported in Section 4.4 where two of the six turned out to be perfectly correlated on the campaign run. Section 5.3 records the assumption among the framework’s stated limits.

Two lifecycle rules apply before aggregation. A node marked DONE has its effective local U_r forced to 0.0, since a completed task carries no further physical risk, and a node marked ABANDONED has it forced to 1.0. These apply identically whether the local urgency comes from the analytic blocks or from the Monte Carlo estimate below.

A stochastic variant of the time block. The analytic u_{time} block assumes a Normal lead time at a single node. On a multi-tier network that assumption breaks, since a node can only start when all its predecessors have delivered, and the maximum of several skewed lead times has no convenient closed form. The module `mc/lead_time.py` therefore replaces the analytic estimate with a stochastic one when enabled.

The simulation works in three steps. Each node draws its lead time L_i from a configurable distribution, either lognormal, normal clipped at zero, or triangular. Completion times then propagate along the DAG in topological order, a node starting when its last predecessor finishes, $S_i = \max_{j \in \text{Pred}(i)} C_j$, and completing after its own lead time, $C_i = S_i + L_i$. Finally the delay probability of each node is estimated as the fraction of draws in which it misses its deadline, $u_{\text{time}} = P(C_i > d_i)$, over $N = 10\,000$ draws by default, with a memory guardrail capping the simulation at $N \times n \leq 5 \times 10^7$ across n nodes.

3.2.2 Why These Six Blocks

The six blocks are not an arbitrary KPI shortlist. The selection started from the question that defines urgency in a supply chain: which mechanisms make the risk of a delivery rupture rise? A rupture, meaning the inability to deliver at the agreed time, place, quantity and quality, is the terminal event the urgency signal must anticipate (Heckmann et al., 2015). Working backwards from that event, and consistently with indicator-based supply risk monitoring (Mokhtar et al., 2019), four direct causal mechanisms and two arbitration variables were identified.

- **Time erosion** (u_{time}): the remaining slack between deadline and realistic lead time. Time is the only non-compressible resource, so a task whose slack reaches zero can no longer be saved (Cohen et al., 2019; Bojko et al., 2021; Sun et al., 2020; Sawik, 2013).
- **System saturation** (u_{cap}): queues, transport capacity, workshop load and bottlenecks. A saturated system introduces hidden waiting delays on top of the nominal lead time (Tomlin, 2006; Hosseini; Tajik, et al., 2019; Taghavi et al., 2024).
- **Critical-resource performance** (u_{perf}): drift in the performance of the resource that produces the part. OEE is used as a composite proxy because precise performance data is rarely accessible on the supplier side (Mokhtar et al., 2019; Hlioui et al., 2017).
- **Adverse-event exposure** (u_{risk}): the expectation of unavailability caused by a disruption. The default incident probability of 0.02 is revised weekly from declared events through the Bayesian operator of Section 3.3.4, building on the DIN supplier risk scorer (Kabouche, 2025) and on probabilistic ripple-effect models (Hosseini; Ivanov; Dolgui, 2020; Liu et al., 2021; Hosseini; Ivanov, 2020).
- **Financial flexibility** (u_{cost}): expediting surcharges, tariff rises and storage costs. Cost drift is not a rupture cause in itself but a stress indicator, and it bounds which corrective actions remain economically viable (Shen et al., 2019; Tomlin, 2006).

- **Carbon overshoot** (u_{co2}): emissions above target. Like cost, it is an arbitration criterion rather than a rupture mechanism, becoming decisive only when a regulatory or ESG constraint genuinely restricts the admissible responses (Hoen et al., 2014; Smart Freight Centre, 2023; International Organization for Standardization, 2023).

Each candidate indicator was screened against the four filter questions of Table 7. Only indicators passing all four were retained, which yields four direct causes plus two arbitration variables and explains why the list has six entries rather than four or eight. Figure 6 draws the resulting structure, separating the four blocks that describe a mechanism leading to rupture from the two that constrain which responses remain available.

Table 7: Selection filter applied to each candidate block

Criterion	Question asked of each candidate indicator
Causal link	Does the indicator have a plausible mechanism linking it to delivery rupture?
Observability	Can it be measured or credibly estimated from client, supplier or planning data?
Actionability	Can an operator act on it, or use it to decide?
Non-redundancy	Does it add information not already carried by a retained indicator?

At this stage the selection remains a conceptual construction. It must still be confirmed by the real availability of the underlying data and by sensitivity tests verifying that each block contributes useful information to the aggregate. Section 4.8 returns to this point, because the campaign measured one block failing that test.

3.2.3 Local Declared Urgency

Symmetrically, declared urgency is not read as a single free-text estimate. It is reconstructed from a structured elicitation so that it can be scored, propagated and compared with $U_{r,local}$ on the same scale. At each node the operator compares four criteria pairwise through the Analytic Hierarchy Process, which yields a consistency-checked weight w_j per criterion, then rates each criterion on a severity scale $s_j \in \{1, \dots, 9\}$. Local declared urgency is the weight-averaged severity rescaled to $[0, 1]$:

$$U_{d,local} = \sum_{j=1}^4 w_j \frac{s_j - 1}{8}, \quad \sum_{j=1}^4 w_j = 1. \quad (5)$$

The elicitation tool, its four criteria and the consistency gate that rejects an incoherent weight vector are application choices described in Section 3.3.2. What matters here is that $U_{d,local}$ and $U_{r,local}$ are both bounded, node-local quantities in $[0, 1]$ before either is propagated.

3.2.4 Network Propagation

Declared urgency propagates downstream, from the final customer at rank 0 toward raw-material suppliers, while real urgency propagates upstream, from suppliers toward the final customer. Both use a leaky product rule that keeps the signals bounded in $[0, 1]$ and monotonically increasing as more neighbours become urgent:

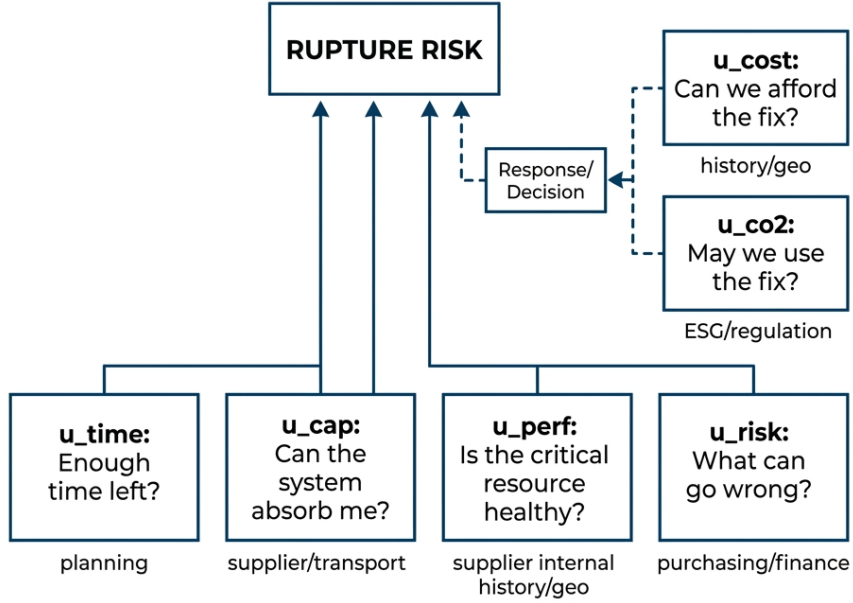


Figure 6: Causal structure behind the six-block selection

$$U_{d,i} = 1 - (1 - U_{d,\text{local},i}) \prod_{k \in \text{Succ}(i)} (1 - \gamma_{ik} U_{d,k}), \quad (6)$$

$$U_{r,i} = 1 - (1 - U_{r,\text{local},i}) \prod_{j \in \text{Pred}(i)} (1 - \beta_{ji} U_{r,j}), \quad (7)$$

where $\gamma_{ik} \in [0, 1]$ is the downstream attenuation coefficient on arc (i, k) and $\beta_{ji} \in [0, 1]$ the upstream amplification coefficient on arc (j, i) . Table 8 defines each term, and Figure 7 shows the two signals travelling in opposite directions across the same graph. Backup arcs are documentary and inert with respect to both propagations, a modelling choice that the campaign scenario exploits deliberately.

When each quantity is computed. The weekly cadence governs when new data *arrives*, not when the model runs, and the two are easy to conflate. The sequence is as follows. A review submitted for one node writes that node's KPI values and questionnaire answers. That write immediately triggers recomputation of the node's own u_m blocks and its two local quantities, $U_{r,\text{local}}$ and $U_{d,\text{local}}$. The new local values then propagate, but only along the directions that can be affected: a change in local real urgency is pushed through the node's downstream cone

Table 8: Terms of the propagation equations

Component	Type	Description
$U_{d,i}, U_{r,i}$	Dependent variables	Propagated declared and real urgency at node i
$U_{d,local,i}, U_{r,local,i}$	Input variables	Local urgency at node i after status-rule adjustment
γ_{ik}	System parameter	Downstream demand attenuation coefficient on arc $i \rightarrow k$
β_{ji}	System parameter	Upstream risk amplification coefficient on arc $j \rightarrow i$
$Succ(i), Pred(i)$	Structural sets	Nominal successors and predecessors of node i

toward the final customer, following Equation 7, and a change in local declared urgency is pushed through its upstream cone toward the suppliers, following Equation 6. Nodes outside those two cones are not recomputed, because their values cannot have changed.

Two consequences matter for reading the results. Nothing is computed on a global weekly timer, so the network is never in a state where all eight nodes were evaluated at the same instant; a node reviewed on Monday and a node reviewed on Thursday hold values one review apart, and the propagated figures reflect whatever the neighbours had most recently submitted. And the campaign turn, not the calendar week, is the unit that makes the state comparable across nodes, which is why every result in Section 4 is indexed by turn.

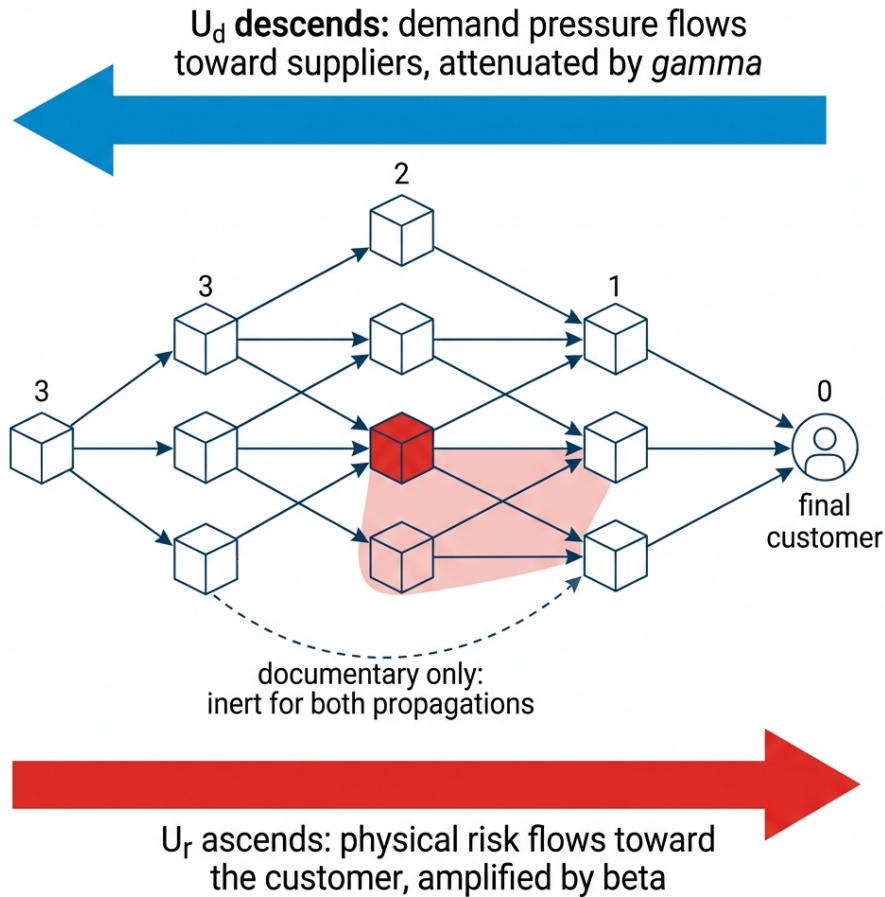


Figure 7: Bidirectional urgency propagation on the DAG

3.2.5 Network Criticality Index

The criticality service answers one question systematically: which node, if it failed completely, would cause the most damage to final-customer delivery? For every active node in turn, the service forces the node's local real urgency to 1.0, re-runs the ascending propagation, and measures two deltas, the rise of U_r at the final-customer nodes and the largest rise anywhere in the network. Nodes are ranked by the first, then the second, then by name. Figure 8 traces a single iteration of that sweep, from the forced saturation of one node to the deltas it produces downstream.

One behaviour of this index matters for interpretation. A node already saturated at $U_r \geq 1.0$ yields a delta of zero and sinks to the bottom of the ranking, since its risk is already realised in the baseline. In a network-wide crisis, where many nodes saturate at once, the index degenerates. This documented behaviour drives the exclusion rule used by hypothesis H3 in Section 3.4.

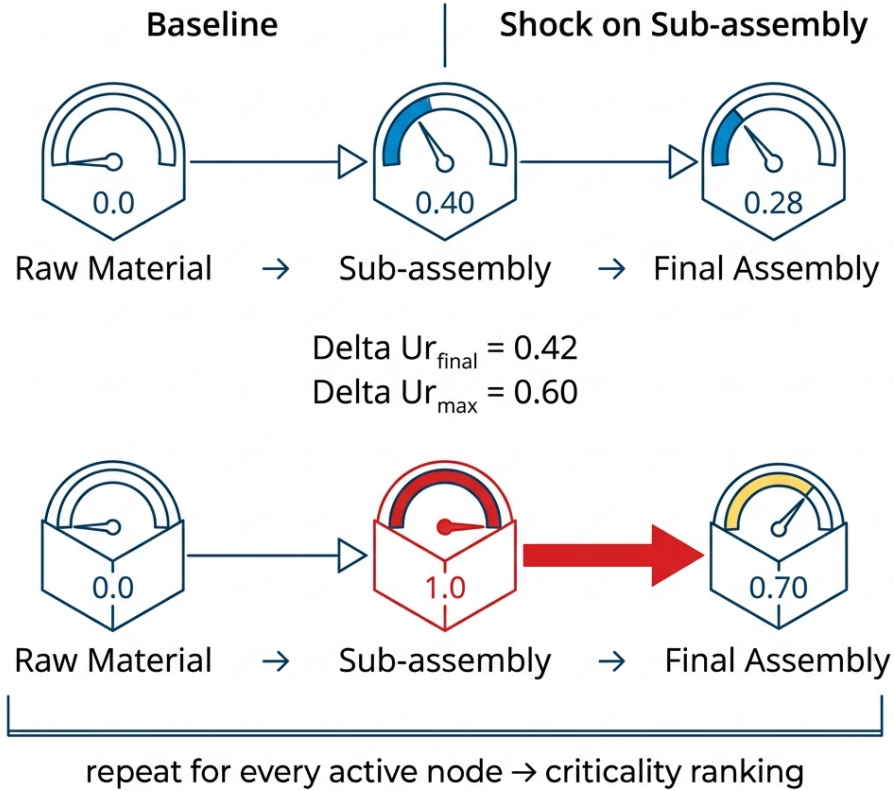


Figure 8: One iteration of the shock simulation

3.2.6 Asymmetric Cognitive Adequacy

The mismatch between declared and real urgency is split into two named pathologies, false urgency F for over-declaration and hidden risk H for under-declaration:

$$F = \max(U_d - U_r, 0), \quad H = \max(U_r - U_d, 0). \quad (8)$$

Drawing on Prospect Theory (Kahneman; Tversky, 1979), the two are penalised asymmetrically and with diminishing marginal sensitivity to large errors:

$$\text{penalty} = \lambda_{\text{under}} H^\alpha + \lambda_{\text{over}} F^\alpha, \quad (9)$$

and the penalty is rescaled into an adequacy score $A \in [0, 100]$ through an exponential mapping bounded by the penalty of a maximal one-sided error, with each term defined in Table 9:

$$A = 100 \times \frac{\exp(-\text{penalty}) - \exp(-\text{penalty}_{\max})}{1 - \exp(-\text{penalty}_{\max})}, \quad \text{penalty}_{\max} = \max(\lambda_{\text{under}}, \lambda_{\text{over}}). \quad (10)$$

Table 9: Terms and parameters of the adequacy equations

Component	Type	Description	Default
F	Dependent variable	False urgency, from over-declaration	n/a
H	Dependent variable	Hidden risk, from under-declaration	n/a
λ_{under}	System parameter	Penalty weight on hidden risk	2.25
λ_{over}	System parameter	Penalty weight on false urgency	1.0
α	System parameter	Psychophysical curvature	0.88
A	Dependent variable	Adequacy score, 100 meaning perfect alignment	[0, 100]

The governing constraint is $\lambda_{\text{under}} > \lambda_{\text{over}}$: a coordinator who panics wastes premium transport capacity, but a coordinator who under-reports risk misses the window to prevent a line stoppage, so under-declaration is priced more expensively. The default ratio of 2.25 reprises the loss-aversion coefficient, and $\alpha = 0.88$ the curvature of the value function, both measured by Kahneman; Tversky (1979) and restated later (Kahneman, 2011). These are literature-derived starting points, not measured properties of supply chain coordinators. Both the aggregation weights ω_m and the penalty parameters are configurable per project rather than hard-coded, so recalibration requires no code change.

3.3 System Implementation

The model above is deliberately silent on tooling. This section describes how each object it defines is elicited from an operator, computed inside the running system, exposed back to a coordinator and kept safe. The delivered system, SupplyScore, comprises 73 Python modules across 10 packages, covered by 1779 automated test functions in 106 test files, including property-based tests.

3.3.1 Architecture

Single facade. All operations go through one orchestrating facade (`services/orchestrator.py`) coordinating the mathematical core, the graph repository and the persistence layers behind a single thread-safe entry point. The web interface, the command-line entry point and the test suite all drive the system exclusively through this facade, and there is no second write path.

Multi-tenant persistence. Data is split between one global registry database holding projects, nodes, arcs, milestones and registry-level audit, and one SQLite database per node holding its private operational history. This isolation prevents write contention between operators reviewing different nodes and enforces the boundary that one supplier’s raw history is not readable through another supplier’s records. It is also what makes the campaign of Section 3.4 auditable per node after the fact.

Gatekeeper writes. Every business write crosses a single mutation service that whitelists the touched fields, validates ranges, diffs against current state so that a no-op write leaves no trace, records the change in the audit log with author and source, and synchronises the in-memory graph. Audit tables are immutable, including from the administration interface.

Graph and propagation. The network lives in an in-memory graph repository backed by networkx. Propagation is incremental, recomputing only the cones of nodes whose local urgency actually changed, which keeps the twin responsive at the scale validated by dedicated performance benchmarks. Figure 9 shows how the facade, the mutation gate and the two persistence layers relate.

Decoupled clocks. Each project carries its own clock: a system clock for live use, a game clock advancing week by week for serious-game sessions, and a fixed clock for deterministic tests. The weekly cycle, milestone pressure and event timestamps all read the project clock rather than the wall clock, which is what makes a compressed campaign possible without touching the model.

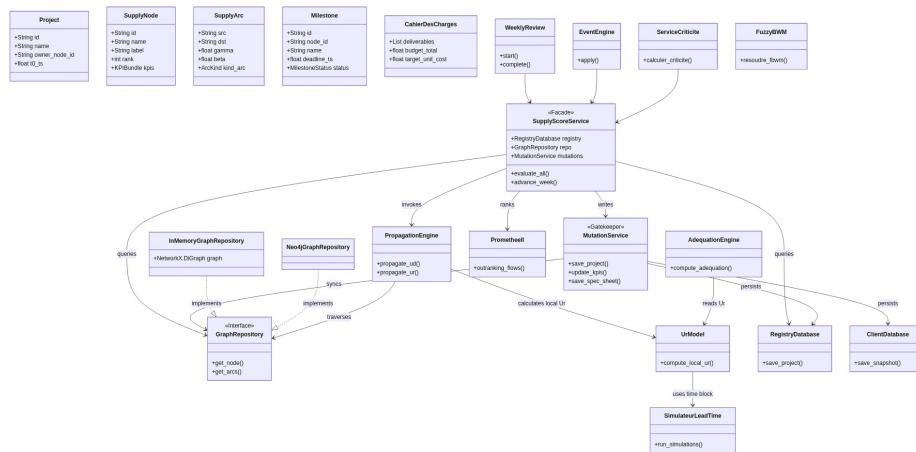


Figure 9: The SupplyScore architecture

3.3.2 AHP Elicitation

The module `core/ahp.py` implements the Saaty eigenvector method for a four-criterion questionnaire administered through the weekly review page, and is the tool that produces $U_{d,local}$. The four criteria are operational impact, meaning the downstream consequence of failure; time window, meaning how long before the situation becomes irredeemable; downstream dependencies, meaning how many tasks are blocked waiting; and recoverability, meaning whether lateness can be caught up, inverted so that lower is worse.

The criteria set follows from two constraints. Cognitively, Saaty’s method degrades as the number of pairwise-compared criteria grows, with consistency hard to maintain beyond roughly seven items and each additional criterion adding $n - 1$ comparisons to the weekly workload (Saaty, 1990; Triantaphyllou; Yanase, 2024). Keeping $n = 4$ limits the questionnaire to six comparisons. Structurally, the four questions mirror how declared urgency is consumed downstream: impact and dependencies capture the severity dimension that propagation amplifies, the time window captures the slack dimension confronted with u_{time} , and recoverability captures the resilience dimension confronted with the recovery term of u_{risk} . Each declared criterion therefore has an operational counterpart in the computed signal, which is what makes the later comparison meaningful rather than a comparison of unrelated quantities. Figure 10 draws those four correspondences. Larger criteria sets are deliberately routed to the Fuzzy BWM weighting of the KPI blocks instead.

The consistency ratio is computed as $CR = \text{ConsistencyIndex}(CI) / \text{RandomIndex}(\text{Saaty})n(RI)$ using Saaty’s Random Index table. If $CR > 0.10$ the interface flags the questionnaire as inconsistent and invites the operator to revise the most contradictory comparison before the assessment is accepted; Figure 11 shows that gate in the interface. Declared urgency is further smoothed with an exponential moving average, $\rho = 0.3$, across consecutive weekly assessments, absorbing transient stress spikes without erasing a sustained change in perception.

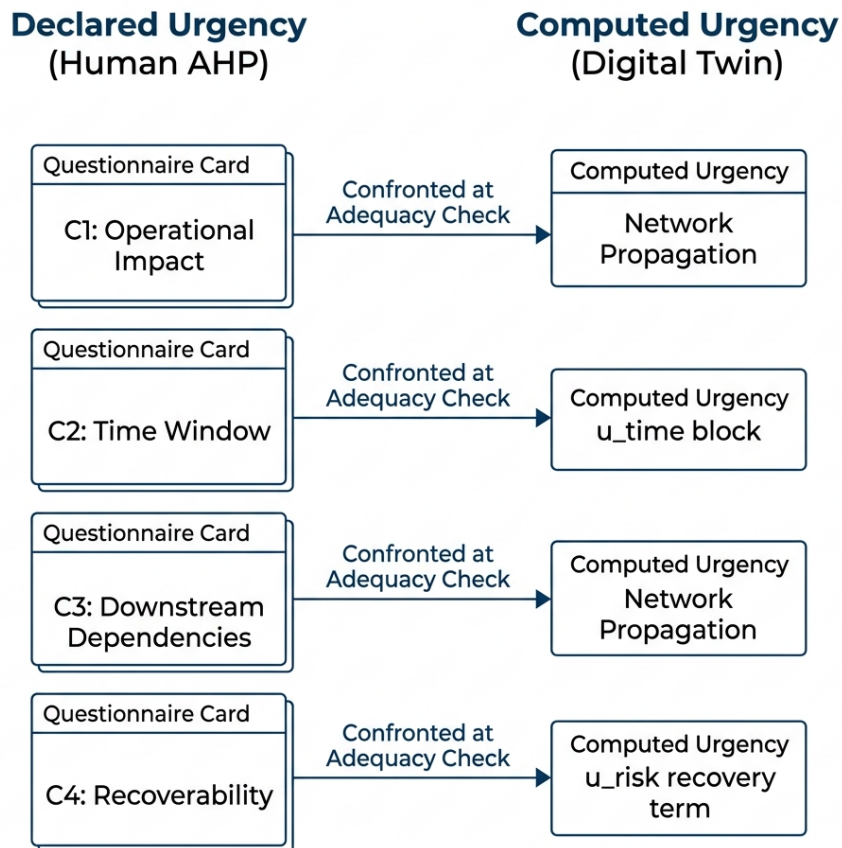


Figure 10: The four AHP criteria and their computed counterparts

www.hebdomadaire'. Below the title, it says 'Projet actif : Programme AERIS — drone cargo AER-200 — Semaine 2026-S32'. The 'Évaluation' section has a dropdown for 'Nœud (à importer quel rang)' with 'Choisir un nœud...' and a text input for 'Identifiant opérateur' with 'ex. jdupont'. There is also a text input for 'Notes (optionnel)' with 'commentaire libre'. The 'Comparaisons par paires (AHP)' section contains six horizontal sliders for pairwise comparisons. Each slider has a label on the left, a scale from -8 to 8 with 'Importance égale' at 0, and a '0' button on the right. The comparisons are: 1. Impact opérationnel vs Fenêtre temporelle, 2. Impact opérationnel vs Dépendances aval, 3. Impact opérationnel vs Récupérabilité, 4. Fenêtre temporelle vs Dépendances aval, 5. Fenêtre temporelle vs Récupérabilité, and 6. Dépendances aval vs Récupérabilité. All sliders are currently set to 0."/>

Figure 11: The AHP questionnaire page

3.3.3 Fuzzy BWM and PROMETHEE II Network Outranking

The modules `mcda/fbwm.py` and `mcda/promethee.py` implement two complementary methods. Fuzzy BWM offers a lower-effort elicitation route than crisp AHP for weighting the six KPI blocks, and PROMETHEE II turns those weights into a network-wide, non-compensatory ranking of nodes.

The linguistic scale maps five qualitative judgements to triangular fuzzy numbers on a 1 to 4.5 scale, each carrying its own maximum consistency index. The consistency ratio is $CR_{BWM} = \xi^*/CI$, where ξ^* is the optimal consistency indicator returned by a constrained non-linear optimisation fitting the fuzzy weights to the elicited comparisons. If the optimisation fails to converge, the implementation falls back to uniform weights rather than raising an error, so a single malformed elicitation cannot break the weekly review. Fuzzy weights are defuzzified with the Graded Mean Integration Representation,

$$w_j = \frac{a_j + 4b_j + c_j}{6}, \quad (11)$$

then renormalised to sum to one. The triangular bounds are written (a_j, b_j, c_j) rather than the more common (l_j, m_j, u_j) to avoid colliding with the risk blocks u_m .

PROMETHEE II ranks the active nodes of a project on the same six weighted urgency blocks that feed local real urgency. For each pair of active nodes and each block, the preference intensity is computed with a Type V linear preference function with an indifference threshold, and the aggregated preference index and net outranking flow follow the standard formulation:

$$\pi(a, b) = \sum_m \omega_m P_m(g_m(a) - g_m(b)), \quad \phi(a) = \phi^+(a) - \phi^-(a). \quad (12)$$

The same weights ω_m used in the local aggregation are reused as the PROMETHEE criteria weights, so that the network ranking and the per-node urgency score remain mutually consistent rather than relying on a separately calibrated weight set.

3.3.4 Weekly Review Cycle and Calibrated Event Engine

The weekly review cycle keeps the twin's KPI state current without free-form manual data entry: every write is structured, signed and auditable. At each new week every active node is flagged as due for review, and the operator completes four sequential sections: the AHP questionnaire, pre-filled with the previous week's values; the week's KPI values, shown against the previous week; progress on active milestones, compared against theoretical planning progress; and any supply chain events that occurred during the week. All four are written as a single signed, atomic transaction, so no partial or anonymous review can corrupt a node's history.

Rather than asking operators to recompute the KPI impact of a disruption by hand, the system exposes a taxonomy of fourteen calibrated event types, each mapped to one or more KPI updates through one of three calibration operators: a Beta-Bernoulli Bayesian update for failure probabilities,

$$p_{\text{post}} = \text{clip}\left(\frac{p \times n_0 + k}{n_0 + n}, 10^{-4}, 0.99\right), \quad (13)$$

with a prior weight $n_0 = 26$, roughly six months of weekly observations; an exponential moving average for continuous variables such as OEE, $x_{\text{new}} = (1 - \lambda)x_{\text{old}} + \lambda x_{\text{obs}}$ with $\lambda \in [0.3, 0.5]$; and a ratchet update for severity, $s_{\text{new}} = \max(s_{\text{old}}, s_{\text{event}})$. Figure 12 follows one declared event through to the KPI fields it revises, showing which of the three operators applies where.

As a worked example, a node declaring a strike lasting $d = 24$ hours and affecting a fraction $p_f = 0.50$ of the workforce implies $H_{\text{lost}} = d \times p_f = 12$ equivalent lost hours over a 168-hour week, so $f_{\text{lost}} \approx 0.0714$ and the observed availability is $A_{\text{obs}} \approx 0.9286$. Updating a baseline availability of 0.95 with $\lambda = 0.5$ gives $A_{\text{new}} \approx 0.9393$, while risk severity updates in ratchet mode to $\max(0.30, 0.50) = 0.50$.

3.3.5 Explainability, Export and Operational Hardening

When the twin flags a node, the coordinator's first question is why. The per-node explanation page answers it with exact decompositions rather than approximations. Declared urgency splits into one additive contribution per AHP criterion, $\kappa_j = w_j(s_j - 1)/8$, summing exactly to $U_{d,\text{local}}$. Local real urgency splits the same way through the logarithm of Equation 4: each block contributes $\ell_m = -\omega_m \ln(1 - u_m)$, the shares $c_m = \ell_m / \sum_k \ell_k$ sum to one, and $U_{r,\text{local}} = 1 - \exp(-\sum_m \ell_m)$ reconstructs the score exactly. Figure 13 shows both decompositions side by side for a single node. This matters for the results reported later, because it is what allows a score to be traced back to the block that produced it without any manual computation.

One action compiles the global registry and per-node history into a twelve-worksheet Excel workbook, so that analyses can run outside the tool on frozen data. The persistence layer enables Write-Ahead Logging, runs an integrity check before every backup, and copies each database through the native `sqlite3.backup` API without locking the running application, with automatic backups every 24 hours and versioned schema migrations.

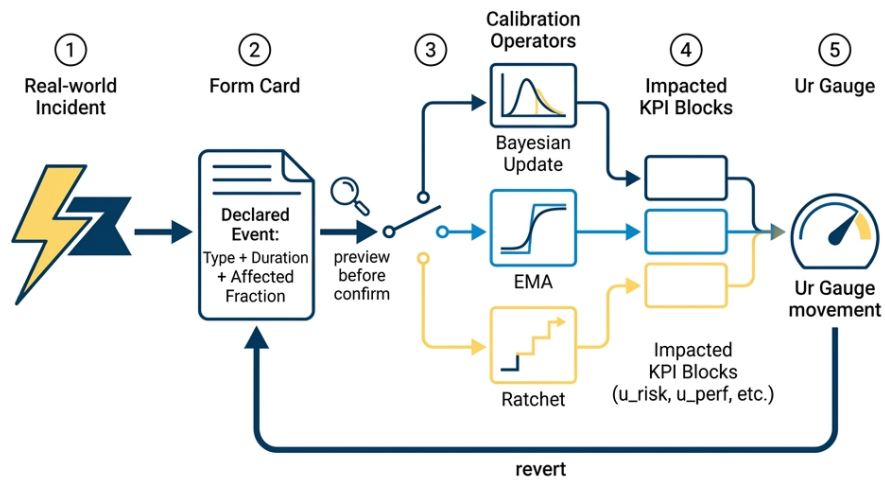


Figure 12: From a declared event to a calibrated KPI update

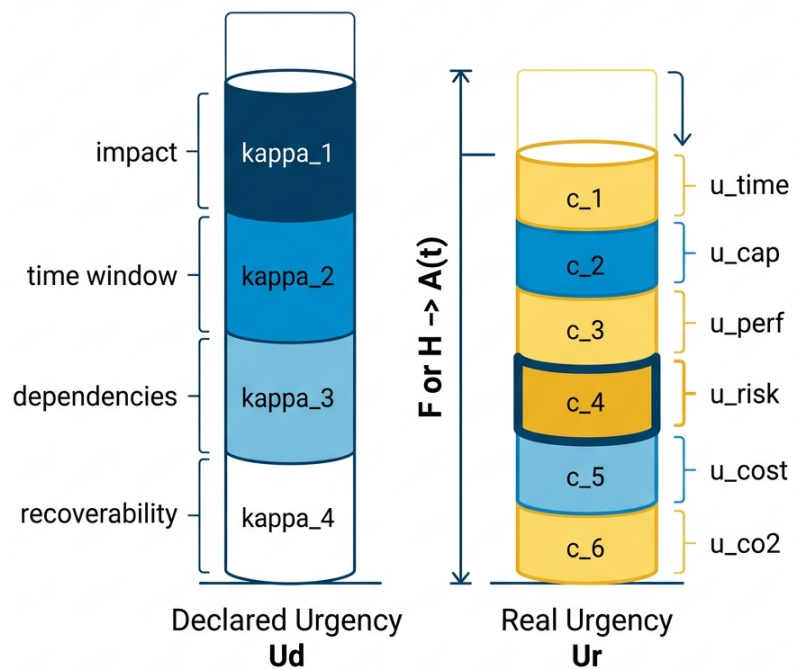


Figure 13: Per-node explainability decomposition

3.4 Experimental Design: The HÉLIOS Campaign

Everything above establishes that the model is implemented, internally consistent and tested. It does not establish whether the adequacy signal says anything true about an operator facing a crisis. This section presents the experiment built for that purpose.

3.4.1 Scenario and Network

The campaign, presented to participants under the cover story *Programme HÉLIOS*, replays the 2020 to 2022 semiconductor shortage over eighteen weekly game turns plus a training turn. The supply network has eight nodes, one per participant, anchored on the real actor types of the crisis (Table 10).

Table 10: The eight HÉLIOS campaign nodes

Node	Role	Rank	Real-world anchor
Orbitalys	Satellite prime, final customer	0	Fictional satellite programme, contractual delivery pressure
AvioSys Intégration	Avionics integrator	1	Qualified-stock squeeze on avionics tier-1 suppliers
Électis EMS	Electronics manufacturing services	2	EMS plants placed under allocation, 2021
CompoDis Europe	Component distributor	3	Distributor allocation and spot-market premiums
TransGlobal Fret	Freight operator	3	Suez Canal blockage, West Coast port congestion
NovaFab Semiconductors	Foundry, expected bottleneck	4	Composite of the Texas winter-storm freeze and the Taiwan drought
Meridian Semi	Secondary foundry, inert backup arc	4	A foundry sold out through 2023, the announced plan B that delivers nothing
SilPure Materials	Raw wafer supplier	5	Roughly 60% of world wafer supply, polysilicon price roughly tripling in 2021

Figure 14 gives the network with its per-arc coefficients, γ attenuating demand as it descends and β amplifying risk as it ascends, together with each node's tier rank. NovaFab at rank 4 is the expected bottleneck. The dashed arc into Meridian is the inert backup arc of Section 3.2.4: it appears in the diagram and carries nothing, which is the point of including it.

The turns follow five acts drawn from the documented chronology: a baseline where early warning signs go unheeded, a tipping point where a foundry freeze, a competitor fire and a canal blockage make the crisis visible, a squeeze where spot prices reach ten to forty times catalogue, a rupture where a major manufacturer cuts production by roughly 40% and the announced backup foundry turns out to have no spare capacity, and a slow, partial recovery that measures whether declared urgency lags the physical de-escalation. One game turn represents one real month, with real durations converted into engine hours at a fixed rate so that documented figures keep their true magnitude inside an eighteen-turn game.

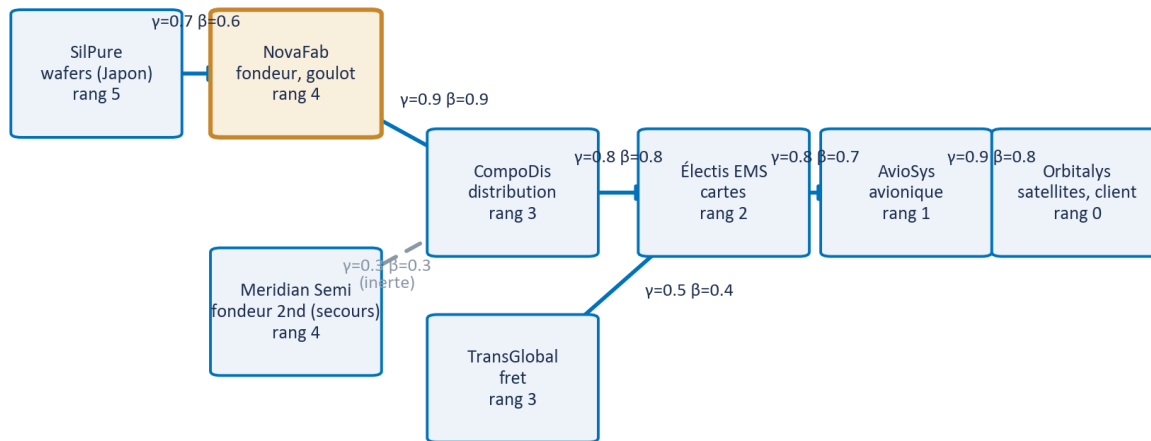


Figure 14: The HÉLIOS network and its per-arc coefficients

The twelve calibrated engine events driving the crisis are listed in Table 11, with the turn numbers as recorded in the campaign databases. One encoding choice is worth noting: the Texas freeze is modelled as a production accident, which triggers the engine’s Bayesian revision of the failure probability, a recovery clock and a severity ratchet, rather than as a capacity loss, which would only reduce storage volume. A hot shutdown of a fabrication line is mechanically closer to the former.

Table 11: The twelve calibrated engine events

Turn	Node	Event type	Real-world anchor
T3	NovaFab	Demand spike	Automotive reorders on sold-out slots, Nov. 2020
T5	CompoDis	Supplier delay	Distributor moves to allocation, Jan. 2021
T6	NovaFab	Accident (critical)	Texas winter-storm fab freeze, Feb. 2021
T7	NovaFab	Demand spike	Reorders after a competitor’s fab fire, Mar. 2021
T7	TransGlobal	Transport disruption	Suez Canal blockage, six days, Mar. 2021
T10	NovaFab	Material shortage	Taiwan drought, ultrapure-water rationing, Jun. 2021
T12	Électis	Capacity loss	Southeast-Asia backend lockdowns, Aug. 2021
T12	TransGlobal	Transport disruption	Shift to saturated air freight, Aug. 2021
T13	CompoDis	Demand spike	Bullwhip over-ordering, Sep. 2021
T14	TransGlobal	Transport disruption	West Coast port congestion, Oct. 2021
T14	CompoDis	Tariff increase	Allocation spot premiums, Oct. 2021
T16	SilPure	Tariff increase	Polysilicon price pass-through, Dec. 2021

3.4.2 Data Provenance

The two urgency signals come from different sources, which is the point of the design. Real urgency is fed from public time series of the actual crisis, namely INSEE production and price indices, WSTS worldwide semiconductor billings and a semiconductor-shortage dataset, frozen in a data pack whose SHA-256 hashes were recorded before the first turn. Declared urgency

comes from the participants through the weekly AHP questionnaire, with the consistency gate active.

Every third-party dataset is passed through an authenticity check before use. A logistics dataset initially considered for the freight node was rejected at this check for the signatures of synthetic training data, namely pre-computed target columns, GPS coordinates scattered outside the region the dataset claimed to cover, and uniformly flat distributions. The semiconductor-shortage dataset passed, showing a secular price decline, a genuine 2021 upturn and sectoral employment consistent with official series. The rejected dataset is retained in the raw-data folder for audit but is never loaded by the analysis pipeline, and the freight node's cost block is driven entirely by calibrated events instead.

Two provenance rules protect the real-urgency series. A single-pilot rule forbids double counting: each KPI field of each node is driven either by a time series or by a calibrated event, never by both, so when an event writes a field on a given turn the series skips that turn. Second, one honest artefact of the real data is kept rather than smoothed away. The INSEE business-failures series falls through 2020 and 2021 under public support schemes, so a failure-rate indicator taken alone would lower external risk exactly while the sector crisis was worsening. Keeping it makes the multi-block argument concrete: a single macroeconomic indicator can tell the opposite of the ground truth.

3.4.3 The Three Arms

The experiment compares three arms on the same frozen scenario. The comparison is what allows the effect of a treatment to be separated from the behaviour of the scenario.

Arm A, blind declaration. Eight isolated agents declare their urgency each turn seeing only their own role card and turn briefing. No forecast is shown. This is the control condition and the source of the prospective forecasts that are scored in Section 4.5.

Arm B, forecast-assisted declaration. The same agents, the same frozen scenario, but from turn 4 onward each agent is additionally shown the predictive analysis of its own node before declaring. The loop is open: nothing changes in the terrain, the scenario stays frozen, so the forecasts remain scorable and the effect of the treatment is measured on perception alone. Each declaration additionally records an `influence_prediction` field taking one of four values, meaning no influence, confirms my assessment, revised upward, or revised downward. That field is the direct measure of the treatment effect.

Arm C, closed loop. The agents act on the terrain through a catalogue of interventions, so the ground truth itself changes. Arm C is designed but not executed, and is reported in Section 5.4 as future work.

The distinction between B and C is one of nature rather than degree. In arm B the ground truth is fixed, so the predictions are scorable with proper scoring rules and the treatment effect is measured on perception. In arm C, action destroys the reference truth, because a rupture avoided thanks to an alert makes a good prediction look wrong, so arm C could only be judged by comparing its final state with arm A.

Two properties of this design are worth stating explicitly, because the results depend on them. First, the agents are language models, one per role, each confined to its own role card and turn briefings with no access to the other roles, the dashboard or future turns. This

anti-lookahead discipline is enforced by the harness rather than by instruction. A single model instance per role has no statistical standing as a substitute for eight independent human judges, and Section 4.8 treats that as the principal threat to external validity. Second, the declaration format in arm B is identical to arm A, which is what makes the two comparable, with the single addition of the influence field.

3.4.4 Scoring Protocol

The forecast layer emits, for each node and each turn from turn 4 onward, a probability p_h that an adverse outcome occurs at horizons $h \in \{1, 2, 3, 4\}$ weeks. Scoring those forecasts requires an outcome definition, and this thesis uses the one the production calibration service itself applies, so that the model is judged against the target it was built for. A node-week is a positive observation when at least one of three conditions holds within the window $[S + 1, S + h]$: a milestone of the node whose deadline falls in the window is abandoned or is still active with its deadline passed; the node is abandoned; or an event of gravity critical or defect is recorded and not subsequently reverted.

Four quantities are computed, following standard forecast verification practice (Brier, 1950; Murphy, 1973; Gneiting et al., 2007):

- **Discrimination**, by the area under the ROC curve, computed by mean ranks so that tied forecasts are handled correctly, with a percentile bootstrap 95% interval over 2 000 resamples. AUC answers whether the forecast orders node-weeks correctly.
- **Calibration**, by the Brier score and its skill against the climatological forecast that always announces the base rate. A negative skill means the forecast is worse calibrated than announcing the base rate.
- **Reliability**, by a banded table comparing the mean announced probability with the observed frequency in each band. The bands are deliberately uneven, because a uniform ten-bin split hides the failure mode that matters here.
- **Operating point**, by the confusion matrix at the pre-registered threshold of 0.5 used by hypothesis H4.

Separating discrimination from calibration is essential rather than decorative (Murphy, 1973): a forecast can order outcomes correctly while announcing the wrong probabilities, and the two failures call for different remedies. Section 4.5 finds exactly that dissociation.

One further check is built into the protocol. Because the outcome definition is a modelling choice rather than a given, the same forecasts are additionally scored against a deliberately naive target, namely any non-reverted event on the node regardless of gravity and ignoring milestones. The difference between the two scorings is reported in Section 4.6 and is treated as a result in its own right.

3.4.5 Protocol Integrity and Controls

The protocol was frozen before the first turn. Participants see only their own role sheet and questionnaire; dashboards and scores stay hidden until the debrief, and no discussion between roles is allowed. Two placebo sheets, a false de-escalation handed to one node at turn 9 and to another at turn 15, control for suggestibility: if a declared urgency follows the placebo, the measure is reflecting the sheet handed over rather than a reading of the situation. A weak-signal turn, where the press mentions a tension with no engine event

behind it, measures who reacts to media noise alone. Roles are assigned so that the two most information-heavy nodes do not go to the most experienced participants, keeping a role effect from being confounded with an expertise effect. Every deviation from protocol is logged the same day in a campaign journal, and any analysis outside the pre-registered hypothesis register of Section 1.5 is labelled exploratory.

3.5 Limitations and Simplifications of the Method

Four simplifications are built into the method itself, as distinct from the limitations of the delivered framework catalogued in Section 5.3.

The six KPI blocks are treated as probabilistically independent so that the noisy-OR aggregation is admissible. They are not independent in reality: a capacity loss raises cost, and a performance drop lengthens lead time. The consequence is that correlated blocks double-count evidence, which the composite-indicator literature identifies as the standard failure mode of weighted aggregation (Greco et al., 2018; Becker et al., 2017). The explainability decomposition of Section 3.3.5 is the mitigation: the coordinator sees per-block contributions rather than only the aggregate, so a correlation is visible rather than hidden.

The network is assumed acyclic, which permits a linear topological sort and makes propagation deterministic and cheap. Real supply networks contain feedback, most obviously in returns and rework loops.

The scenario is scripted, which is what makes the ground truth knowable and the forecasts scorable, but it also means the events are the only source of adverse outcomes. The consequence, quantified in Section 4.5, is that the positive class is extremely small, and any conclusion about discrimination is correspondingly weak.

The declarants are language models. This buys perfect protocol compliance, exact reproducibility and no scheduling cost, and it loses the property that matters most for hypothesis H4, namely a genuinely imperfect human perception of a crisis. This is discussed at length in Section 4.8 rather than treated as a detail.

3.6 Reproducibility

Every numerical result in Section 4 is regenerated from raw artifacts by a single script, `Utilities/figures/derive.py`, which writes a JSON file that both the figures and the text of this thesis read. No number in this document is transcribed by hand from an intermediate report.

The inputs to that script are the two campaign databases, one per arm, and the raw per-agent declaration and forecast logs. The outcome labels are not reimplemented: the script calls the production `CalibrationService` directly, so the thesis scores the model against the same target definition the running system applies. Random procedures are seeded, specifically the bootstrap interval on the AUC and the Monte Carlo lead-time simulation.

Two provisions guard against silent drift. The script recomputes the arm A against arm B forecast comparison on every run and reports whether the two arms carry identical prediction-layer output, which they must, since the replay is deterministic and only the declarations differ.

It also reports the outcome base rate at every horizon, so that a change in the underlying database would be visible immediately rather than absorbed into a summary statistic.

One earlier analysis file in the campaign folder, `analysis/rapport_experience.md` with its companion JSON, contains a superseded run whose participant-influence counts disagree with the raw logs. It is deliberately not used anywhere in this thesis, and the derivation script names it in an exclusion list so that the choice is recorded in the output rather than left implicit.

The satellite observation system of Section 4.9 is a separate codebase with its own verification script. Its network calls are cached on disk, so a re-run does not depend on the availability of the public services it draws from, and its calibrated constants are stated in the source with the sample size behind each.

3.7 Ethical Considerations

The research was conducted under the Cranfield University Research Ethics System. The study as executed involves no human participants: the eight declarants are language-model agents, and no personal data was collected, processed or stored at any point. The approval documentation is reproduced in Appendix A.

The protocol was originally designed for eight human consultants, and that version would have required participant consent for two specific features. The first is the cover story: participants would be told they are playing a fictional satellite programme without being told that the scenario replays a documented historical crisis, since knowing it would destroy the measure. The second is the two placebo sheets, which deliberately hand a participant false information. Both are standard incomplete-disclosure designs and both would be resolved by a full debrief at the end of the campaign, in which the real scenario is revealed and each participant is asked at which turn they recognised it. Were that campaign to be run, informed consent, the right to withdraw, and the debrief would need to be in place before the first turn.

Two further considerations apply to the work as it stands. The industrial data behind the scenario is public, namely INSEE and WSTS series and an open dataset, and the commercial material belonging to ALTEN is covered by the confidentiality statement at the front of this thesis. For the satellite observation work, all imagery and building data come from public French open-data services under their published licences, no imagery of private individuals is collected or retained, and the analysis operates on industrial buildings rather than on people. The regulatory database used there serves only to calibrate a constant and is never used to infer a result about a specific site, which is a deliberate constraint rather than an incidental property.

3.8 Conclusion

The method combines a probabilistic multi-block urgency model, a structured elicitation of declared urgency, a bidirectional propagation over the supply network, and an asymmetric scoring of the mismatch between the two, implemented as a tested, auditable and persistent system. Around that system, a pre-registered three-arm experiment on a frozen historical scenario allows the resulting forecast to be scored against recorded outcomes with proper scoring rules, and allows the effect of showing that forecast to the declarant to be isolated

from the behaviour of the scenario itself.

What this method can establish is bounded by its own design, and those bounds have been stated rather than left implicit. [Section 4](#) reports what it measured.

4 Results and Discussion

4.1 Introduction

This section reports what was measured, in five parts. The delivered system and its mechanical validation come first, establishing that the pipeline runs end to end. The blind declaration arm follows, giving the pre-registered hypothesis verdicts. The forecast layer is then scored against recorded outcomes, which produces the central negative result of this thesis and, through it, a diagnosis of the mechanism behind it. The forecast-assisted arm measures what happens when that same forecast is placed in front of the declarant. Finally, a short opening reports a separate system that addresses the freshness limitation the rest of this work documents about itself.

Every number below is regenerated from raw artifacts by the derivation script described in Section 3.6.

4.2 The Delivered System

SupplyScore is delivered as a standalone Python application: 73 modules across 10 packages, covered by 1779 automated test functions in 106 test files, driving a multi-page Dash web interface over a multi-tenant SQLite persistence layer. All of the mathematical objects of Section 3.2 are implemented and exercised: the six-block noisy-OR aggregation, the AHP elicitation with its consistency gate, the Fuzzy BWM weighting, the bidirectional DAG propagation, the asymmetric adequacy score, the Monte Carlo lead-time estimate, the PROMETHEE II network outranking and the systematic shock-simulation criticality index.

Three operational capabilities matter for what follows rather than being peripheral. The per-node explainability page reconstructs any score term by term, which is what allowed the diagnosis in Section 4.5.4 to be made from the application rather than by hand. The Excel export produces frozen data for analysis outside the tool, which the campaign relies on. The hardened persistence layer, with write-ahead logging, pre-backup integrity checks and automatic snapshots, is what makes a campaign database an auditable record weeks after the run.

Two pages carry most of the operator’s interaction with the system. Figure 15 shows the weekly review, whose four sections commit as one signed transaction so that no partial review can enter a node’s history. Figure 16 shows the ranking the shock simulation produces, which is the view a coordinator would use to decide where to look first.

4.3 Mechanical Validation: The Synthetic Dry Run

Before any declaration data existed, the full campaign mechanics were exercised end to end on a disposable database. Five iterations calibrated the scenario and eliminated engine artefacts, and the final iteration replayed all nineteen turns with synthetic respondents standing in for the eight declarants.

This is an engineering check, not a scientific one. The reason is in how the synthetic declarations were made: each one is anchored on the node’s own local real urgency, following $1 + 8 \times U_{r,\text{local}}$ with a fixed panic, under-report or neutral bias per role. A declaration built that way cannot lag the signal it is derived from. It cannot under-report that signal for long,

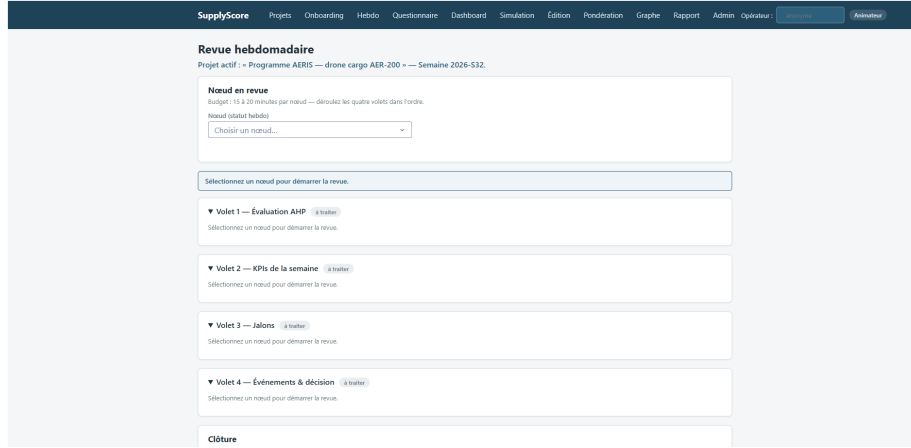


Figure 15: The weekly review page

and it cannot react at all to a press event it never reads. Run against the pre-registered tests, it therefore returns not confirmed or not applicable on all six hypotheses by construction. That is the expected outcome, and an informative one.

What it does establish is that injection, weekly decay, milestone drift, event calibration, propagation, criticality, snapshotting and backup all run correctly under the two-turns-per-day cadence the campaign requires. It also calibrated the null baseline detector, a raw threshold $U_{r,local} \geq 0.5$, at precision 0.16 and recall 0.27: the bar the adequacy signal must clear to demonstrate value beyond a raw KPI reading.

Figure 17 shows the circularity visually. Real urgency is drawn solid and synthetic declared urgency dashed, for all eight nodes across the full replay. The two track each other closely at every node rather than one lagging or decoupling from the other, which is the signature of the construction and not a property the campaign is expected to reproduce. It is also why H1, H2, H5 and H6 cannot be read off this run: each of them asks about a divergence that the generator cannot produce.

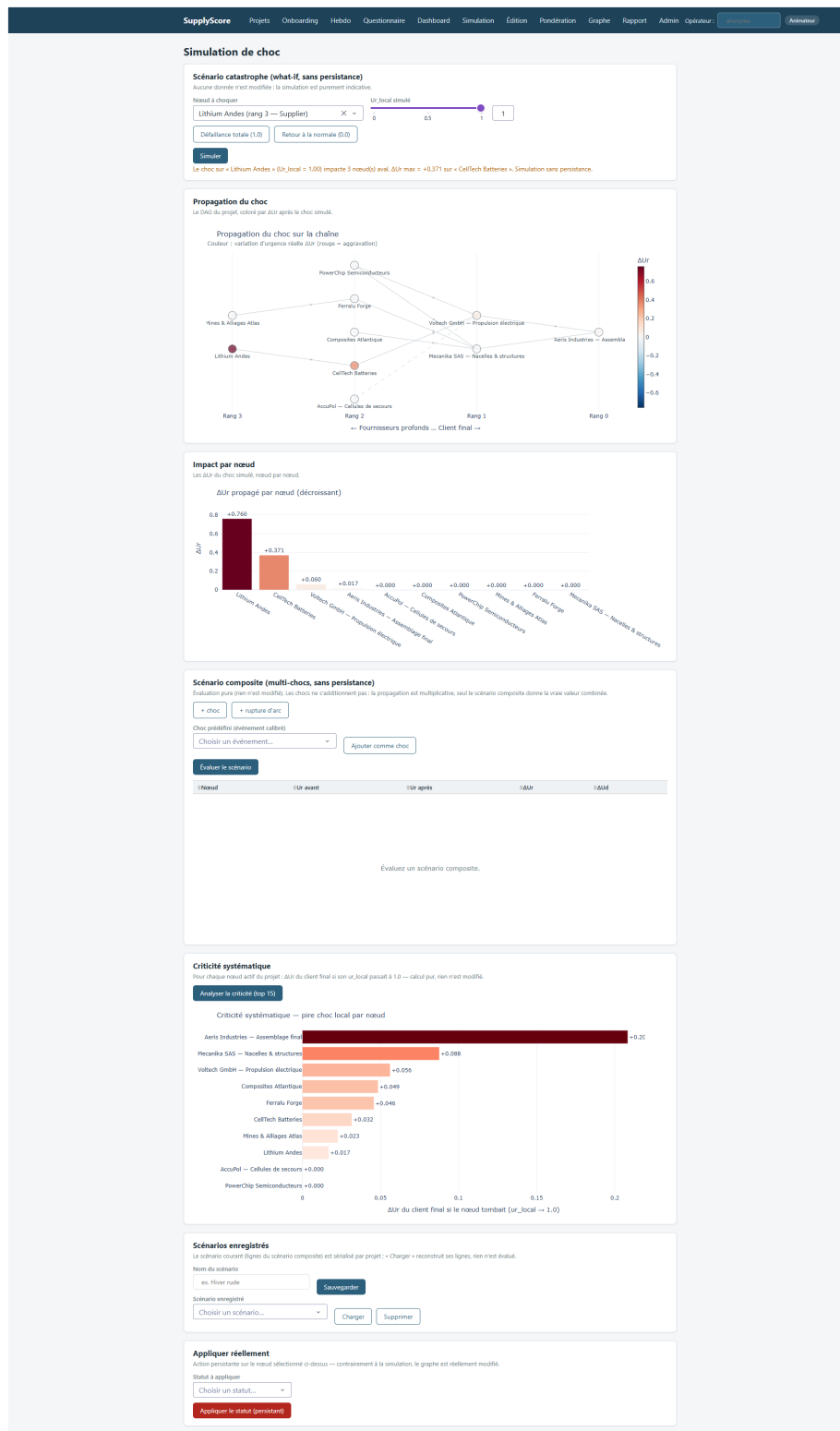
4.4 Arm A: Blind Declaration by Eight Isolated Agents

Arm A replaced the synthetic generator with eight isolated language-model agents, one per role. Each was confined to its own role card and turn briefings, with no sight of the other roles, the dashboard, or any future turn. Each read its briefing in character, then answered the real AHP questionnaire of Section 3.3.2; pairwise comparisons and severity scores passed through the production consistency-ratio code untouched.

The difference from the dry run matters. This arm drove the full production pipeline. Campaign setup, turn injection and questionnaire submission all went through the same weekly-cycle service a browser-based participant would reach.

Table 12 gives the verdicts on the six pre-registered tests.

Two results stand out. The urgency-contagion hypothesis H5 is confirmed: declared urgency at nodes that are not physically affected rises by 0.079 on press-event turns against 0.005 on calm turns, matching the pre-registered direction and ordering. This is the mere-urgency effect (M. Zhu et al., 2018) reproduced at network scale. The latency hypothesis H1 moves



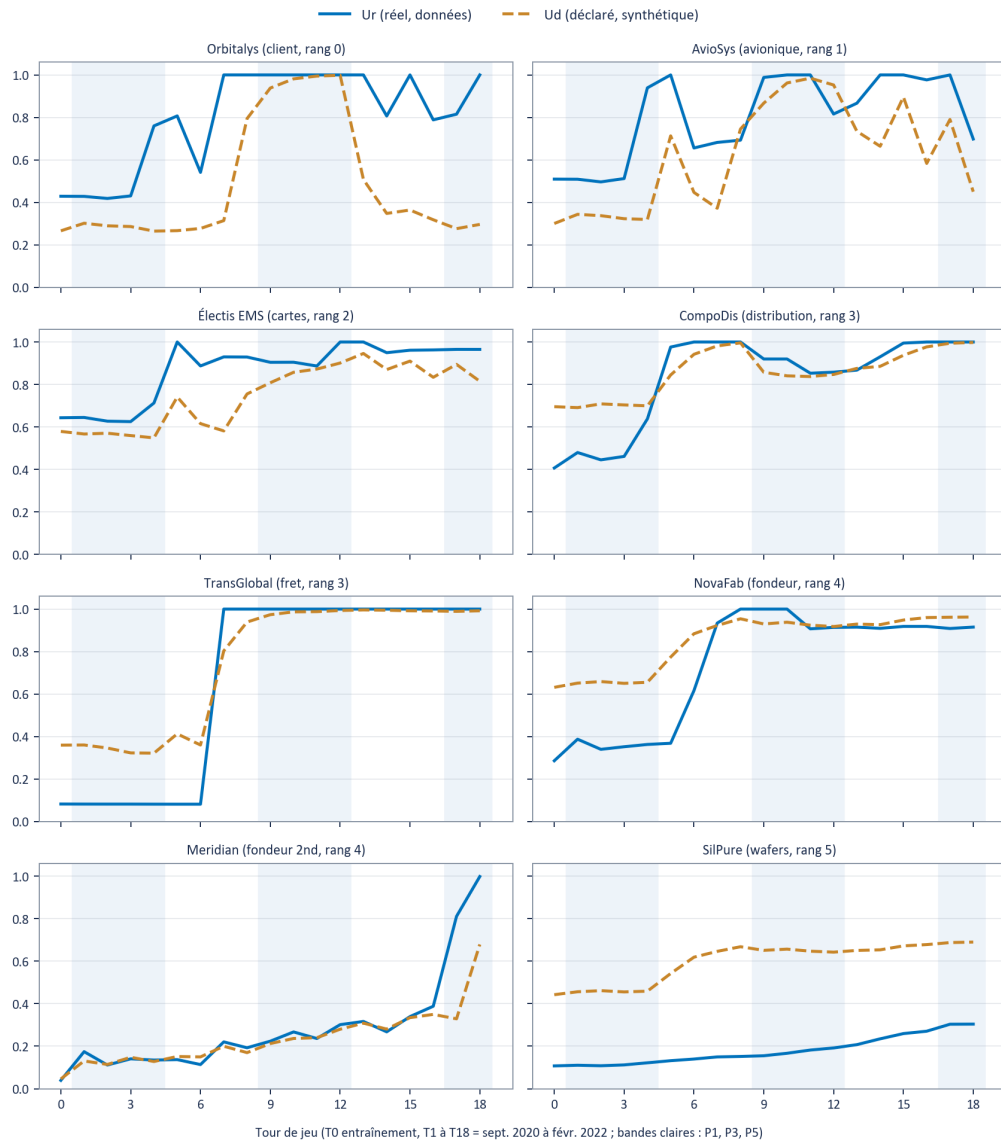


Figure 17: Real against synthetic declared urgency, dry run

Table 12: Arm A verdicts on the six pre-registered hypotheses

Test	Verdict	Measured value
H1 latency	Not confirmed	3 of 8 nodes at lag ≥ 1 turn (threshold: 5 of 8)
H2 early detection	Not confirmed	H (NovaFab) peaks at 0.017 on turns 6 to 12 (threshold: 0.3)
H3 criticality	Not confirmed	NovaFab top-1 on 7 of 10 informative turns, 70% (threshold: 80%; saturated turns excluded per Section 3.2.5)
H4 predictive validity	Not confirmed	precision 0.00, recall 0.00 at $H \geq 0.5$
H5 contagion	Confirmed	non-affected nodes: $\Delta U_d = +0.079$ on press turns against $+0.005$ on calm turns
H6 hysteresis	Not applicable	real urgency shows no measurable decline over the recovery turns

from 1 of 8 nodes on the synthetic dry run to 3 of 8 at a lag of one turn or more, closer to but still short of the threshold of 5.

H4 stays not confirmed for a specific and informative reason. These agents read their briefing carefully and declared urgency close to what the briefing in fact described, so their U_d tracks U_r too closely to leave the gap that H is built to catch. The tracking is even tighter than the fixed-bias dry run, with $H(\text{NovaFab})$ peaking at 0.017 against 0.070. H3 is unchanged from the dry run because it depends only on the network’s real-urgency criticality ranking, not on U_d at all.

The arm left behind a live project, reloadable in the same dashboard a coordinator would use in production. Read directly from that project after the eighteenth turn, the network-average adequacy is 65.5, four of the eight nodes carry a hidden-risk reading above 0.1, one carries a false-urgency reading above 0.1, and the worst-scoring node by a wide margin is Orbitalys at $A = 15.3$. Table 13 gives the full per-node snapshot.

Table 13: Final per-node snapshot of the arm A project

Node	U_d	U_r	A	F	H
Orbitalys	0.409	1.000	15.3	0.000	0.591
AvioSys Intégration	0.727	0.698	95.2	0.029	0.000
Électis EMS	0.914	0.965	82.9	0.000	0.052
CompoDis Europe	0.882	1.000	67.5	0.000	0.118
TransGlobal Fret	0.819	1.000	56.0	0.000	0.181
Meridian Semi	0.841	0.998	60.1	0.000	0.157
NovaFab Semiconductors	0.912	0.915	98.5	0.000	0.003
SilPure Materials	0.876	0.304	48.9	0.572	0.000

Orbitalys’ own explanation page traces the chain from a single scripted event to its score without manual computation. Its next milestone had slipped past its deadline by turn 18, which forces u_{time} to 1 under the rule of Table 6; with the node’s real urgency already saturated by this cause alone, the page attributes the whole of it to the local block and none to the upstream network. The agent playing Orbitalys declared $U_d = 0.41$, so the hidden risk is $H = 0.59$, penalised through Equation 9 at $2.25 \times 0.59^{0.88} = 1.42$, giving $A = 100 \times (e^{-1.42} - e^{-2.25}) / (1 - e^{-2.25}) = 15.3$, the same number the dashboard reports. The network-wide PROMETHEE II ranking places NovaFab and Orbitalys at the top of the priority list, at $\phi = +0.220$ and $+0.215$, which are the expected bottleneck and the node its own missed milestone makes most urgent. The same ranking carries a caveat the tool raises on its own: on this run the time and performance blocks are perfectly correlated, $\rho = 1.00$, which is exactly the composite-indicator risk flagged in Section 2.8.

4.5 Arm A Rescored: Discrimination Without Calibration

The hypothesis verdicts above test the adequacy signal at a single threshold. The forecast layer that the system also emits, a probability of an adverse outcome at horizons of one to four weeks, deserves a fuller treatment, because a single operating point cannot distinguish a forecast that orders outcomes badly from one that orders them well but announces the wrong numbers. This subsection scores all 120 prospective forecasts of arm A, covering turns 4 to 18 across eight nodes, against the production outcome definition of Section 3.4.4.

4.5.1 Discrimination

Table 14 gives the results at each horizon.

Table 14: The arm A forecast layer, scored by horizon

Horizon	Positives	Base rate	AUC [95% CI]	Brier	Brier skill
1 week	2	0.017	0.555 [0.470, 0.639]	0.130	−6.93
2 weeks	4	0.033	0.372 [0.258, 0.492]	0.166	−4.14
3 weeks	5	0.042	0.428 [0.315, 0.546]	0.191	−3.77
4 weeks	6	0.050	0.485 [0.389, 0.583]	0.207	−3.36

The discrimination result is that there is none to speak of. The AUC ranges from 0.372 to 0.555 and every 95% interval contains 0.5, so on this campaign the forecast does not order node-weeks better than chance. That statement must be made with its sample size attached: the positive class contains between 2 and 6 observations depending on horizon, which is far too few to support a confident claim in either direction. What can be said is that the campaign provides no evidence of discrimination, not that discrimination has been refuted.

4.5.2 Calibration

The calibration result is much stronger, because it does not depend on the small positive class in the same way. The Brier skill against climatology is between −3.36 and −6.93, meaning the forecast is between four and eight times worse, in mean squared error, than a trivial forecaster that announces the base rate every time. A negative skill of that magnitude is not a marginal shortfall.

The reliability table shows where it comes from. Table 15 bands the four-week forecasts by announced probability and compares the mean announced value with the observed frequency in each band.

Table 15: Reliability of the arm A four-week forecasts

Announced band	<i>n</i>	Mean announced	Observed	Reading
[0.0, 0.1)	74	0.070	0.081	Well calibrated
[0.1, 0.2)	24	0.128	0.000	None materialised
[0.2, 0.5)	1	0.354	0.000	None materialised
[0.5, 0.9)	7	0.859	0.000	None materialised
[0.9, 1.0]	14	0.992	0.000	None materialised

The pattern is unambiguous, and Figure 18 shows it, with each marker's area proportional to the number of forecasts in its band and the diagonal marking perfect calibration. Where the model announces a low probability, it is right: 74 forecasts announcing a mean of 0.070 were followed by an adverse outcome 8.1% of the time. Where it announces a high probability, it

is wrong every single time. Twenty-one forecasts announced a mean probability of 0.948, and none of the twenty-one was followed by the outcome it predicted. The confusion matrix at the pre-registered threshold of 0.5 records this as 0 true positives, 21 false positives, 6 false negatives and 93 true negatives, which is why H4 returns precision 0.00 and recall 0.00.

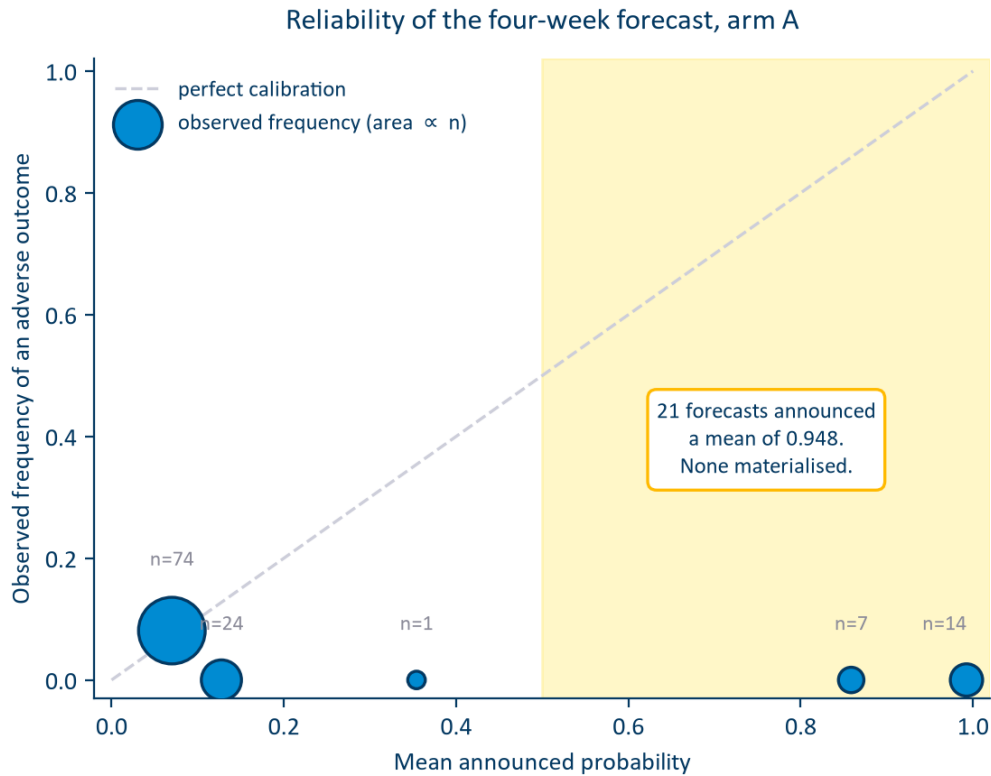


Figure 18: Reliability of the four-week forecast, arm A

The two failures are therefore of different kinds, and this is the dissociation that Murphy's decomposition anticipates (Murphy, 1973): the ordering is uninformative, and the level is badly wrong at the top of the range while being close to right at the bottom.

4.5.3 Ruling Out Right-Censoring

An obvious objection is that forecasts made near the end of the campaign have observation windows extending past the last turn, so their outcomes could not be recorded even if they occurred. If the confident forecasts were concentrated at the end, censoring rather than miscalibration would explain the result.

They are not. The twenty-one confident forecasts are spread over turns 5 to 18 and across seven of the eight nodes. Repeating the scoring with every forecast whose window would run past the final turn removed leaves $n = 88$ forecasts, still 0 true positives against 16 false positives, and the Brier skill becomes worse rather than better, moving from -3.36 to -7.82 . Censoring is not the explanation.

4.5.4 The Mechanism

The confident tail has a single source, and the explainability layer of Section 3.3.5 makes it visible without instrumentation. Across all twenty-one confident forecasts, the four-week

probability and the milestone-miss probability never differ by more than 0.020. The confident tail is the milestone channel, and nothing else contributes to it.

Set against the registry, this is decisive. Of the eighteen milestones in the campaign, ten completed and eight remained active, and none was abandoned. Exactly one was actually missed at its deadline, the Orbitalys delivery of Table 13. The model announced near-certain misses for AvioSys, CompoDis, Électis, NovaFab, TransGlobal and Meridian, and all of them delivered.

The same defect is visible directly in the state snapshots, from the opposite side. Reading the u_{time} block at turn 6, the turn on which the scenario's one critical event lands, gives the values of Table 16.

Table 16: The time block at turn 6

Node	u_{time}
Orbitalys	7.3×10^{-9}
AvioSys Intégration	1.2×10^{-8}
Électis EMS	0.0
CompoDis Europe	0.99996
TransGlobal Fret	1.7×10^{-4}
Meridian Semi	2.9×10^{-9}
NovaFab Semiconductors	0.0216
SilPure Materials	0.0

Six of the eight nodes read as numerically zero in the middle of a crisis, and the seventh reads as saturated. The block behaves as a step function rather than as a risk gradient: it is either off or fully on, with almost nothing in between. The cause is structural. The block compares a lead-time distribution against the time remaining to the next deadline, and the scripted events act on capacity, cost and failure probability without acting on the lead time itself, so a shock never moves the quantity that the milestone probability is computed from. The block is blind to the crisis by construction, and because that block drives the entire confident tail, so is the forecast.

This is a defect in the model rather than a property of the scenario, and it is the single most actionable finding of the campaign. It also invalidates, for this run, the assumption recorded in Section 3.2.2 that each block contributes useful information to the aggregate. One block did not.

4.6 The Target-Definition Trap

The scores above depend on a choice that is easy to overlook: what counts as an adverse outcome. Section 3.4.4 uses the production definition, requiring a missed milestone, an abandoned node, or an event of gravity critical or defect. A plausible alternative, and the one an evaluator without access to the model's own definition would most likely reach for, is simply any event recorded on the node.

Scoring the identical forecasts against that naive target gives Table 17.

The model is identical, the forecasts are identical, and the horizon is identical. Only the definition of truth changes. The measured base rate moves from 5.0% to 38.7%, and the

Table 17: The same forecasts under two outcome definitions

Outcome definition (4-week horizon)	n	Base rate	AUC	Brier skill
Production definition (missed milestone, abandonment, or critical event)	120	0.050	0.485	−3.36
Naive definition (any recorded event)	75	0.387	0.484	−0.64

Brier skill improves by a factor of more than five, from -3.36 to -0.64 . An analyst choosing the easier target would report a system that appears five times better calibrated without a single line of the model having changed.

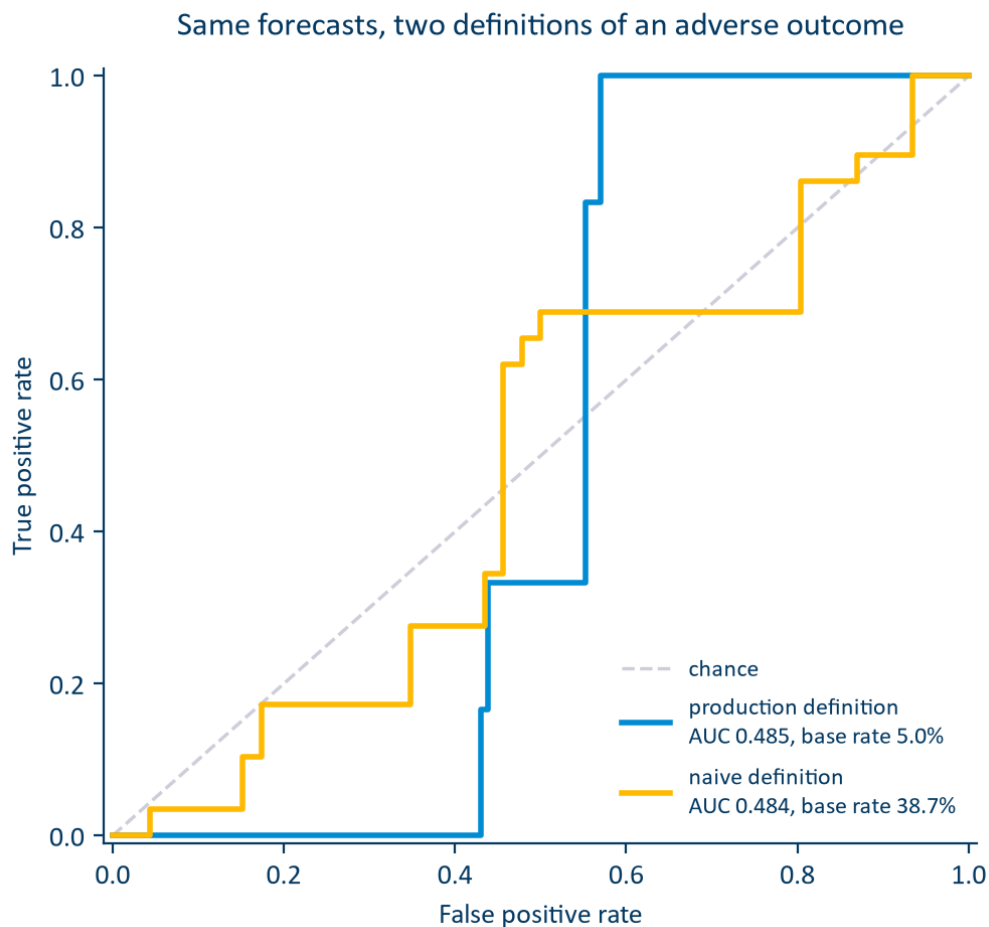


Figure 19: ROC under both outcome definitions

Discrimination, by contrast, is almost untouched, moving from 0.485 to 0.484 (Figure 19). That contrast is itself informative: the ordering of node-weeks is a property of the forecast, while the apparent quality of the probabilities is substantially a property of how often the target fires. A frequent target flatters a forecaster who announces high probabilities, because more of those high announcements happen to land on a positive.

The practical consequence is that a calibration figure reported without its outcome definition is close to meaningless. This has a direct bearing on how the framework of this thesis should be evaluated in future work: the outcome definition belongs in the pre-registration alongside the thresholds, and it is not currently part of the H1 to H6 register.

4.7 Arm B: What Happens When the Forecast Is Shown

Arm B replays the identical frozen scenario with the identical eight agents, changing one thing: from turn 4 onward, each agent is shown the predictive analysis of its own node before declaring its urgency. The loop stays open, so the terrain does not change and the forecasts remain scorable.

One property of the design must be established before any interpretation. The prediction layer is deterministic, so the two arms should carry identical forecasts on the turns they share, and they do: all 72 shared node-turn forecasts are numerically identical between the arms. Any behavioural difference therefore comes from the act of displaying the forecast, not from a different forecast being displayed.

The dataset comprises 97 declaration records, reduced to 96 after removing one duplicate submission by a single node at turn 1. Each carries an influence field recording whether the displayed forecast had no influence, confirmed the agent’s own assessment, or caused an upward or downward revision. Table 18 gives the distribution turn by turn.

Table 18: Arm B: influence of the displayed forecast, by turn

Turn	No influence	Confirms	Revised up	Revised down	Note
T0	8	0	0	0	forecast hidden
T1	8	0	0	0	forecast hidden
T2	8	0	0	0	forecast hidden
T3	8	0	0	0	forecast hidden
T4	0	3	0	5	first exposure
T5	0	5	3	0	first upward revisions
T6	1	3	2	2	critical accident event
T7	0	4	3	1	
T8	1	1	3	3	
T9	2	5	1	0	
T10	2	3	2	1	
T11	0	3	1	4	
Total	38	27	15	16	$n = 96$

The first exposure is the result, and Figure 20 shows it. At turn 4, when the forecast becomes visible for the first time, all eight agents record an influence, and the direction is entirely one-sided: five revised their declared urgency downward and three had their existing assessment confirmed. Not one revised upward. The first upward revision anywhere in the campaign appears at turn 5, and the scenario’s single critical event, the foundry accident, lands at turn 6, two turns after the moment every declarant was either reassured or confirmed.

Over the full campaign the distribution is close to balanced, with 15 upward against 16 downward revisions, so the one-sided effect is specific to first contact rather than a persistent bias. That distinction matters for interpretation: what the data shows is a strong first-exposure effect, not a permanent suppression of declared urgency.

Two aggravating factors follow from Section 4.5. First, the forecast being displayed at turn 4 is drawn from the same layer whose confident predictions never materialise, and whose u_{time} block is blind to the very shock that is about to arrive. Second, the forecast is displayed as a

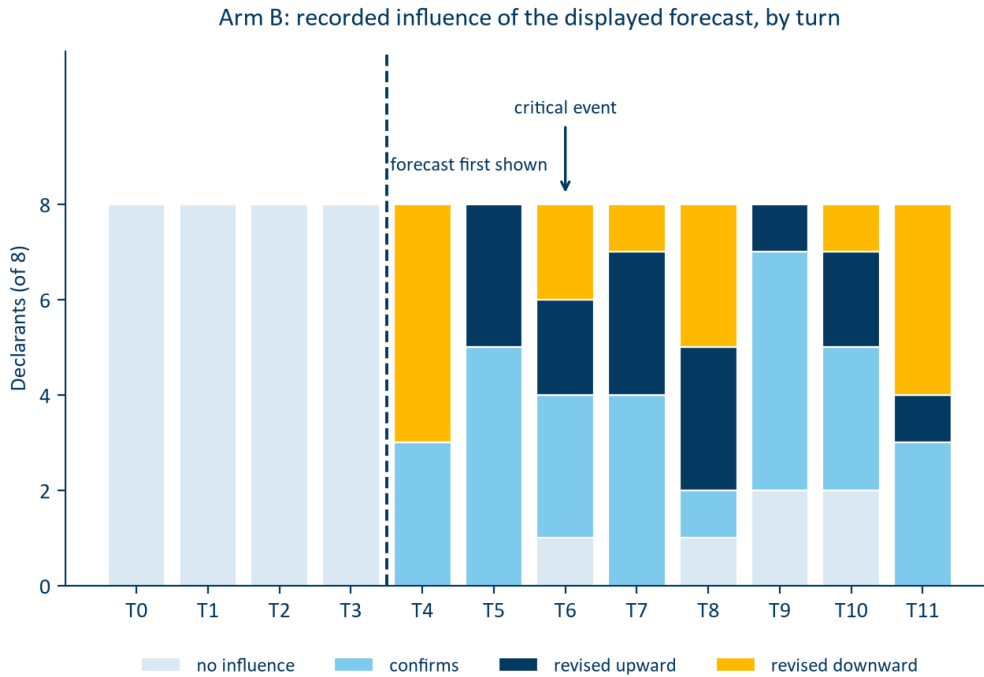


Figure 20: Arm B: influence of the displayed forecast, by turn

numeric probability with a confidence interval, a presentation format the automation-bias literature identifies as increasing reliance (Goddard et al., 2012; Bansal et al., 2021). The system was, in effect, presenting an unfounded reassurance in the most persuasive available form, immediately before the event it had failed to anticipate.

One qualification is important and cuts against the strongest reading. The influence field records the effect on the declared urgency level, not on the decision taken. Inspection of the free-text notes shows that agents continued to take protective actions at the same turn, including negotiating allocation, alerting the customer and ordering ahead of schedule. The measured effect is on what was declared, not on what was done. Given that the declared value is precisely the input the adequacy layer consumes, this is a consequential effect on the measurement system even where it is not, on this evidence, an effect on behaviour.

4.8 Discussion

The circularity problem is the central methodological finding. Across the dry run, arm A and arm B, the same difficulty recurs in three different forms. In the dry run, synthetic declared urgency was anchored on real urgency by construction, so the gap the system exists to measure could not appear. In arm A, agents read their briefings carefully and declared close to what the briefing described, so the gap again did not appear, with H peaking at 0.017 where the hypothesis required 0.3. In arm B, showing the model's own output to the declarant closed the loop explicitly, so the declaration is no longer independent of the model at all. A system that measures the divergence between a human judgment and a computed state is only measuring anything when those two quantities have independent sources, and preserving that independence is harder than it appears.

Ordering and level are separate problems needing separate fixes. The scores of Section 4.5 do not support a single verdict on the forecast layer. On this campaign it neither orders well nor announces well, but the evidence for those two claims is of very different strength: discrimination rests on between two and six positive observations and is therefore inconclusive, while the calibration failure rests on 21 confident forecasts none of which materialised and is not plausibly a sampling artefact. The remedies differ accordingly. Miscalibration of this shape is routinely corrected by a post-hoc layer without touching the underlying model (Niculescu-Mizil et al., 2005; C. Guo et al., 2017). Poor discrimination is not, and would require the model to change.

The specific defect is repairable and general. The mechanism identified in Section 4.5.4 is not a scenario artefact. The u_{time} block computes milestone risk from a lead-time distribution against a deadline, while the event engine writes to capacity, cost and failure probability. No path connects a shock to the lead time, so the block cannot respond to a disruption. Any deployment of this framework would inherit that gap. The repair is to make the calibrated events write a lead-time impact, which is a change to the event mapping rather than to the mathematics.

Communicating ignorance is a design requirement, not a nicety. The arm B result reframes what an advisory layer owes its reader. Faced with an exogenous shock, a weather event, a fire, a geopolitical decision, the model has no basis for prediction at all. Displaying a low probability in a confident format is then worse than displaying nothing, because it converts an absence of information into an apparent reassurance. Two mitigations follow directly and neither requires new science: make the horizon salient, so that a low four-week probability cannot be read as a general statement of safety, and state explicitly which classes of risk the model does not cover.

Threats to validity. Four limit what the above can support. The declarants are language models, so the external validity of any claim about human declaration behaviour is unestablished; this is the principal threat, and it is why arm B is reported as a measured effect on this population rather than as a finding about operators. The campaign is a single scenario, so the outcome base rate, the event mix and the milestone structure are all properties of one script. The positive class is very small, which bounds every discrimination claim. Finally, arm B is not blinded, since the agents are told they are being shown a predictive analysis; a design in which some turns show a deliberately uninformative forecast would separate the effect of the content from the effect of the display.

Against the pre-registered thresholds. Five of the six hypotheses are not confirmed and one, H5 on urgency contagion, is confirmed. That is a largely negative result against the register, and it is reported as such. The value of the campaign lies less in those verdicts than in the diagnosis they made possible: the register asked whether the signal works, and the answer, with its mechanism, is that one identifiable block prevents it from working and that showing the resulting forecast to the declarant is actively counterproductive until that block is fixed.

4.9 Opening: Grounding Real Urgency in Observed Capacity

4.9.1 Motivation

The framework above has a structural dependency that the campaign brought into focus. Every KPI reaching the six blocks arrives either through a manual weekly review or through a scripted event, so a node's score is never fresher than its last submitted review, and the capacity block in particular depends on the same operator whose declaration the adequacy score exists to audit. The independence that Section 4.8 identifies as the precondition for the whole measurement is therefore not structurally guaranteed.

A second system, developed alongside the main work, addresses one part of that dependency. It estimates the physical capacity of a warehouse from public satellite imagery, which is an observation no operator supplies and none can bias. This subsection reports it at the depth that claim requires. It is an opening rather than a second contribution: the two systems have not been connected, and no claim is made that they have.

4.9.2 Method and Calibrated Constants

The pipeline takes a coordinate, retrieves the building footprint from the French national topographic database, retrieves 20 cm orthophotography and, where available, a LiDAR surface model, and derives a storable volume and a transfer capacity. Its four constants are calibrated rather than assumed, and each is reported with the sample behind it in Table 19.

Table 19: Calibrated constants of the capacity model

Constant	Value	Basis
Fill factor	0.80	Median of declared volume over footprint $\times$ height across 122 regulated warehouses, IQR [0.52, 1.22]
Dock pitch	3.8 m	Median 3.83 m, coefficient of variation 9.6% over 19 independently measured facades
Default height	11.6 m	Median of the corpus, IQR 9.8 to 13.7 m, used only when the topographic database gives no height
Ground sample distance	0.25 m	Measured optimum for the pre-trained detector; at 0.50 m the model collapses

The storable volume follows as $0.80 \times \text{footprint} \times \text{height}$ and the transfer capacity as the equipped facade length divided by 3.8 m. The coefficient of variation of 9.6% on the dock pitch is the result that makes the second of these generalisable: because the pitch is stable across sites, capacity depends only on how much facade is equipped, and no per-site measurement of the pitch is needed.

One constraint is enforced throughout and is worth stating because it is unusual. The regulatory database of classified installations is used only to calibrate the fill factor, never to infer a result for a specific site. Many warehouses fall below the classification thresholds and do not appear in it at all, so a system that inferred from it would silently fail on exactly the smaller sites a supply chain twin most needs to see.

One of these four constants does not survive audit, and the discrepancy is reported here rather than left in the code. The fill factor of 0.80 is the value implemented in production and attributed in the source to a 122-site calibration, but the detailed output of that run was not retained. The corpus that is still on disk contains 68 matched sites, and recomputing the

same ratio over it gives a median of 0.97 with an interquartile range of [0.80, 2.00], with only 47% of sites falling inside the [0.7, 1.4] band against the 70% the calibration script itself sets as its acceptance target (Figure 21). Two readings are possible and the available data does not separate them: either the retained corpus is a biased remnant of the original sample, or the production constant understates the true fill factor by roughly twenty per cent. Either way the constant is not currently reproducible from the artifacts, which is why recollecting the regulatory corpus cleanly and rederiving the factor is listed in Section 5.4 rather than presented as settled. The volumetric figures in this section should be read with that uncertainty attached; the dock capacity results, which depend on the pitch constant rather than on the fill factor, are unaffected.

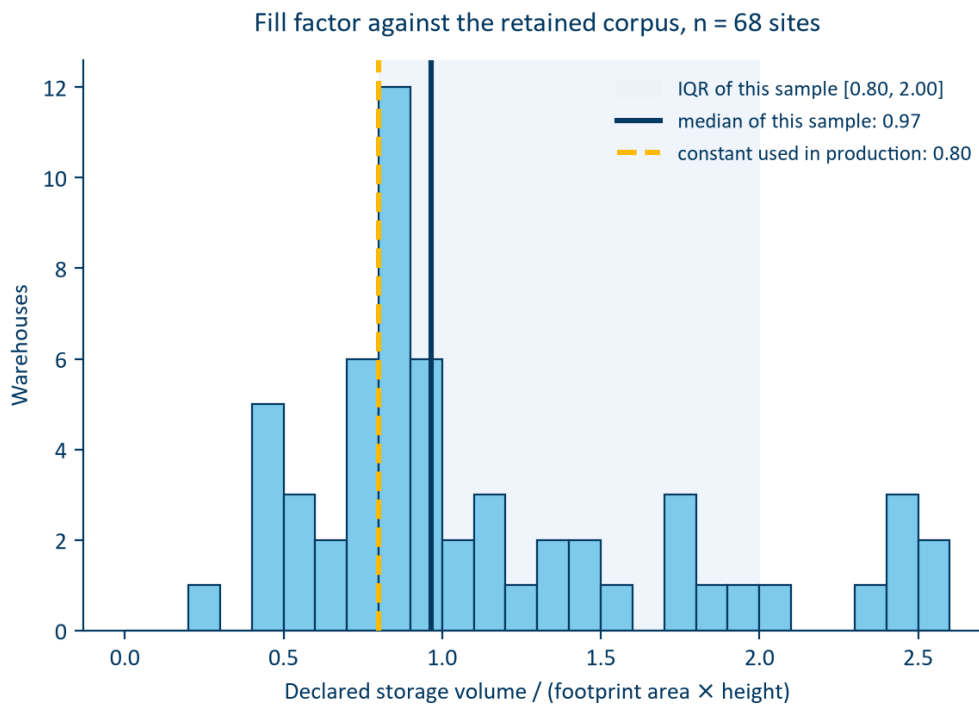


Figure 21: The fill factor against the auditable corpus

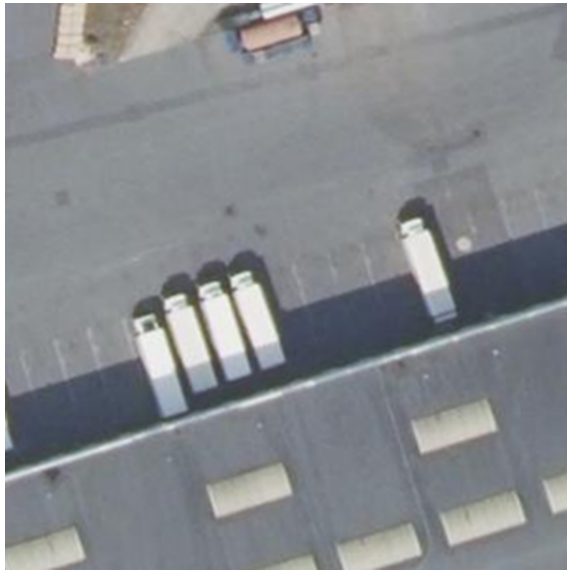
4.9.3 The Physical Limit and the Reframing

The original objective was to count loading docks. That objective is unreachable, and establishing why is the most transferable result of this work.

An empty dock door under a canopy produces no measurable signal from directly overhead. Measured contrast at verified empty bays is 0.18σ , 0.22σ and 0.01σ across a full sweep of assumed depths, while occupied bays on the same facades register at 5.3σ to 6.4σ . The measurement is therefore sensitive enough; the signal is absent. Repeating the test on 8 cm very-high-resolution imagery leaves the doors equally invisible, which shows the cause to be occlusion by the canopy roof rather than insufficient resolution. Any method claiming to count empty doors one by one from a nadir view is recovering an artefact.

Figure 22 shows the same equipped facade at both resolutions, 20 cm on the left and 8 cm on the right. The trailers, the apron and the dark canopy band along the building edge are legible at 20 cm and sharper still at 8 cm, yet no individual door resolves inside the canopy

band at either scale. A gain of $2.5\times$ in ground sample distance recovers no additional door. What improves with resolution is everything except the quantity of interest.



(a) 20 cm orthophotography



(b) 8 cm very-high-resolution imagery

Figure 22: The same equipped facade at 20 cm and at 8 cm

That finding splits the problem in two, and the distinction should not be blurred:

- counting each door individually is not observable at nadir, and no amount of resolution changes it;
- recognising that a facade is equipped for docking, and delimiting its extent, is observable, because the canopy, the apron and the bay rhythm remain visible whether or not a vehicle is present.

The second formulation is learnable, and it is sufficient, because capacity is a function of equipped length rather than of individually identified doors. Figure 23 is the direct evidence. It shows three facades, of 135 m, 105 m and 15 m, on which the automatic trailer detector found no vehicle at all. Individual doors are not resolvable on any of them, yet the regular bay markings and the canopy edge remain plainly legible, which is exactly the distinction the reframing turns on.

One reservation is recorded honestly. The proportion of sites showing no usable dock signature was measured on a subsample that required vehicles to be present to establish ground truth, and is therefore biased toward sites with vehicles. It should be re-measured on an independent draw.

4.9.4 Approaches Tested and Rejected

Five approaches were tried and abandoned, each for a measured reason. Table 20 records them, because the reasons constrain what a successful method can look like.

A related finding constrains the zero-shot approach that preceded the learned model. Parked trailers are not a proxy for docks: median trailer distance to the building reaches 126 m on large platforms, and at one site only 8 of 89 detected trailers lay within 26 m of the building, the

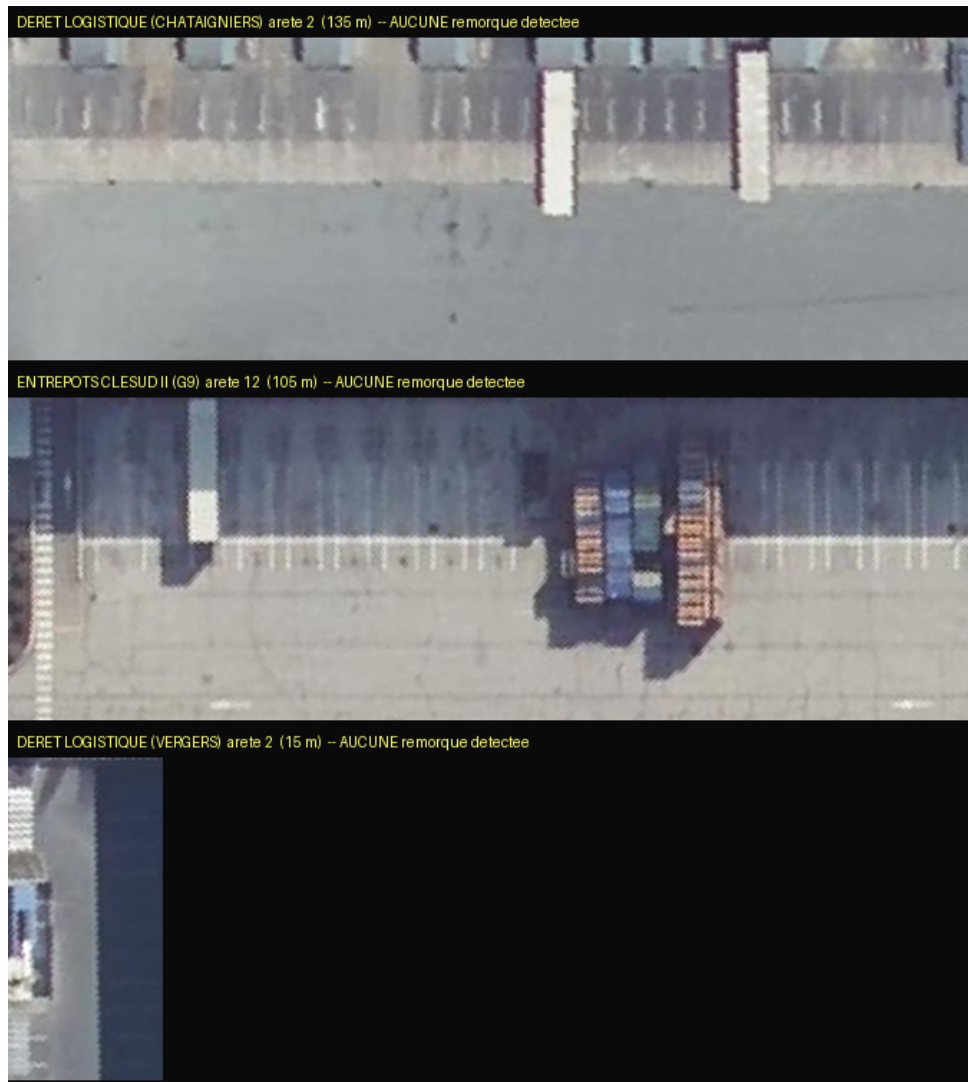


Figure 23: Equipped facades on which no trailer was detected

Table 20: Approaches tested and rejected

Approach	Measured outcome
Photometric thresholding of the apron	Recall of 5 of 6 on true aprons, but roads, an aeronautical tarmac and small-business forecourts all pass the same test
Vehicle density to reject car parks	Fails on a car park that was empty at the time of capture
Open-vocabulary detection from a text prompt	Confidence of 0.11 to 0.12, at the level of noise; the model is trained on ground-level imagery
Region growing from detected trailers	Returns 222 docks where 57 exist: the paved surface of an industrial estate is a single connected component, and the region spills onto neighbouring roofs of similar tone
Frequency analysis to count bays	Three real rhythms coexist on one facade, at 3.3 to 4.6 m for bays, 5.5 m for canopy posts and 8.9 to 10.7 m for portal frames. A frequency analysis returns a frequency, never its meaning

rest being in holding yards. With a proximity and perpendicularity filter applied, the zero-shot pipeline produced a capacity figure for only 36% of sites, 25 of 70 measured. Elsewhere it returned nothing, which is a gap in coverage rather than a wrong answer.

4.9.5 Learned Dock-Zone Segmentation

The reframed problem was addressed by training a segmentation model to delimit equipped facade zones directly. Table 21 gives the progression across four training runs, read at the best epoch of each by mask mAP50-95.

Table 21: Segmentation training progression

Run	Train / val	Image size	Precision	Recall	mAP50	mAP50-95
1	86	640	0.461	0.404	0.363	0.155
2	203 / 50	640	0.635	0.539	0.610	0.283
3	351 / 91	640	0.702	0.606	0.618	0.305
4	351 / 91	1280	0.652	0.547	0.570	0.272

Two observations follow, and Figure 24 shows both. Annotated data drove the gain: quadrupling the training set from 86 to 351 images roughly doubled mask mAP50-95. Resolution did not: run 4 is identical to run 3 except for a doubled input size, and it is worse on every metric. At a fixed data budget and model capacity, on a CPU-only training budget, the additional pixels cost more than they bought.

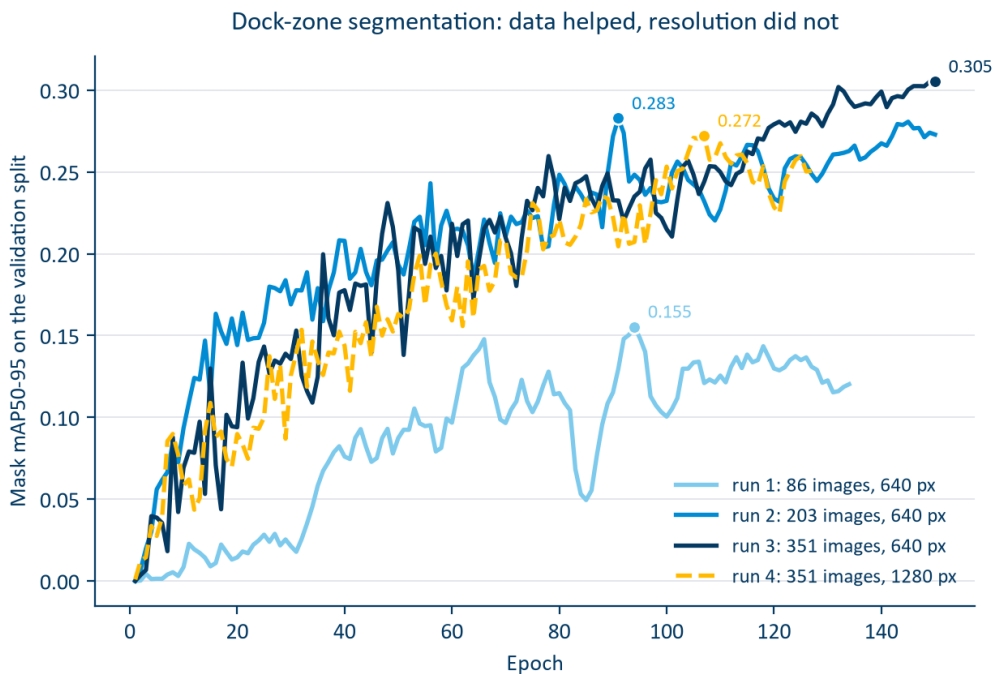


Figure 24: Validation mask mAP50-95 across the four runs

A second methodological point concerns what is measured. Standard detection metrics score shape overlap, but the operational quantity here is capacity, which depends only on the length of the equipped zone. A polygon drawn too deep, or with imprecise edges, still yields the

correct dock count. The project therefore scores itself on the error in the derived dock count rather than on mAP, and that business metric, not the detection metric, is the one used to decide whether a model is good enough to deploy. On the reference run that metric reported a global bias of -35% , a median error of 62% on images containing docks, and 89% of true negatives correctly left empty, which is a system that is directionally useful and quantitatively not yet trustworthy. Figure 25 shows predictions on held-out imagery. The model delimits equipped facades on buildings of varying shape and orientation, rotation invariance having been obtained by augmentation rather than by rectifying the geometry first.

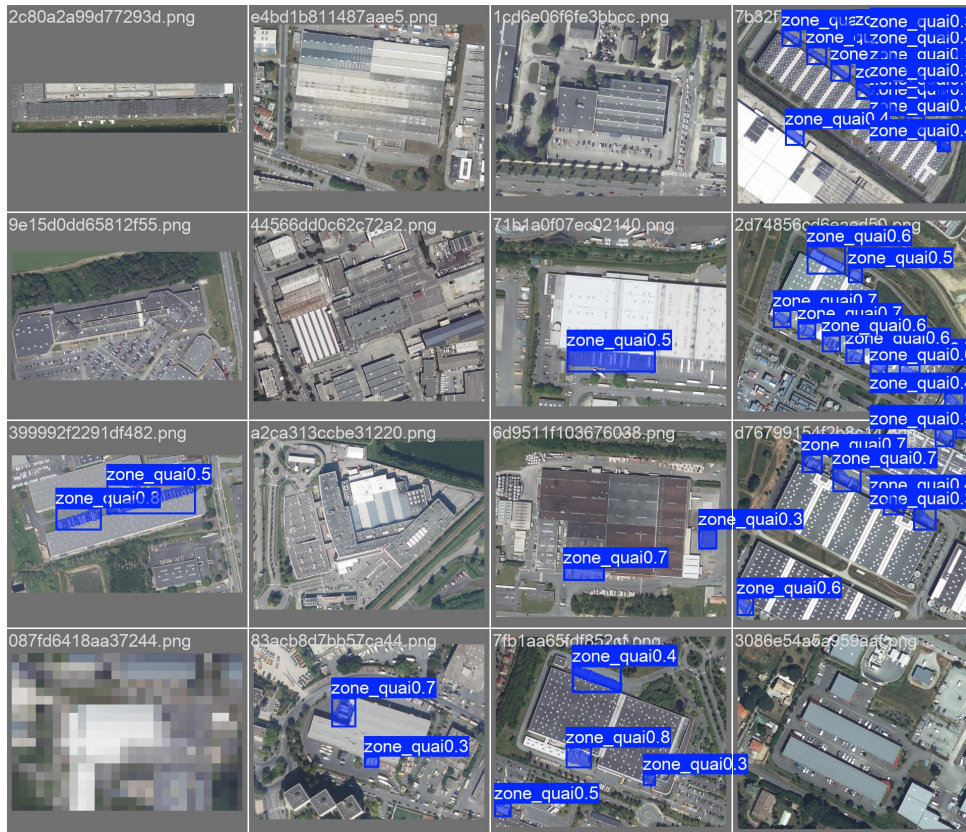


Figure 25: Predicted dock zones on held-out imagery

4.9.6 Annotation Integrity

One episode in this work bears directly on research method and is reported for that reason. An automated third-party annotation process produced 442 annotations and reported its own output as verified and complete. The annotations were unusable: the process had placed a dock zone along every edge of every building without examining the imagery at all, then subdivided each zone at the nominal pitch, producing an implausible 88% positive rate.

All of it was discarded and the dataset was re-annotated by hand. An audit of the manual work found 11 genuine problems across 687 annotated zones, a defect rate of 1.6% . A verification tool was then written and placed ahead of training, checking format, geometry and consistency against the project's own measured thresholds, so that no external annotation enters the pipeline unaudited. That tool caught a defect in its own first implementation, where distance was measured to the largest building in a scene rather than to all buildings, generating 26 false alarms on correct annotations of neighbouring buildings.

The transferable point is that a confident self-assessment from an automated process is not evidence, and that the cost of the verification step was small relative to the cost of training on corrupted labels.

4.9.7 What This Can and Cannot Feed

The pipeline runs end to end on a geographic area, retrieving footprints, presenting each to the trained model, assigning predicted zones to the correct building and writing capacity figures (Figure 26). One defect found during that deployment is worth recording, because it illustrates the difference between a model error and a systems error.

The national topographic database fragments a single large warehouse into several polygons, so a 434 m dock apron was initially attributed in full to a 5 149 m² fragment, yielding an implausible 221 docks per hectare against a corpus median of 7.9. The fix was geometric rather than statistical: each predicted zone is now clipped so that a building receives only the portion facing its own facade, and the quality control was changed from a density threshold, which must be calibrated per site type, to the ratio of dock length to building perimeter, which is dimensionless and cannot legitimately exceed one.

The honest boundary is as follows. The system supplies a storable volume, a pallet-position estimate and, where a dock zone is detected, a transfer capacity and an occupancy ratio. Those map onto the capacity block of Equation 4 and onto nothing else: it says nothing about lead time, cost or emissions. Coverage is metropolitan France, bounded by the public data services it draws on. The trained segmentation model is not yet wired into the production capacity path, which still uses the zero-shot trailer method and its 36% coverage. And the imagery is a snapshot, so the resulting figure is a structural capacity rather than a live occupancy.

What it does change is the provenance of one input. A capacity figure derived from imagery is not supplied by the operator whose declaration the adequacy score audits, which is precisely the independence Section 4.8 identifies as the precondition for the measurement to mean anything. Figure 27 states where that input would enter and, equally important, which blocks it would leave untouched.

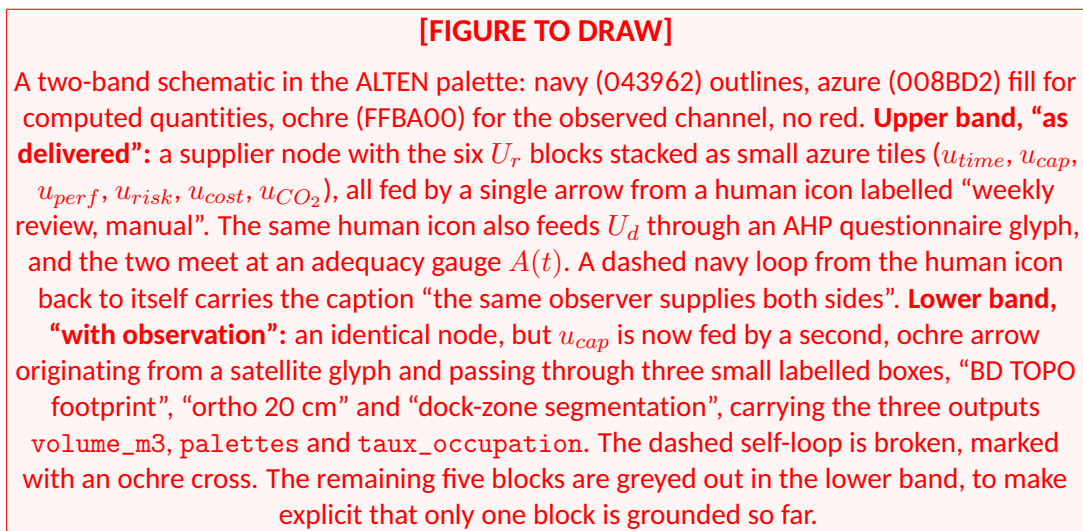


Figure 27: Where satellite-observed capacity would enter the real-urgency model, and which blocks it does not reach.



Figure 26: End to end output on one deployment site

4.10 Synthesis Against the Research Questions

RQ1, multi-block real urgency. A six-block noisy-OR aggregation with bidirectional DAG propagation was specified, implemented and exercised on a nineteen-turn campaign across eight nodes. The construction works as designed and is exactly decomposable, which is what allowed its own defect to be found. The answer is qualified rather than clean: five of the six blocks behaved as intended, and the sixth, the time block, was shown to be structurally unable to respond to a disruption, because the event engine writes to no quantity it reads.

RQ2, cognitive adequacy formulation. The asymmetric Prospect-Theory formulation produces a bounded, interpretable score whose every term can be reconstructed, as demonstrated on the Orbitalys node. Predictive validity was not demonstrated: at the pre-registered threshold the signal returns precision and recall of zero, and the underlying forecast layer shows no discrimination above chance on this campaign while being substantially worse calibrated than a base-rate forecaster. Section 4.5.4 attributes that failure to one identified and repairable cause rather than to the formulation itself.

RQ3, network prioritisation and criticality. AHP and Fuzzy BWM elicitation, PROMETHEE II outranking and systematic shock simulation were combined into a working ranking that placed the expected bottleneck at the top. Two documented behaviours bound it: the criticality index degenerates when many nodes saturate at once, which is common in precisely the network-wide crisis where it would be most useful, and the outranking inherited a perfect correlation between two of its criteria on this run, which the tool flagged on its own.

RQ4, operational validation and the effect of showing the forecast. The framework was operationalised as a persistent digital twin and exercised through a pre-registered weekly-cadence campaign with a control arm and a treatment arm on a frozen scenario. Five of six hypotheses are not confirmed and H5 on urgency contagion is confirmed. On the second half of the question the answer is unambiguous within its population: at first exposure, all eight declarants were either reassured or confirmed by the displayed forecast, none revised upward, and the scenario's critical event arrived two turns later.

5 Conclusion

5.1 Contributions

This research set out to determine whether the gap between declared and computed urgency can be measured reliably enough, inside a working digital twin, to be used as a decision aid rather than as a descriptive statistic. On the evidence gathered the answer is not yet, and the value of the work lies in establishing why with enough precision to act on.

Four contributions are claimed.

An implemented adequacy architecture. A multi-block noisy-OR real urgency model, propagated bidirectionally across a supply chain DAG and confronted with a consistency-gated, structurally elicited declared urgency through an asymmetric Prospect-Theory penalty, is specified, implemented and operable. It fills the combination that Section 2.14 identifies as absent from the reviewed literature. Its decomposition is exact rather than approximate, which is what made the diagnosis below possible from the application rather than by hand.

A diagnosed failure with an identified and repairable cause. The forecast layer is shown to be badly miscalibrated on the campaign, with a Brier skill between -3.4 and -6.9 against a base-rate forecaster and twenty-one confident predictions of which none materialised. That failure is traced to a single block: the time block computes milestone risk from a lead-time distribution while the event engine writes only to capacity, cost and failure probability, so no path connects a shock to the quantity the milestone probability reads. Mid-crisis the block reads as numerically zero at six of eight nodes. This is a property of the model, not of the scenario, and the repair is a change to the event mapping rather than to the mathematics.

A measured first-exposure effect of showing a forecast to the declarant. On the turn the predictive analysis first became visible, all eight declarants either revised their declared urgency downward or had it confirmed, and none revised upward, two turns before the scenario's one critical event. Because the two arms are shown to carry numerically identical forecasts, the effect is attributable to the display rather than to a different prediction. The finding is reported for a population of language-model agents, and its extension to human operators is unestablished.

A demonstration that the outcome definition dominates the reported calibration. Scoring identical forecasts against a plausible but naive outcome definition instead of the production one moves the base rate from 5.0% to 38.7% and improves the apparent Brier skill by a factor of more than five, while leaving discrimination essentially unchanged. A calibration figure reported without its outcome definition carries little information.

A fifth result, reported as an opening rather than a contribution, establishes that individual loading docks are not observable from directly overhead because of canopy occlusion rather than insufficient resolution, and that the reframed problem of delimiting an equipped facade is both learnable and sufficient for capacity estimation.

5.2 Status of the Research Hypotheses

Table 22 gives the status of each pre-registered hypothesis across the three runs. It is the single authoritative statement of those verdicts in this thesis.

Table 22: Status of the six hypotheses across the executed runs

Hypothesis	Dry run	Arm A	Reading
H1 latency	Not confirmed	Not confirmed	1 of 8 nodes on the dry run, 3 of 8 in arm A against a threshold of 5. The direction is right and the level is short.
H2 early detection	Not confirmed	Not confirmed	H peaks at 0.070 then 0.017 against a threshold of 0.3. The declared signal tracks the computed one too closely for a gap to open.
H3 criticality	Not confirmed	Not confirmed	Top-1 on 7 of 10 informative turns, 70% against a threshold of 80%. Identical across runs, since the test does not depend on U_d .
H4 predictive validity	Not confirmed	Not confirmed	Precision and recall both 0.00 at the threshold. Section 4.5.4 attributes this to the time block rather than to the adequacy formulation.
H5 contagion	Not confirmed	Confirmed	$\Delta U_d = +0.079$ on press turns against $+0.005$ on calm turns. The mere-urgency effect reproduced at network scale.
H6 hysteresis	Not applicable	Not applicable	Real urgency shows no measurable decline over the recovery turns, so the comparison is not meaningful on this scenario.

One hypothesis of six is confirmed. Two qualifications belong with that count. H4 fails for an identified mechanical reason rather than because the adequacy concept is unsound, and its verdict should be revisited once the time block is repaired. H6 was never testable on this scenario, because the script does not de-escalate far enough for a decay rate to be compared, which is a design fault in the scenario rather than a result.

5.3 Limitations

Table 23 lists the limitations judged material enough to state directly, with their operational implication.

5.4 Future Work

The work divides into what must happen before the framework can be evaluated fairly, and what becomes worth doing afterwards.

Repair the time block first. Nothing else should be measured until the calibrated events write a lead-time impact alongside their existing capacity, cost and failure-probability effects. Until then the forecast layer cannot respond to a disruption, H4 cannot be tested fairly, and

Table 23: Documented limitations and their implications

Limitation	Implication
The time block is blind to shocks	Measured, not suspected: the calibrated events write to capacity, cost and failure probability but never to lead time, so milestone risk cannot respond to a disruption and reads as numerically zero mid-crisis at six of eight nodes. This is the cause of the H4 failure and the first thing to repair.
Validation rests on agent declarants	The campaign was executed with eight isolated language-model agents, not human consultants. Every statement about declaration behaviour is a statement about that population. The human campaign, for which the protocol is frozen, has not been run.
A single scenario, with a very small positive class	All results come from one nineteen-turn scenario in which between two and six node-weeks are positive depending on horizon. Discrimination claims are correspondingly weak in both directions.
No probabilistic guarantee	The framework produces probabilities, scores and confidence intervals, not certainties, and the campaign showed those probabilities to be poorly calibrated at the top of their range.
Score freshness follows input cadence	Without an ERP connector, a node's score is never more current than its last submitted weekly review. Manual entry is an accepted trade-off, not an oversight, and Section 4.9 reports one route out of it for a single block.
Blocks are assumed independent	The noisy-OR aggregation requires it, and the campaign produced a counter-example: two blocks were perfectly correlated on the arm A run, which the tool flagged on its own.
AHP is limited to four criteria	Saaty's method degrades past roughly seven pairwise compared criteria, which is why declared urgency stays on four while the larger set of KPI blocks is weighted through Fuzzy BWM instead.
Event calibration priors are defensible, not measured	The Beta-Bernoulli prior weight and the EMA reactivity coefficients follow standard conjugate Bayesian revision but are not empirically fitted.
Event reversibility is conditional	Reverting a declared event is guaranteed correct only if no other write has touched the same KPI in the meantime; otherwise the system raises an explicit conflict rather than silently rebasing later writes.
Double counting is possible	An operator can both declare a calibrated event and manually correct the same KPI in the same week. The interface warns but does not forbid it, since the human operator remains the final authority.
The satellite capacity model is not connected	It reaches one block of six, covers metropolitan France only, produces a structural capacity rather than a live occupancy, and its trained segmentation model is not yet wired into the production capacity path.
The fill factor is not reproducible from the retained data	The production constant of 0.80 comes from a 122-site calibration whose detailed output was not kept. Recomputing over the 68 sites still on disk gives a median of 0.97. Volumetric outputs therefore carry an unquantified bias of order twenty per cent until the corpus is recollected. Dock capacity, which depends on the pitch constant instead, is unaffected.

any campaign run against it will reproduce the same result. This is the single highest-value change identified by this thesis and it is small.

Add a calibration layer, trained off-scenario. The reliability profile of Section 4.5 is the shape that post-hoc calibration addresses (Niculescu-Mizil et al., 2005; C. Guo et al., 2017), and the correction is a thin layer rather than a model change. One methodological constraint should be respected: the calibration must be fitted on data other than the evaluation scenario, otherwise it is tuned to the test and the system's generality is lost.

Make the forecast state its own ignorance. Given the arm B result, an advisory layer that cannot anticipate exogenous shocks should not present a low probability in a confident numeric format. Two changes follow directly: make the horizon salient in the display, so a low four-week figure cannot be read as general safety, and state explicitly which risk classes the model does not cover.

Re-run the comparison with the display controlled. Arm B is not blinded. A design in which some turns display a deliberately uninformative forecast would separate the effect of the content from the effect of the display, and would establish whether the first-exposure effect is a response to the numbers or to being given a number at all.

Run the human campaign, then arm C. The frozen protocol, the hashed data pack and the operating scripts are ready. Only the human campaign can decide the hypotheses as they were written, and it should be run only after the time block is repaired, since running it against a known-defective forecast would waste the participants. Arm C, the closed loop in which declarants act on the terrain, follows: it cannot be scored by prediction accuracy, because an averted rupture makes a correct prediction look wrong, so it must be judged by comparing its final state against arm A.

Put the outcome definition in the pre-registration. Section 4.6 shows the outcome definition moving the reported skill by a factor of five. It belongs alongside the numeric thresholds in the H1 to H6 register, where it currently is not.

Connect the observed capacity channel. Three steps stand between the satellite work and a usable input: wire the trained segmentation model into the production capacity path in place of the trailer-span method, re-measure the no-signature rate on an unbiased draw, and recollect the regulatory corpus cleanly to re-derive the fill factor. Beyond those, the surface model from LiDAR is the most promising unexplored route, because the canopy step is several metres lower than the main roof and would give both the canopy extent and the true roof edge, correcting the 0 to 14.4 m overhang that currently displaces the apron boundary. One governance question must also be settled rather than deferred: the detection library is distributed under a copyleft licence, which is immaterial for internal use but becomes determinative the moment the capability is distributed or exposed over a network.

Broaden the elicitation and couple the platforms. Two directions identified but deliberately descoped remain open: replacing single-operator AHP elicitation with a multi-expert protocol aggregating several judgments per node, and coupling SupplyScore with the DISCO platform

it currently runs alongside rather than inside. Neither is a design commitment, and both depend on what a repaired system shows.

5.5 Personal Conclusion

5.5.1 Technical Skills Developed

The project drew on four distinct areas, and the difficulty of the subject lay mostly in making them work together rather than in any one of them.

- **Multi-criteria decision analysis:** pairwise elicitation through AHP and Fuzzy BWM, consistency validation, and outranking aggregation through PROMETHEE II.
- **Probabilistic modelling:** noisy-OR aggregation, Monte Carlo simulation, conjugate Bayesian revision in a sparse-data regime, and, added during the campaign analysis, probabilistic forecast verification through proper scoring rules and reliability decomposition.
- **Graph algorithms:** propagation over a multi-rank directed acyclic graph, topological ordering, shock simulation and network criticality.
- **Software engineering:** architecture of a multi-tenant Python system, hardened SQLite persistence with write-ahead logging, backup and migrations, a Dash web interface, and a milestone-based quality process with blocking test coverage and automated end-to-end tests.

The satellite work added a fifth: geospatial data engineering against public web services, dataset construction and annotation, and the training and evaluation of a segmentation model under a strict CPU budget.

5.5.2 What I Learned

The hardest problem was not mathematical. It was recognising, three separate times, that the system was measuring itself. The dry run anchored declared urgency on real urgency by construction; the agents in arm A read their briefings so carefully that their declaration reproduced the computed state; and arm B closed the loop deliberately by showing the model's own output to the declarant. Each time the symptom looked like a modelling problem and each time the cause was circularity in the measurement. I would now design the independence of the two signals first, before designing either of them.

The second lesson concerns what counts as a result. For a long stretch of the project I treated a negative outcome as a failure to be fixed before reporting. The most useful findings in this thesis are negative: the forecast layer is badly calibrated, the confident tail is entirely false, one block is structurally blind, and showing that forecast to a declarant is counterproductive. None of those would have been found by a project optimising for a positive result, and each of them is actionable in a way that a marginal improvement would not have been. The related discipline, which the annotation episode of Section 4.9 taught bluntly, is that a confident self-report is not evidence, whether it comes from an automated process or from oneself, and that the cheap verification step should always come before the expensive one.

With hindsight I would change two things. I would have scored the forecast layer against recorded outcomes far earlier, since doing so took a few hundred lines and revealed the time block defect immediately, where months of implementation had not. And I would have

written the outcome definition into the pre-registration alongside the thresholds, having since measured how much it moves the answer.

The skill that progressed most was not a technical one. It was the willingness to state a result that does not flatter the work, in enough detail that someone else can act on it. The subject of this thesis is, after all, the gap between what people declare and what is actually true, and it would be a poor outcome to have measured that gap in a supply chain while reproducing it in the reporting.

Professionally, the project confirmed a direction. I want to work where a model meets the person who has to act on it, because that boundary is where the interesting failures live: not in the mathematics, which can be checked, but in what a number does to the judgment of the person reading it.

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Acronyms

AHP Analytic Hierarchy Process. 14

BWM Best-Worst Method. 14

CI Consistency Index. 30

CR Consistency Ratio. 14, 30, 31

DIN Direction Innovation Numérique. 1

DT Digital Twin. 13

F-BWM Fuzzy Best-Worst Method. 14

GLEC Global Logistics Emissions Council. 1, 17

HGV Heavy Goods Vehicle. 2

IoT Internet of Things. 1

IPG Itinerary Performance Index. 2

LPPL Log-Periodic Power Law. 10

MCDM Multi-Criteria Decision-Making. 8, 12

MTS Manchester Triage System. 16

PI Physical Internet. 8

RI Random Index (Saaty). 30

SC Supply Chain. 13

TFN Triangular Fuzzy Number. 14

TMT Temporal Motivational Theory. 11

TTR Time To Recovery. 2

TTS Time To Service. 2

VECTO Vehicle Energy Consumption calculation TOol. 17

A Ethics Documentation

The study as executed involves no human participants. The eight declarants of the HÉLIOS campaign are language-model agents, and no personal data was collected, processed or stored at any stage. On that basis the project was submitted to the Cranfield University Research Ethics System as a low-risk study, and the approval documentation is filed with this thesis.

Two features of the protocol would require participant consent if the campaign were rerun with the eight human consultants for whom it was originally designed, and both are recorded here so that the requirement is not lost.

Incomplete disclosure through the cover story. Participants are told they are playing a fictional satellite programme. They are not told that the scenario replays the documented 2020 to 2022 semiconductor shortage, because a participant who recognised the historical chronology could anticipate events rather than react to them, which would destroy the latency measurement that hypothesis H1 depends on. The debrief reveals the real scenario and asks each participant at which turn they recognised it, which converts the principal threat to the measurement into a recorded quantity.

Deliberate misinformation through the placebo sheets. Two participants receive a false de-escalation notice, one at turn 9 and one at turn 15. These sheets control for suggestibility: a declared urgency that follows the placebo is reflecting the sheet rather than a reading of the situation. Participants are not told in advance which sheets are genuine.

Both are standard incomplete-disclosure designs in behavioural operations research, and both are resolved by the full debrief. Were the human campaign to run, the following would need to be in place before the first turn: written informed consent covering the use of incomplete disclosure; the unconditional right to withdraw at any point, with the participant's data removed on request; the debrief protocol, delivered to every participant regardless of whether they completed the campaign; and confirmation that no performance data is passed to line management, since the participants would be colleagues and the campaign records their judgment quality week by week.

Data ethics for the satellite observation work. All imagery and building data come from public French open-data services under their published licences. No imagery of private individuals is collected or retained, and the analysis operates on industrial buildings rather than on people. Vehicle detections are used only as an intermediate geometric cue for locating dock aprons; no vehicle is identified, tracked or associated with an operator, and the detections are not retained beyond the derivation of a facade extent. The regulatory database of classified installations is used solely to calibrate one constant across an aggregate of 122 sites, and never to infer or publish a result about a named site.

B Source Review Grid

Following the structured review method required by the programme's writing guidelines, the grid below records for each stream reviewed in Section 2 its principal sources, the method-

ological approach those sources take, what this thesis takes from them, and a critical appraisal of the limitation that prevents the stream from answering the research question on its own. The final column is the one that matters for the argument: it is the accumulation of those limitations that defines the gap stated in Section 2.14.

Table 24: Source review grid

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Deadline-sensitive value and risk propagation <i>Foundation</i>	(K. Chen et al., 1996; J. Chen et al., 2025): analytical utility-accrual scheduling. (Ivanov; Dolgui, 2020; Ivanov; Dolgui, 2019; Li et al., 2020): conceptual and simulation models of the ripple effect	Urgency treated as time-varying and network-propagated rather than node-local; the noisy-OR choice over weighted averaging	Time-value work stops at a single isolated task. Ripple-effect work models propagation but not the node-level aggregation of heterogeneous blocks that must precede it. Neither addresses a declared signal at all
Cognitive bias in urgency <i>Foundation</i>	(M. Zhu et al., 2018): five controlled experiments. (Steel et al., 2006): formal discounting model. (Rusou et al., 2020; DiwasSingh et al., 2020): field studies	The justification for treating declared urgency as a biased signal; the asymmetric penalty structure; hypothesis H5	Diagnoses the bias but prescribes no operational correction. Effect sizes come from laboratory and clinical settings; the magnitude in industrial supply chains is uncalibrated, which this thesis inherits as an open parameter rather than resolves

Table 24, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Closest antecedents <i>Antecedent</i>	(Sawik, 2013; Sawik, 2014; Sawik, 2016): mixed-integer programming. (Cohen et al., 2019; Afrasiabi et al., 2022): MCDM supplier prioritisation	Confirmation that multi-criteria, disruption-aware supplier evaluation is established, and precise identification of what it omits	Risk is evaluated once, at the point of decision, not maintained per node and propagated. Where a client-declared signal exists it is consumed directly, never validated against an operational counterpart. The optimisation machinery targets allocation, not measurement
Bayesian risk updating <i>Enabling</i>	(Pearl, 1988; Koller et al., 2009): theory. (Hosseini; Ivanov, 2020): survey. (Huang et al., 2020; Deng et al., 2020; Lockamy et al., 2011; Qazi et al., 2017): applied models	Beta-Bernoulli conjugate revision of the failure probability in the risk block, at $n_0 = 26$	Suited to the sparse-evidence regime, but says nothing about whether the resulting probabilities are well calibrated once a human reads them. This thesis measured that gap directly and found it large
Digital twins and decision support <i>Enabling, primary gap source</i>	(Grieves, 2014; Tao et al., 2018; Fuller et al., 2020): concept and survey. (Ivanov; Dolgui, et al., 2021; Ivanov; Dolgui, 2020; Niloofar et al., 2023): supply chain twins and cognitive twins	The architectural family the delivered system belongs to; the human-in-the-loop framing	No retrieved system combines a declared input, a computed operational urgency and their propagated mismatch. A second gap matters here: these systems are evaluated on state estimation accuracy and assume their inputs simply arrive, so input circularity is not treated as a measurement problem

Table 24, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Multi-criteria decision methods <i>Enabling</i>	(Saaty, 1980): eigenvector method. (Rezaei, 2015; S. Guo et al., 2017): best-worst and fuzzy extensions. (Brans; Vincke, et al., 1986; Brans; Mareschal, 2005): outranking. (Greco et al., 2018; Becker et al., 2017; Borgonovo et al., 2021): composite-index critique	The elicitation of U_d with its consistency gate; the Fuzzy BWM weighting of the six blocks; the PROMETHEE II network ranking	Static tools, rarely compared against a computed counterpart. The composite-index critique is usually offered as a caveat rather than instrumented; this thesis instruments it through the explainability decomposition, and the campaign then produced exactly the predicted failure, two perfectly correlated criteria
Forecast-assisted decision and automation bias <i>Foundation for the treatment arm</i>	(Parasuraman et al., 1997): taxonomy. (Skitka et al., 1999; Cummings, 2004): controlled studies. (Goddard et al., 2012): systematic review. (Dietvorst et al., 2015; Logg et al., 2019; Bansal et al., 2021): direction and explanation effects	The interpretive frame for the arm B result; the design requirement that an advisory layer communicate its own ignorance	Dominated by single-decision laboratory and clinical tasks. Does not cover a repeated multi-turn loop in which the declaration feeds back into the model, nor the case where the displayed forecast is known to be miscalibrated in a specific direction, which is the situation measured here

Table 24, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Probabilistic forecast verification <i>Enabling</i>	(Brier, 1950): score. (Murphy, 1973): decomposition. (Gneiting et al., 2007): proper scoring rules. (Niculescu-Mizil et al., 2005; C. Guo et al., 2017): post-hoc calibration	The scoring protocol of Section 3.4.4; the separation of discrimination from calibration that carries the central result	Assumes the predicted outcome is unambiguously defined. In an operational supply chain it is a modelling choice, and Section 4.6 measures that choice moving the reported skill by a factor of five
Serious games <i>Validation rationale</i>	(Stermann, 1989): the Beer Game. (Prensky, 2001; Deterding et al., 2011): design theory. (Mohaddesi et al., 2020; Bellotti et al., 2013; Lafkihi et al., 2025): methodology and applications	Justification of a weekly-cadence game as an empirical instrument for observing declaration bias	Usually scored on learning outcomes or aggregate performance, rarely used as a controlled comparison replaying one frozen scenario with and without a treatment on the declarant, which is the design adopted here
Medical triage <i>Analogy only</i>	(Mackway-Jones et al., 2013): protocol. (Kuligowski et al., 2025): comparative study, weighted $\kappa = 0.51$. (Kim et al., 2020): admission control	Vocabulary of over- and under-triage; the principle of asymmetric risk behind $\lambda_{\text{under}} > \lambda_{\text{over}}$	Analogy only, and flagged as such wherever used. Clinical triage has an outcome observable within hours and a trained, certified assessor; neither holds in a supply chain, so no numeric parameter is transferred from this stream

Table 24, continued

Stream and role	Principal sources and method	Taken into this thesis	Critical appraisal
Physical Internet and carbon accounting <i>Context</i>	(Montreuil, 2011; W. Guo et al., 2020; Li et al., 2020): network context. (Smart Freight Centre, 2023; European Commission, 2017; ADEME, 2014; International Organization for Standardization, 2023): emissions frameworks	The motivation for treating urgency as a network-level quality of service variable; the carbon block as a separate arbitration criterion	Provides context and regulatory grounding, not method. The carbon block was the least exercised of the six on the campaign, so its inclusion remains justified by design intent rather than by measured contribution
Earth observation of built assets <i>Foundation for the opening</i>	(Xia et al., 2018): oriented detection benchmark. (Maggiori et al., 2017; Van Etten et al., 2018): footprint benchmarks. (Kirillov et al., 2023): promptable segmentation. (Henderson et al., 2012; Jean et al., 2016): activity proxies from imagery	The pre-trained detector and its vehicle classes; the sampling of candidate sites from imagery alone; the precedent that imagery can substitute for slow official statistics	The economics work measures aggregate activity at regional scale and produces nothing per-facility. The detection work stops at footprints and vehicles without converting them into an operational capacity. The conversion, and its calibration, is what Section 4.9 had to construct

C HÉLIOS Campaign Protocol Extract

This appendix records the elements of the frozen protocol that a reader would need in order to reproduce or audit the campaign, beyond the design already given in Section 3.4.

Cadence and Sequence

Nineteen turns are indexed T0 to T18. T0 is a training turn used to verify that every declarant can complete the questionnaire inside ten minutes with the consistency gate active; its declarations are retained in the record but excluded from every analysis. The eighteen campaign turns run at two per day, morning and late afternoon, over nine working days. One game turn represents one real month, from September 2020 to February 2022, with real durations converted into engine hours at a fixed rate so that documented figures such as twelve to twenty-five week supplier lead times or a six-day canal blockage retain their true magnitude inside an eighteen-turn game.

Frozen Artifacts

Four artifacts were frozen before T0 and are the reference against which any later analysis is checked: the scenario definition with its twelve calibrated events and their turn assignments; the data pack of public time series, with a SHA-256 hash recorded per file; the role cards, one per node, each stating only what that declarant may know; and the hypothesis register H1 to H6 with its numeric thresholds. Any analysis not derivable from the register is labelled exploratory in the reporting.

Controls

- *Isolation.* Each declarant sees only its own role sheet, its own turn briefing and its own questionnaire. Dashboards, scores, the network view and other declarants' material stay hidden until the debrief. In the executed campaign this is enforced by the harness, which gives each agent a private working directory, rather than by instruction.
- *Placebo sheets.* A false de-escalation notice is handed to the avionics integrator at T9 and to the wafer supplier at T15. A declared urgency that follows the placebo is reflecting the sheet rather than a reading of the situation.
- *Weak-signal turn.* At T4 the press briefing mentions a tension with no engine event behind it, which measures who reacts to media noise alone. This is also the turn on which the arm B treatment first becomes visible, so the two effects are deliberately not separated on that turn and the arm A control is what isolates them.
- *Role assignment.* The two most information-heavy nodes, the foundry and the distributor, are assigned away from the most supply-chain-experienced participants, so that a role effect cannot be confounded with an expertise effect.
- *Journal.* Every deviation from protocol is logged the same day.

Contingencies Arbitrated in Advance

Three contingencies were decided before the campaign rather than improvised under time pressure. The campaign truncates at T16 if it overruns by more than two turns, rather than rushing the remaining responses. It drops to one turn per day if the questionnaire exceeds ten minutes at the training turn. Any absence, or any unresolved consistency failure, is carried forward unchanged and excluded from the reactivity analyses.

Declaration Record Format

Each declaration is one JSON record per node per turn, carrying the turn index, the six bipolar pairwise comparisons, the four severity scores, the derived criteria weights, the principal eigenvalue with its consistency index and consistency ratio, a flag for whether the consistency gate passed, the number of attempts, the raw and smoothed declared urgency, and a free-text note. Arm B adds one field, `influence_prediction`, taking one of four values: no influence, confirms, revised upward, or revised downward. The format is otherwise identical between arms, which is what makes them comparable.

Deviations Recorded in the Executed Campaign

Two are on record. One node submitted turn 1 twice; the first submission is retained and the duplicate discarded, which is why the arm B analysis reports 96 declarations from 97 raw records. Arm B ran to T11 rather than T18, so its forecast log covers turns 4 to 12 against turns 4 to 18 in arm A; all arm B comparisons in this thesis are therefore made over the turns actually recorded, and the arm A scoring uses its own full range.

D Full Scoring Tables

Every value in this appendix is written by the derivation script of Section 3.6 and is reproduced here in full, so that the summary tables of Section 4 can be checked against the complete output.

Turn to ISO Week Mapping

The campaign runs from ISO week 2026-S31 at T0 to 2026-S49 at T18, one turn per week. The outcome window of a forecast issued at turn t with horizon h therefore spans weeks $t + 1$ to $t + h$ inclusive.

Arm A, Production Outcome Definition

Table 25 gives every quantity behind the summary of Section 4.5, at all four horizons, including the confusion matrix at the pre-registered threshold and the climatological Brier score the skill is measured against.

Arm A, Naive Outcome Definition

The naive definition counts any non-reverted event recorded on the node, regardless of gravity, and ignores milestones entirely. It is scorable on the 75 node-turns for which the node has at least one event in its history. Table 26 repeats the scoring under that definition, and the contrast with Table 25 is the measurement reported in Section 4.6.

Right-Censoring Sensitivity

Dropping every forecast whose observation window would extend past the final campaign turn leaves the four-week scoring on 88 observations with 2 positives. The result does not

Table 25: Complete arm A scoring, production outcome definition

Quantity	h = 1	h = 2	h = 3	h = 4
Observations	120	120	120	120
Positive outcomes	2	4	5	6
Base rate	0.017	0.033	0.042	0.050
AUC	0.555	0.372	0.428	0.485
AUC 95% CI, lower	0.470	0.258	0.315	0.389
AUC 95% CI, upper	0.639	0.492	0.546	0.583
PR-AUC	0.027	0.031	0.041	0.054
Brier score	0.130	0.166	0.191	0.207
Brier, climatology	0.016	0.032	0.040	0.047
Brier skill	−6.93	−4.14	−3.77	−3.36
True positives at 0.5	0	0	0	0
False positives at 0.5	15	18	20	21
False negatives at 0.5	2	4	5	6
True negatives at 0.5	103	98	95	93

Table 26: Complete arm A scoring, naive outcome definition

Quantity	h = 1	h = 2	h = 3	h = 4
Observations	75	75	75	75
Positive outcomes	11	19	25	29
Base rate	0.147	0.253	0.333	0.387
AUC	0.603	0.601	0.545	0.484
Brier score	0.155	0.248	0.337	0.389
Brier skill	−0.24	−0.31	−0.52	−0.64

improve: the operating point remains 0 true positives against 16 false positives, and the Brier skill moves from −3.36 to −7.82. Censoring is not the explanation for the confident tail.

The Confident Tail, Forecast by Forecast

Table 27 lists every arm A forecast announcing a four-week probability of at least 0.5. The rightmost column is the milestone-miss probability that the model reports alongside it. The two never differ by more than 0.020, which is the evidence that the confident tail is the milestone channel and nothing else. None of the twenty-one was followed by the predicted outcome.

Set against the registry, ten of the eighteen campaign milestones completed, eight remained active, and none was abandoned. Exactly one was missed at its deadline. Every other node for which a near-certain miss was announced delivered.

Arm B, Per-Node Influence Record

Table 28 records the first-exposure turn node by node, which is the observation behind the result of Section 4.7.

Table 27: Every arm A forecast at or above 0.5 at four weeks

Turn	Node	p (4 weeks)	p (milestone miss)	Outcome
T5	AvioSys Intégration	0.898	0.880	none
T6	CompoDis Europe	0.996	0.996	none
T7	CompoDis Europe	1.000	1.000	none
T8	CompoDis Europe	1.000	1.000	none
T8	NovaFab Semiconductors	0.866	0.846	none
T8	Orbitalys	1.000	1.000	none
T9	NovaFab Semiconductors	1.000	1.000	none
T9	Orbitalys	1.000	1.000	none
T10	AvioSys Intégration	0.900	0.884	none
T10	NovaFab Semiconductors	1.000	1.000	none
T10	Orbitalys	1.000	1.000	none
T11	AvioSys Intégration	0.898	0.884	none
T11	Orbitalys	1.000	1.000	none
T11	TransGlobal Fret	0.854	0.838	none
T12	Orbitalys	1.000	1.000	none
T13	Électis EMS	1.000	1.000	none
T15	AvioSys Intégration	0.896	0.884	none
T17	AvioSys Intégration	0.894	0.884	none
T17	CompoDis Europe	0.998	0.998	none
T18	CompoDis Europe	1.000	1.000	none
T18	Meridian Semi	0.704	0.690	none

Table 28: Arm B at turn 4, node by node

Node	Recorded influence of the displayed forecast
AvioSys Intégration	Confirms existing assessment
CompoDis Europe	Revised downward
Électis EMS	Confirms existing assessment
Meridian Semi	Confirms existing assessment
NovaFab Semiconductors	Revised downward
Orbitalys	Revised downward
SilPure Materials	Revised downward
TransGlobal Fret	Revised downward
Total	5 downward, 3 confirmed, 0 upward, 8 of 8 responding

NovaFab is the node that carries the scenario's critical accident event two turns later, at turn 6.

E Satellite Capacity Estimation: Protocol and Data

This appendix records the operational detail behind the opening of Section 4.9, so that its measurements can be audited or repeated.

Data Sources

All sources are public French open-data services, accessed over standard web map and feature services and cached on disk so that a re-run does not depend on service availability. Table 29 lists each one with the role it plays in the pipeline.

Table 29: Data sources of the capacity pipeline

Source	Use
Orthophotography, 20 cm	Primary imagery for detection and segmentation
Orthophotography, 5 to 10 cm	Partial coverage, used only for the resolution control of Figure 22
LiDAR surface model	Building height where available, delivered as 32-bit float raster
National topographic database	Building footprints and, where populated, height
Classified installations register	Declared storage volume, used only to calibrate the fill factor

All geometry is handled in the national metric projection, in which the ground sample distance is exactly the bounding-box width in metres divided by the image width in pixels rather than an approximation.

Calibration of the Fill Factor

The fill factor relates declared storage volume to the geometric envelope. For each site in the regulatory register, the nearest building footprint is matched, the geometric volume is computed as footprint area times eaves height, and the ratio of declared to geometric volume is recorded. Ratios outside $[0.05, 20]$ are discarded as either an aberrant declaration or a mismatched building. The production constant of 0.80 is attributed in the source to a median over 122 matched sites with an interquartile range of $[0.52, 1.22]$.

That run cannot be reproduced from the retained artifacts, and the discrepancy is recorded here in full. The calibration script writes its detailed output to a file that is not present in the repository, so the only auditable sample is the 68-site corpus retained for other purposes. Recomputing the identical ratio over those 68 sites gives a median of 0.966, an interquartile range of $[0.80, 2.00]$, and 47% of sites inside the $[0.7, 1.4]$ band, against the 70% the calibration script itself prints as its acceptance target. The available evidence does not distinguish between a biased remnant sample and an understated production constant. Recollecting the register cleanly, with the departmental loop and pagination the script provides, and rederiving the factor, is the resolution.

One matching rule is worth recording because it changes the answer materially. Only the largest building within the search radius is matched, and neighbouring buildings are not summed: summing them moves the median ratio from 0.80 to 0.58, because a declared volume belongs to one warehouse while the footprints of a whole industrial estate do not.

Annotation Protocol

The dataset comprises 442 georeferenced image chips, each retained at a single location, a single resolution and a single vintage, with a sidecar file recording its geographic transform. Of these, 351 form the training split and 91 the validation split.

Annotators mark one class, the continuous extent of facade served by a dock apron, and the following rules apply.

- Mark an equipped facade even where no vehicle is parked. The apron and canopy remain visible, and a facade annotated only when a vehicle is present teaches the model to detect vehicles.
- Do not annotate a trailer holding yard that is not adjacent to a wall.
- Do not annotate an aeronautical tarmac or a small-business forecourt. Both are paved and deep but serve no docks.
- A facade without docks is annotated empty. That is a useful negative, not an omission, and roughly 40% of the target sample is negative by design.
- Ignore everything above the facade line, which is roof. Rooflights and sawtooth roofs are perfectly periodic and are the principal source of false positives.
- Annotate every visible dock zone in a chip regardless of which building owns it. Ownership is resolved downstream by geometric clipping, so the model is never asked to infer it from pixels, and annotating a zone in one chip but not in the overlapping neighbour would give the model two contradictory labels for the same ground.

Sampling is stratified by footprint area across three bands, roughly 1 500 to 5 000 m², 5 000 to 20 000 m² and above 20 000 m², and across five site categories covering small enterprises, aerospace, refrigerated, very large and retail distribution. An unstratified random draw over-represents large warehouses, and the model inherits that bias.

Training Configuration

The model is a segmentation network trained from pre-trained weights at a 640 pixel input size, with rotation augmentation over the full circle, horizontal and vertical flips, and scale jitter. Rotation invariance is obtained by augmentation rather than by geometric rectification of the facade, which is what allows the model to accept any building shape and orientation without requiring a footprint as input. Training ran on CPU throughout, with no GPU available, which is the practical reason the input size was not raised further and the context for the negative resolution result of Table 21.

Measured Facts Recorded for Reuse

- Empty bay contrast at verified locations: 0.18σ , 0.22σ and 0.01σ , against 5.3σ to 6.4σ at occupied bays on the same facades, over a full sweep of assumed apron depths from 0.3 m to 20 m.

- Dock pitch coherence between facades of the same building: 0.03 to 0.05 m, which provides a free quality control.
- Roof overhang beyond the footprint edge: 0 to 14.4 m. This is why the footprint edge is not the start of the apron, and it was the cause of an earlier defect in which rooflights were counted as bays.
- Trailer distance to building: median up to 126 m on large platforms; at one site only 8 of 89 detected trailers lay within 26 m. Trailers are not a proxy for docks.
- Zero-shot coverage before the learned model: 25 of 70 sites, or 36%.
- Prompt sensitivity of the promptable segmentation model used for site sampling: a prompt naming an industrial warehouse returns 88 752 m², some 45% of the view, where a prompt naming a warehouse roof returns the 13 478 m² of the roof itself. The first is useful for locating a site, the second for measuring it, and they are not interchangeable.

Licensing Constraint

The detection and segmentation library used is distributed under a strong copyleft licence. This has no consequence for internal research use, but it becomes determinative the moment the capability is distributed or exposed over a network. The options are a commercial licence from the vendor or replacement of the backbone with a permissively licensed equivalent. This is recorded as a governance decision to be taken deliberately rather than discovered at deployment.